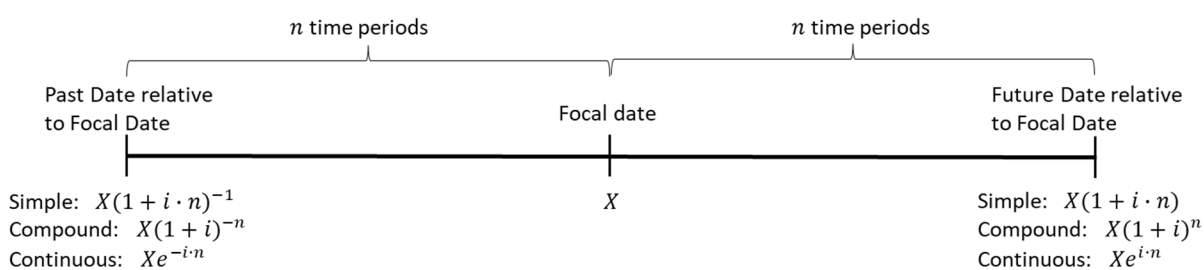


Financial Mathematics

A Concise Guide and Handbook

3rd Edition



by **Stephen Fratini**

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Preface to Third Edition

There are two kinds of forecasters: those who don't know, and those who don't know they don't know. — John Kenneth Galbraith

Never test the depth of a river with both feet. — attributed to Warren Buffett

This book is intended for readers who want to understand the mathematical foundations of fundamental financial concepts—interest, annuities, loans, bonds, and capital budgeting—without requiring an extensive background in mathematics. It is written for a broad audience, including students of business and finance, individual investors, small business owners, and mathematically trained readers who wish to see how familiar tools apply in a financial setting.

The mathematical prerequisites are intentionally modest. A working knowledge of basic algebra is sufficient, along with a light exposure to limits and elementary ideas from calculus. Where formulas are introduced, they are typically derived in the text, not merely stated, so that the reader can see both the structure and the reasoning behind them. Readers seeking a more comprehensive or formal treatment of financial mathematics may wish to consult standard references such as Brown and Kopp [2], Broverman [3], or Alhabeeb [4].

An important theme of this book is that, in many cases, the *financial interpretation* of a problem is more subtle than the mathematics used to solve it. The formulas themselves are often straightforward; understanding when and how to apply them requires greater care. To highlight this distinction, I occasionally include **Author's Remarks**—brief comments that step outside the main exposition to emphasize intuition, clarify assumptions, or offer perspective.

What's New in the Third Edition

This edition expands the book's scope with three new chapters and introduces several refinements aimed at improving clarity and usability:

- Summary tables have been added throughout the text to consolidate key formulas and provide quick reference points.
- Additional explanations have been incorporated in selected sections to clarify concepts that may benefit from further discussion.
- Minor typographical errors from earlier editions have been corrected.
- Section 8 introduces the valuation of stocks, covering the constant dividend model and the Gordon Growth Model. These connect equity pricing directly to the perpetuity formulas of Section 4 and extend the present-value framework to uncertain, growing cash flows.
- Section 9 covers derivatives and structured products, including options, futures, forwards, and interest rate swaps, as well as the structured credit instruments — mortgage-backed securities, collateralized debt obligations, and credit default swaps — that played a central role in the 2008 financial crisis. Where appropriate, the mathematics draws on the compound interest and present-value tools developed in earlier chapters.
- Section 10 brings together the mathematical tools from across the book through three interconnected case studies. Following Maria and Anthony DeLuca, a married couple in Wilkes-Barre, Pennsylvania, the studies apply compound interest, annuities, loan

amortization, capital budgeting, and stock valuation to the realistic problems of home purchase, business formation, and retirement planning.

Together, these additions and refinements extend both the scope and the accessibility of the book, while preserving the gradual, motivated approach to financial mathematics that characterizes earlier editions.

In addition to the theoretical development, many of the concepts in this book are supported by practical computational tools. A collection of Python programs has been developed to illustrate and implement the key ideas, and these are freely available online:

https://github.com/sfratini33/art-of-managing-things-external/tree/master/fin_math/code

These programs are designed to be accessible even to readers with little or no programming experience. While brief descriptions are provided, this book is not intended to serve as a tutorial in Python. The code was developed using the open-source Anaconda distribution and Jupyter Notebook environment, though it can be adapted to other platforms. Readers interested in learning Python will find many excellent online resources.

For certain topics—particularly in capital budgeting—Microsoft Excel is also used as a computational aid. Functions such as NPV, IRR, and MIRR are referenced where appropriate, and supporting spreadsheets corresponding to examples in the text are available alongside the Python programs.

This third edition reflects a continued effort to balance clarity, rigor, and practical applicability. The goal is not only to present formulas, but to develop a coherent framework for thinking about financial problems—one that emphasizes equivalence of cash flows, the role of time, and the consistent use of interest-rate models.

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Other Books by the Author:

- *The Art of Managing Things (2nd edition)*, self-published on Amazon, <https://www.amazon.com/Art-Managing-Things-Stephen-Fratini-ebook/dp/B07N4H4YWH/>, January 2019.
- *Mathematical Thinking: Exercises for the Mind (2nd Edition)*, self-published on Amazon, <https://www.amazon.com/dp/B0CL34FRP1>, October 2023.
- *Math in Art, and Art in Math*, self-published on Amazon, <https://www.amazon.com/dp/B091D1F8MB>, March 2021.
- *Algebra through Discovery and Experimentation*, self-published on Amazon, <https://www.amazon.com/dp/B09B5L9WLS>, July 2021.
- *The Struggle Against Chaos*, self-published on Amazon, <https://www.amazon.com/dp/B09BLPQ86Q>, July 2021.
- *Mathematical Vignettes: Number theory, stochastic processes, game theory, cryptography, linear programming and more*, self-published on Amazon, <https://www.amazon.com/Mathematical-Vignettes-stochastic-cryptography-programming-ebook/dp/B0BBP1PBJQ/>, August 2022.
- *Learning Math through Puzzles: Number properties, counting, sequences and series, algebra, functions, and mathematical reasoning*, self-published on Amazon, <https://www.amazon.com/dp/B0BZFRZP5B>, March 2023.
- *Mathematical Vignettes: Volume II: Topics from combinatorial design, magic squares, finite geometry, abstract algebra, error correcting codes, geometric packing problems and much more*, self-published on Amazon, <https://www.amazon.com/dp/B0CM1CLSK8>, October 2023.
- *Shape Up and Solve It!: Learn Geometry Through Puzzles*, self-published on Amazon, <https://www.amazon.com/dp/B0CRS7DRWF>, January 2024.
- *Mathematical Vignettes III (2nd edition): Introductions to non-Euclidean geometry, topology and complex analysis*, self-published on Amazon, <https://www.amazon.com/Mathematical-Vignettes-III-Introductions-non-Euclidean/dp/B0F1N6522B>, March 2025.
- *The Shape of Space: A Guided Tour of Vectors, Matrices, Tensors and Markov Chains*, self-published on Amazon, <https://www.amazon.com/dp/B0FQCGRNW1>, September 2025.
- *Beyond Limits: Infinitesimals, Abstract Algebra, Category Theory, Synthetic Differential Geometry, and Automatic Differentiation*, self-published on Amazon, <https://www.amazon.com/Beyond-Limits-Infinitesimals-Differential-Differentiation/dp/B0G7FXMQ6V>, January 2026.
- *A Symphony of Patterns: Endless Variations on a Theme*, self-published on Amazon, <https://www.amazon.com/dp/B0GRBSXRDT>, March 2026.

Electronic versions of my books (in pdf) are available (free of charge) at

- https://github.com/sfratini33/art-of-managing-things-external/tree/master/free_books

My Video Series:

- *Linear Algebra*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_UniSLwX82IFFqN1OsloWPK
- *Matrices - Some Less Common Properties*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_Umu0emwEcVtjDQs7O4ioZgr
- *Vector and Matrix Calculus*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_UlI8xLnrTOQaNe-KX6v0EDp
- *Tensors*, https://www.youtube.com/playlist?list=PLSaIWQAk7_UltOIPJUX7nRHxIZ1RMYJXN
- *Non-negative matrices, Markov chains and Least Squares*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_UlvuQ49JQUmV6GZGU7U1voD
- *Puzzles to Exercise the Mind*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_UmCBhNtgSpmgss0XAJq392j
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- *Complex Analysis*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_UkA9q_2Ptfz9mK9yzmngTRe
- *Abstract Algebra*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_UmjFOzc39_O5U5o953dJ22N
- *Number Theory Problems*, https://www.youtube.com/playlist?list=PLSaIWQAk7_UmEL-RoC5MqszhLIUGP1ft
- *Combinatorics*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_UmH81dp3K9ApvFviltCaimW
- *Graph Theory*, https://www.youtube.com/playlist?list=PLSaIWQAk7_UICE6yqQeEYjl-P5zzkzb-d
- *Financial Mathematics*,
https://www.youtube.com/playlist?list=PLSaIWQAk7_Ukha7O8LAXIJw8IEIUAj9q

If you are working from a paper copy of my book, it may be easier to visit my YouTube channel rather than typing the long URLs.

www.youtube.com/@sfratini

1 Introduction

Our favorite holding period is forever. — Warren Buffett

An economist is an expert who will know tomorrow why the things he predicted yesterday didn't happen today. — Laurence J. Peter

1.1 Purpose

The primary purpose of this book is to present the mathematical foundations underlying several fundamental financial concepts, including interest compounding, annuities, loans, bonds, and capital budgeting. While these topics are often introduced procedurally, this text emphasizes *why* the formulas work, not just how to apply them.

A secondary objective is to provide practical computational tools—primarily in Python—to support and reinforce the concepts developed throughout the book. These tools are intended to bridge the gap between theory and application.

Author's Remark: While the mathematical techniques used in this book are generally straightforward, the financial interpretations can be subtle. Careful attention should be given to assumptions, timing of cash flows, and the meaning of equivalence between different financial arrangements.

1.2 Intended Usage

This book is designed to be flexible in how it is used. In particular, it may serve:

- as a self-study guide for readers seeking to understand the mathematics behind common financial concepts
- as a supplementary text for advanced high school or undergraduate courses in business, finance, or applied mathematics
- as a concise reference for key formulas and methods in financial mathematics
- Readers using the book for self-study are encouraged to work through the examples and exercises, as much of the understanding comes from actively applying the concepts.

1.3 Prerequisites

The mathematical prerequisites are minimal. A solid understanding of basic algebra is sufficient for most of the material. Some exposure to elementary calculus concepts—particularly limits and exponential functions—may be helpful but is not strictly required.

A recurring theme in the book is the summation of sequences and series, especially geometric progressions. The necessary background is introduced in Section 2.

From a financial perspective, no prior experience is assumed. However, readers should be aware that some concepts—particularly those involving equivalence of cash flows and time value of money—require careful thought and may not be immediately intuitive.

To use the accompanying Python programs, the reader should be able to install and run a Python environment such as Anaconda. The programs are designed with graphical interfaces and are

generally usable without prior programming experience. In some cases, output is written to CSV files, which can be opened in spreadsheet applications such as Microsoft Excel.

1.4 Conventions

This book adopts standard U.S. numerical conventions, using a period as the decimal separator and a comma as the thousands separator (e.g., \$24,000.98). Readers should be aware that other conventions are used internationally. All monetary amounts are stated in U.S. dollars.

Interest rates are always expressed as decimals in formulas, regardless of how they are described in the surrounding text. A rate stated as "6% per year" enters every formula as 0.06. This distinction is important: substituting 6 instead of 0.06 will produce answers that are off by a factor of 100.

Unless stated otherwise, rates and time periods are assumed to be annual. When a monthly, quarterly, or other non-annual period is used, this is stated explicitly, and the rate is converted accordingly before being substituted into any formula.

Annuity payments are assumed to occur at the end of each period — the ordinary annuity convention — unless a specific exception is noted. Annuities due, deferred annuities, and other variants are always identified as such when they arise.

In cash flow analysis, inflows (revenues, receipts, loan proceeds) are treated as positive quantities and outflows (costs, repayments, initial investments) as negative. This convention is applied consistently throughout the capital budgeting material of Section 7 and the case studies of Section 10.

Dollar amounts in final answers are rounded to two decimal places. Intermediate values are carried to additional decimal places to limit accumulated rounding error. Interest rates and other dimensionless quantities are generally expressed to four decimal places (e.g., 0.0542) unless greater precision is warranted.

The symbol P is used in two distinct contexts: it denotes the principal of a loan in Sections 3 through 6, and the price of a bond or stock in Sections 6, 8, and 9. The intended meaning is clear from context in each case, but readers moving between chapters should be aware of this dual usage.

Repeating decimals are represented using an overhead bar notation. For example, a repeating decimal such as 0.333... is written using a bar over the repeating digit or digits. This notation extends naturally to longer repeating sequences.

The symbol $\approx$ denotes an approximate equality — one that results from rounding or from a formula that is itself an approximation. An exact equality is always written with $=$.

In general, notation is chosen to prioritize clarity and consistency over compactness. Where multiple conventions exist in the literature, the one that best supports understanding in context is used.

1.5 Outline

The book is organized as follows:

- **Section 1** introduces the purpose, structure, and conventions of the text.
- **Section 2** provides mathematical background on progressions and series, which form the foundation for many financial formulas.

- **Section 3** develops the core concepts of interest, including simple, compound, and continuous interest, as well as the time value of money—a central theme for the remainder of the book.
- **Section 4** focuses on annuities, including accumulated and present value formulations.
- **Section 5** examines loan repayment structures, including amortization and refinancing.
- **Section 6** introduces bonds and their valuation.
- **Section 7** covers capital budgeting techniques such as Net Present Value (NPV) and Internal Rate of Return (IRR).
- **Section 8** introduces the valuation of stocks. Starting from the perpetuity formula of Section 4, it derives the constant dividend model and the Gordon Growth Model for stocks with growing dividends, and discusses the relationship between dividend yield, capital gains, and required return.
- **Section 9** covers derivatives and structured products. It introduces options (including put-call parity), futures and forwards (priced via the compound interest formula), and interest rate swaps. It also examines mortgage-backed securities, collateralized debt obligations, and credit default swaps, and discusses their role in the 2008 financial crisis.
- **Section 10** presents three interconnected case studies following the same couple across three stages of their financial lives: saving for a home and their children's education, opening their own law firm, and planning for retirement. Each study draws on multiple tools from earlier sections, illustrating how the concepts of this book combine in practice.

Throughout the book, summary tables are provided to consolidate key formulas and serve as quick reference points. Supporting Python programs and Excel-based tools are available online to assist with computations and to illustrate the concepts in practice.

The code for the Python programs used in this book is available online at https://github.com/sfratini33/art-of-managing-things-external/tree/master/fin_math/code.

2 Background

The safest way to double your money is to fold it over and put it in your pocket. — Kin Hubbard

2.1 Progressions

Progressions are sequences where each term is based on mathematical operations applied to one or more previous terms. Progressions are essential in finance concerning topics such as interest accumulation.

2.1.1 Arithmetic Progressions

An arithmetic progression is an ordered list (i.e., sequence) of terms where each term is the sum of the previous terms plus some common amount (which can be positive or negative). For example,

- 1, 4, 7, 10, 13, ... is an arithmetic progression with difference 3 between successive terms
- 10, 5, 0, -5, -10, ... is an arithmetic progression with difference -5
- -0.125, 0.125, 0.375, 0.625, 0.875, ... is an arithmetic progression with difference 0.25.

In general, an arithmetic progression with initial term x_1 and difference d can be represented as

$$x_1, x_1 + d, x_1 + 2d, \dots$$

where the n^{th} term is $x_n = x_1 + (n - 1)d$.

The sum of the first n terms (call it S_n) can be computed by writing the progression forward and backward, and then adding, i.e.,

$$S_n = x_1 + (x_1 + d) + (x_1 + 2d) + \dots + (x_n - 2d) + (x_n - d) + x_n$$

$$S_n = x_n + (x_n - d) + (x_n - 2d) + \dots + (x_1 + 2d) + (x_1 + d) + x_1$$

which implies (after some cancellation of terms)

$$2S_n = n(x_1 + x_n)$$

or

$$S_n = \frac{n}{2}(x_1 + x_n)$$

2.1.2 Geometric Progressions

A geometric progression is an ordered list of terms where each term is a multiple of the preceding term. For example,

- 3, 15, 75, 375, 1875 ... is a geometric progression with a common multiple of 5
- -10, 1.25, -0.15625, 0.01953125, ... is a geometric progression with a common multiple -0.125.

In general, a geometric progression with initial term a and multiplier r can be represented as

$$a, ar, ar^2, ar^3, \dots$$

where the n^{th} term is $x_n = ar^{n-1}$ where $n = 1, 2, 3, \dots$

The sum of the first n terms is

$$S_n = a + ar + ar^2 + \dots + ar^{n-1} \quad (1)$$

Multiplying by r on both sides of the equation gives

$$rS_n = ar + ar^2 + ar^3 + \dots + ar^n \quad (2)$$

Subtracting equation (2) from (1) yields

$$\begin{aligned} S_n - rS_n &= a - ar^n \\ S_n &= \frac{a(1 - r^n)}{1 - r} \end{aligned}$$

In subsequent parts of this book, we use a slightly modified version of the above formula (gotten by multiplying the top and bottom of the fraction by -1)

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

As an example, consider the sum of the first 5 terms in the geometric sequence with multiplier 2 and with initial term $a = 100$. Using former version of the summation formula for geometric series, we get the following

$$S_5 = \frac{100(1 - 2^5)}{1 - 2} = -100(-31) = 3,100$$

For $-1 < r < 1$, the term r^n approaches zero for n approaching infinity. Apply this observation to the formula for S_n gives the following result:

$$a + ar + ar^2 + ar^3 + \dots = \frac{a}{1 - r}, -1 < r < 1$$

For example, the sum of the series with first term 2 and multiplier $\frac{1}{3}$ is

$$2 + \frac{2}{3} + \frac{2}{3^2} + \frac{2}{3^3} + \dots = \frac{2}{1 - \frac{1}{3}} = 3$$

2.1.3 Recursive Progressions

A recursive progression is a progression in which each term is defined using one or more of the previous terms where the first several terms are given. For example, $a_n = a_{n-1} + a_{n-2}$ with $a_1 = a_2 = 1$ defines the famous Fibonacci sequence. When expanded, the recursion gives the following progression:

1	1	2	3	5	8	13	21	34	55	89	144	...
---	---	---	---	---	---	----	----	----	----	----	-----	-----

For another example, consider the tetranacci numbers which start with four predetermined terms, with each term afterwards being the sum of the preceding four terms, i.e., $a_n = a_{n-1} + a_{n-2} + a_{n-3} + a_{n-4}$ with $a_1 = a_2 = a_3 = 0$ and $a_4 = 1$. When expanded, the recursive formula yields the following:

0	0	0	1	1	2	4	8	15	29
56	108	208	401	773	1,490	2,872	5,536	10,671	20,569
39,648	76,424	147,312	283,953	547,337	1,055,026	...			

The Fibonacci and tetranacci progressions are examples of difference equations.

[Author's remark: Recursive progressions are not discussed further in this book. They are mentioned here for completeness with regard to the types of progressions.]

2.2 Supporting Programs

2.2.1 Arithmetic Progression Calculator

File name: arith_prog.py (see the preface for location of this file and instructions about running Python programs)

This program generates an arithmetic progression, given a starting value (it can be a decimal, and it can be positive or negative), an increment (it can be a decimal, and it can be positive or negative) and a desired number of terms. If the number of terms is large (fills the output window), then a scroll bar will appear on the right.

There is another button that allows one to request a given term in an arithmetic progression without computing the entire progression up to that term. For example, the program can compute the 101st term of the arithmetic progression with initial term 71 and increment 19035, which is 1,903,571.

There is a button to clear the input so that another progression can be computed. There is also a button to quit the program. (For the record, the clear and quit buttons are present in the other programs provided in support of this book.)

For those looking to learn the tkinter module of Python, this program may be helpful as an example. The tkinter module focuses on the user interface.

There are several tkinter tutorials on the Internet, e.g., <https://www.geeksforgeeks.org/python-tkinter-tutorial/> and the YouTube video entitled “Tkinter Course - Create Graphic User Interfaces in Python Tutorial” at <https://youtu.be/YXPyB4XeYLA>.

2.2.2 Geometric Progression Calculator

File name: geo_prog.py

This program generates a geometric progression, given a starting value (it can be a decimal, and it can be positive or negative), an increment (it can be a decimal, and it can be positive or negative) and a desired number of terms. The sum of the progression is also computed and displayed.

The output is listed in the GUI, with one number per line.

There is another button that only generates the n^{th} term in a given progression.

2.3 Exercises

1. Show that the sum of an arithmetic progression can also be represented as $S_n = \frac{n}{2}(2x_1 + (n-1)d)$.
2. Find the 335th term in the arithmetic progression 3, 9, 15, 21, ... and also find the sum of the first 335 terms.

Hint: Use the Python program on arithmetic progressions.

3. If the 57th term in an arithmetic progression is 677 and the difference between terms is 12, what is the first term.

Hints: One approach is to use the equation for the n^{th} term in an arithmetic progression and solve for x_1 . Another approach is to use the Python program with 677 as the initial term, -12 as the difference and 57 as the desired term (basically working backward with negative steps).

4. Find the 237th term in the geometric progression with initial term 2 and multiplier 1.1.

Hint: Use the Python program on geometric progressions.

5. Find the sum of the first 10 terms in the geometric progression with initial term 25 and multiplier 7.

Hint: Use the Python program on geometric progressions.

6. If the 8th term in a geometric progression is 2,470,629 and the multiplier is 7, what is the first term?

Hint: Work backwards by dividing by 7 (eight times).

7. The Lucas progression is given by 2, 1, 3, 4, 7, 11, 18, ... Determine a recursive formula that yields the Lucas progression.

Hint: The initial two terms (2 and 1) are given. The recursive formula should start with the third term.

8. Determine $\sum_{n=0}^{\infty} \sum_{m=0}^n \frac{1}{2^{m+n}}$.

Answer: First rewrite the inner sum: $\sum_{m=0}^n \frac{1}{2^{m+n}} = \frac{1}{2^n} \sum_{m=0}^n \frac{1}{2^m}$

The finite geometric sum is $\sum_{m=0}^n \frac{1}{2^m} = \frac{1 - (\frac{1}{2})^{n+1}}{1 - \frac{1}{2}} = 2 \left(1 - \frac{1}{2^{n+1}}\right)$

Therefore, $\sum_{m=0}^n \frac{1}{2^{m+n}} = \frac{1}{2^n} \cdot 2 \left(1 - \frac{1}{2^{n+1}}\right) = \frac{2}{2^n} - \frac{1}{2^{2n}}$

So, the original double sum becomes $\sum_{n=0}^{\infty} \left(\frac{2}{2^n} - \frac{1}{2^{2n}}\right)$

Now compute each geometric series:

$$\sum_{n=0}^{\infty} \frac{2}{2^n} = 2 \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = 2 \cdot \frac{1}{1-\frac{1}{2}} = 4, \quad \sum_{n=0}^{\infty} \frac{1}{2^{2n}} = \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}.$$

$$\text{Thus, } \sum_{n=0}^{\infty} \sum_{m=0}^n \frac{1}{2^{m+n}} = 4 - \frac{4}{3} = \frac{8}{3}.$$

3 Interest

The stock market is designed to transfer money from the Active to the Patient. — Warren Buffett

If all the economists were laid end to end, they'd never reach a conclusion.

— George Bernard Shaw

3.1 Simple Interest and Comparison to a Balloon Loan

Simple interest is computed based on the amount of principal and time duration, but without any compounding. Simple interest can be applied to a loan or to money saved in a bank account.

In its most basic form, simple interest entails an interest rate r (which is typically stated per year), a given amount of principal (P) and the term (t) when the principal and interest are paid back to the lender. The term is typically stated in years or fractions thereof. Under these assumptions, the amount of simple interest plus the original principal is given by the formula:

$$P + Prt$$

which can be written as

$$P(1 + rt)$$

For example, Mr. Smith purchases a car from a car dealer. The car costs \$28,440. Mr. Smith agrees to make an initial (upfront) payment of \$20,000 and the car dealer agrees to allow Mr. Smith to pay the remaining amount, with 6% annual interest, in 7 years. At the end of 7 years, Mr. Smith must pay the principal (\$8,400) plus simple interest over the 7 years, i.e., $\$8,400 * .06 * 7 = \$3,528$.

In comparison to the above example, let's say that Mr. Smith pays the principal back to the car dealer at the end of 7 years but he must make monthly interest payments in the amount of $\$8,400 * \left(\frac{.06}{12}\right) = \42 . The accumulated amount of interest paid to the car dealer is an arithmetic progression, i.e., 42, 84, 126, ..., 3,528. This is an example of what is called a **balloon loan** that allows for the entire principal to be paid back at the end of the loan's term. To be clear, this is not simple interest. Balloon loans are discussed further in Section 5.1.4.

3.2 Compound Interest

Compound interest is computed on the sum of the original principal plus accrued interest. Compound interest can be thought of as "interest on interest," and grows faster than simple interest of the same rate.

The term nominal annual interest rate or just **nominal interest rate** as used here is the interest rate "as stated," without adjustment for the effect of compounding. An interest rate is called nominal if the frequency of compounding (e.g., monthly or daily) is not identical to the basic time unit in which the nominal rate is quoted (normally a year). For example, a bank may advertise a nominal yearly interest rate of 6% compounded daily which (as will be shown at the end of Section 3.4) is effectively about 6.183% when daily compounding is applied.

Let P be the initial principal (e.g., money placed in a bank account), r be the nominal annual interest rate, n be the compounding frequency per year, and t be the length of time the interest is applied (using the same units as r). Table 1 shows the computation of principal after the 1st, 2nd and

final compounding intervals. The entries in the right-hand column constitute a geometric progression, with initial term P and multiplier $\left(1 + \frac{r}{n}\right)$.

Some financial books and articles use the notation $i^{(n)}$ to indicate a nominal annual interest rate that is compounded n times per year. Thus, our r is $i^{(n)}$. **[Author's remark:** This notation is confusing. First, it is not a power of i . Secondly, if the nominal annual interest rate is i , we would say that $i^{(n)} = i$ for all values of n (also confusing). So, I have avoided using this notation as much as possible and have taken the approach of writing out in prose what is meant, i.e., “nominal annual interest of 6% compounded monthly” rather than $i^{(12)} = .06$.]

Table 1. Computation of Final Principal using Compound Interest

Initial principal	P
Principal after first compounding interval	$P + \frac{r}{n}P = P\left(1 + \frac{r}{n}\right)$
Principal after second compounding interval	$P\left(1 + \frac{r}{n}\right) + P\left(1 + \frac{r}{n}\right)\left(\frac{r}{n}\right)$ $= P\left(1 + \frac{r}{n}\right)\left(1 + \frac{r}{n}\right)$ $= P\left(1 + \frac{r}{n}\right)^2$
...	...
Principal after nt compounding intervals	$P\left(1 + \frac{r}{n}\right)^{nt}$

Alternately, we can let $i = \frac{r}{n}$ be the interest rate per compounding period and $s = nt$ be the number of compounding intervals. These substitutions yield an equivalent formula for the accumulated principal after s compounding intervals, i.e., $P(1 + i)^s$.

As an example, consider a scenario where Ms. Ortega deposits \$10,000 into a bank account that advertises a 2% nominal annual interest rate which is computed monthly. Assuming no withdrawals have been made, how much is the account worth after 3 years and 6 months? Using the formula for compound interest with $P = \$10,000$, $r = .02$, $n = 12$ and $t = 3.5$, the final principal is $\$10,000\left(1 + \frac{.02}{12}\right)^{42} = \$10,724.46$.

3.3 Continuous Interest

At first thought, one might think there is no bound to the amount of interest that can be accrued if the compounding is done infinitely often. However, that is not the case. As the number of compounding intervals approach infinity, the formula converges.

[Author's remark: The following derivation entails calculus. If you are not familiar with calculus, don't worry. The key point here is that if the number of compounding periods is infinite, the formula for accumulated interest is simply Pe^{rt} where $e \cong 2.71828$.]

In terms of calculus, we want to determine

$$\lim_{n \rightarrow \infty} P\left(1 + \frac{r}{n}\right)^{nt}$$

where r is the nominal annual interest rate (compounded n times per year where n approaches infinity) and t is duration of interest accumulation in years.

To this end, let $x = n/r$ and substitute into the above expression to get

$$\lim_{n \rightarrow \infty} P \left(1 + \frac{r}{n}\right)^{nt} = \lim_{x \rightarrow \infty} P \left(1 + \frac{1}{x}\right)^{xrt} = P \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{x}\right)^x \right]^{rt}$$

From calculus, we have that $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$ where $e \cong 2.71828$ (known as Euler's number). Thus, the formula for continuous interest is given by Pe^{rt} .

Returning to the example of Ms. Ortega's bank account, assume the same interest rate and time period (3.5 years) but with continuous interest. In this case, the account would be worth $\$10,000e^{(.02)(3.5)} = \$10,725.08$ which is only slightly more than monthly compounding!

3.4 Effective Rate of Interest

As defined by Brown and Kopp in their book *Financial Mathematics: Theory and Practice* [1]

The **effective rate of interest** is the amount of money that one unit of capital, invested at the beginning of a period, will earn during the period, where interest is paid at the end of the period.

[Other sources define effective interest rate as pertaining to a period of one year. For example, Wikipedia provides the following definition:

The effective interest rate (EIR), effective annual interest rate, annual equivalent rate (AER) or simply effective rate is the interest rate on a loan or financial product restated from the nominal interest rate and expressed as the equivalent interest rate if compound interest were payable annually in arrears.

This is perhaps a more common definition but less general than the definition from Brown and Kopp.]

For example, if Ms. DeMarco buys a house (as an investment) at \$200,000 and sells the house 18 months later (assuming no capital investment in the house) at \$275,000, then the gain is \$75,000 over 18 months. What is the percentage gain? In other words, what percentage (call it x) of 200,000 equals 75,000? This can be converted into a precise equation, i.e.,

$$\frac{x}{100} \cdot 200,000 = 75,000$$

which yields, $x = .375$ or 37.5% over 1.5 years, which equates to $\frac{37.5}{1.5} = 25\%$ per year (assuming simple interest).

Take the same problem as above but assume the house was held 30 months before being sold. In this case, the effective interest rate per year is $\frac{37.5}{2.5} = 15\%$ (assuming simple interest).

Again, using the same base problem, but now assuming the owner incurs \$25,000 in capital improvements and nets \$20,000 from renting the house during the 2.5 years the house is owned by Ms. DeMarco. The net capital gain is now $75,000 + 20,000 - 25,000 = \$70,000$. So, the effective interest rate per year is

$$\frac{\left(\frac{(70,000)(100)}{200,000}\right)}{2.5} = \frac{35\%}{2.5} = 14\%$$

In general, if the value of an investment is $V(t)$ at time t and $V(\hat{t})$ at some later time $\hat{t}$, then the effective interest rate from time t to $\hat{t}$ is given by the formula

$$\frac{V(\hat{t}) - V(t)}{V(t)}$$

In words, the effective interest rate formula indicates the increase or decrease (as a fraction) of an investment from time t to a later time $\hat{t}$.

The effective interest rate provides a way to compare the growth of simple and compound interest. Assume the starting principal is P and the interest rate per time period is i .

In case of compound interest, the effective interest rate in period n is

$$\frac{P(1+i)^n - P(1+i)^{n-1}}{P(1+i)^{n-1}} = \frac{1+i-1}{1} = i$$

In the case of simple interest, the effective interest rate in period n is

$$\frac{P(1+ni) - P(1+(n-1)i)}{P(1+(n-1)i)} = \frac{i}{1+(n-1)i}$$

The effective interest rate for compound interest remains constant over time. However, the effective interest rate for simple interest decreases over time (i.e., as n gets larger) because there is no compounding.

It is common for banks to state the effective yearly interest rate when principal is compounded at a nominal yearly rate of r . This is referred to as **Annual Percentage Yield (APY)**. The APY (represented as the variable y below) is derived by inserting the compound interest formula (with $t = 1$) into the effective interest rate formula to get:

$$y = \frac{P\left(1 + \frac{r}{n}\right)^n - P}{P} = \left(1 + \frac{r}{n}\right)^n - 1$$

For example, if a bank advertises a nominal yearly interest rate of 6% compounded daily, then the APY is $\left(1 + \frac{.06}{365}\right)^{365} - 1 \cong .06183$ or about 6.183%.

It is also possible to solve for the nominal yearly compound interest rate that is equivalent to a given APY, i.e.,

$$r = n((y + 1)^{1/n} - 1)$$

For example, what nominal interest rate (compounded monthly) is equivalent to an APY of 8%?

Using the above equation with $n = 12$ and $y = .08$, we get $r = 12\left(1.08^{\frac{1}{12}} - 1\right) = .077206$ or 7.7206%.

A common use of effective interest rate is to normalize other types of interest accumulation with respect to simple interest. It is also possible to relate (via formulas) different types of interest (this is covered in Section 4.7).

3.5 Present Value

Present value refers to the sum of money which if invested now at a given rate of interest will accumulate to a specified amount at a particular future date.

For example, what is the present value of \$25,000 three years from now assuming 5% interest compounded monthly? In other words, how much money needs to be invested now at 5% interest compounded monthly (with no additional investments or withdraws) to have \$25,000 in three years? In terms of the compound interest formula, the following needs to be solved for P

$$\$25,000 = P \left(1 + \frac{.05}{12} \right)^{(12)(3)}$$

Solving this equation, we get $P = \$21,524.41$. Someone might do such a calculation if he or she wants to buy a \$25,000 car three years from now and wants to know how much to put into a given fixed interest investment (such as a government bond) right now.

Let PV be the present value and FV be the future value of PV over s compounding periods with interest rate i per compounding period. In terms of the compound interest formula, we have

$$FV = PV(1 + i)^s$$

Solving for FV yields the following formula for PV in the case of compound interest

$$PV = FV(1 + i)^{-s}$$

[The other formulation for compound interest, i.e., $P \left(1 + \frac{r}{n} \right)^{nt}$, could also have been used in the above calculation.]

In the cases of simple interest and continuous interest, similar formulas can be derived, i.e.,

- $PV = FV(1 + rt)^{-1}$ (present value for simple interest)
- $PV = FV(e^{-rt})$ (present value for continuous interest).

In both of the above formulas, r is the nominal annual interest rate.

3.6 Discounting

Discount is a concept similar to interest, i.e.,

- Discount is paid/due at the beginning of a payment period and is calculated on the balance at the end of the period.
- Interest is paid/due at the end of a payment period and calculated on the balance at the beginning of the period.

There are some cases where the anticipated amount of money owed at the end of a loan is deducted at the start of the loan, i.e., the “interest” is paid up front. This is known as “discounting a loan” and the rate of discount is represented by the symbol d . (Note that the term “discount rate” also refers to the interest rate that the United States Federal Reserve charges banks for short term loans. However, that is not the topic of this section.)

For example, a bank makes a short-term (9 months) \$10,000 loan to Ms. Peeples at a 7% annual discount rate. The bank deducts $\frac{9}{12}(7) = 5.25\%$ (\$525) at the beginning of the loan. Ms. Peeples

receives $\$10,000 - \$525 = \$9,475$ up front, and needs to repay $\$10,000$ at the end of 9 months.

The effective interest rate per 9 months (as seen by the bank) is $\frac{525}{9,475} \cong .05541$ or 5.541%.

The effective rate of discount (denoted as d) from time t to some later time $\hat{t}$ is the ratio of the change in value of an investment (known as the amount of discount) to the value at the end of period (at time $\hat{t}$), i.e.,

$$d = \frac{V(\hat{t}) - V(t)}{V(\hat{t})}$$

where $V(s)$ is the value of an investment at time s . This is almost the same as the effective interest rate formula except that the denominator is the value at the end of the time period rather than at the beginning. The situation is depicted in Figure 1.

In the previous example, $d = \frac{10,000 - 9,475}{10,000} = \frac{525}{10,000} = .0525$ or 5.25%.

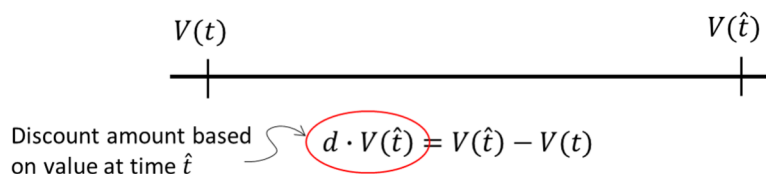


Figure 1. Effective Rate of Discount for a Given Time Period

In the case of compound interest, the effective rate of discount is constant for the repayment periods as can be seen from the calculation below. (Note that i is the interest rate per compounding period.)

$$d = \frac{V(t_n) - V(t_{n-1})}{V(t_n)} = \frac{P(1+i)^n - P(1+i)^{n-1}}{P(1+i)^n} = \frac{1+i-1}{1+i} = \frac{i}{1+i}$$

Solving for i in the above equation, we get $i = \frac{d}{1-d}$.

In the case of simple interest, the effective rate of discount depends on both the duration of the loan (t) and the annual interest rate (r). Since there is no compounding, there is only one repayment period which starts at time 0 and ends at time t . The formula for the effective rate of discount in this case is derived as follows:

$$d = \frac{V(t) - V(0)}{V(t)} = \frac{P(1+rt) - P}{P(1+rt)} = \frac{rt}{1+rt}$$

What is the effective rate of discount for a simple interest discount loan of $\$10,000$ for 18 months

at 10% per annum? We are given $r = .1$ and $t = 1.5$. Thus, $d = \frac{.15}{1+.15} = .130435$ or 13.0435%.

Note that the discount is 13.0435% of the final value: $13.0435\% \times \$11,500 = \$1,500$, so the

borrower receives $\$11,500 - \$1,500 = \$10,000$. In summary, an 18-month loan of $\$11,500$ at a

discount rate of 13.0435% would result in the borrower getting \$10,000 at the beginning of the loan with an obligation to pay the bank \$11,500 in 18 months.

While uncommon, it is possible to accumulate a sum of money with a discount rate. Consider an initial investment of x_0 that is compounded at a discount rate of d for n periods:

- At the beginning (time 0), x_0 is calculated as the value of the investment at time 1 (call it x_1) minus the discount, i.e., dx_1 (note that juxtaposition is meant to imply multiplication). So, $x_0 = x_1 - dx_1$ which implies $x_1 = x_0(1 - d)^{-1}$.
- Similarly, $x_1 = x_2 - dx_2$ which implies $x_2 = x_1(1 - d)^{-1} = x_0(1 - d)^{-2}$.
- Continuing in this manner, $x_n = x_0(1 - d)^{-n}$.

Growth of the initial amount (x_0) is depicted in Figure 2.

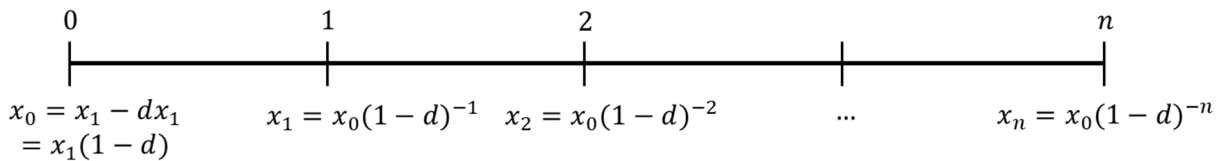


Figure 2. Accumulating with a Discount Rate

For example, what amount must be invested, at a discount rate of .45% per month, to have \$50,000 at the end of 10 years under a monthly compound discount approach? We are given $d = .0045$ and $n = 10 \times 12 = 120$ and are looking for x_0 such that $x_{120} = 50,000$. Using the formula for compound discounting, we have

$$x_{120} = \$50,000 = x_0(1 - .0045)^{-120}$$

$$x_0 = \$50,000(1 - .0045)^{120} \cong \$29,101.93.$$

As another example, consider a discounted 5-year loan at a discount rate of .4% per month where the loanee initially receives \$100,000 from a bank. What is the actual size of the loan, i.e., the amount of money to be paid back to the loaner at the end of 5-years? We are given $d = .004$, $n = 5 \times 12 = 60$ and $x_0 = 100,000$ and are being asked to find x_{60} . Using the formula for compound discounting, we have

$$x_{60} = (100,000)(1 - .004)^{-60} \cong \$127,186.11$$

In other words, a discounted loan in the amount of \$127,186.11 is requested. The discount amount $\$27,186.11 = \$127,186.11 - \$100,000$ is deducted from the loan by the bank and the loanee receives \$100,000. At the end of 5 years, the loanee must pay back \$127,186.11 to the bank.

The equivalent interest rate is $i = \frac{d}{1-d} = \frac{.004}{1-.004} = .004016064$ (monthly). Compounding at i for 60 months with an initial value of \$100,000 yields a future value (in 60 months) of

$$\$100,000(1 + .004016064)^{60} = \$127,186.11$$

3.7 Time Value of Money

Regarding financial decisions, the value of an amount of money (relative to a given time frame) needs to be considered. For example, receiving \$100 from an investment today is not the same as receiving \$100 ten years from now, assuming the \$100 could be making interest over the ten-year span.

Figure 3 shows the value of X at a focal date (could be the present or some other date of importance) and equivalent values in the past and future based on simple, compound, and continuous interest. For example, if the amount $X(1+i)^{-n}$ is compounded for n time periods, then we will have $X(1+i)^{-n}(1+i)^n = X$. If the amount Xe^{-in} is continuously compounding for n time periods, then it would grow to $(Xe^{-in})e^{in} = X$ at time n . We can also work in the opposite direction. For example, how much is needed at the focal date to have $X + in$ at n time periods in the future. From the diagram, we see the required amount is X .

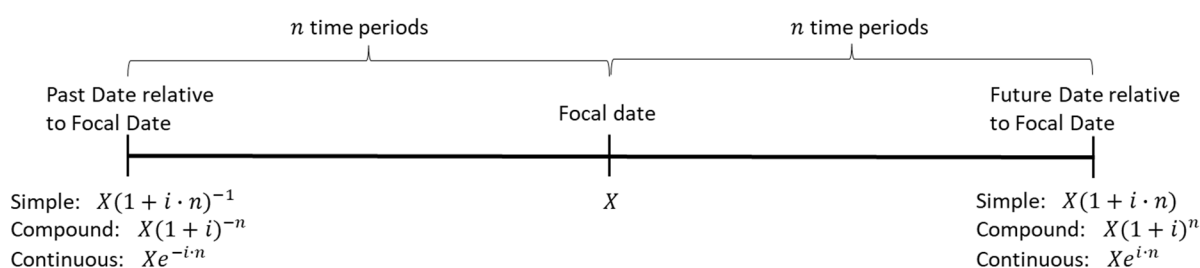


Figure 3. Equivalent Values to X in the Past and Future

Consider a personal loan of \$5000 between friends that is due at the end of a three year period. Determine an equivalent amount of debt due in 1 year and in 5 years if they agree to annual nominal interest rate of 6% compounded monthly.

Select the focal date to be when the debt is due, i.e., in three years. The equivalent value of the debt two years before the focal date (i.e., one year from the present) is

$5,000 \left(1 + \frac{.06}{12}\right)^{-24} = \$4,435.93$. In this case, the debt is paid early. The equivalent value of

the debt two years after the focal date is $5,000 \left(1 + \frac{.06}{12}\right)^{24} = \$5,635.80$. In this case, the debt is later than the agreed repayment time (which we assumed to be the focal date in the calculations).

As a second example, assume Mr. Jasper owes Ms. Dubois \$1,000 in 6 months and another \$1,500 in 2 years from now. What single payment, in the present, will liquidate these debts if both parties agree money is worth 3% (compounded monthly). Same question if the debt is paid in a year.

For the first question, we compute the equivalent value of the two future payments if made at the present time, i.e.,

$$X = 1,000 \left(1 + \frac{.03}{12}\right)^{-6} + 1,500 \left(1 + \frac{.03}{12}\right)^{-24} = \$2,397.88$$

For the second question, notice that the first expected payment is being paid late and will thus be greater than \$1000.

$$X = 1,000 \left(1 + \frac{.03}{12}\right)^6 + 1,500 \left(1 + \frac{.03}{12}\right)^{-12} = \$2,470.82$$

A common technique in financial mathematics is to replace one set of payments with an equivalent set under a given interest-rate model. Two payment streams are said to be **equivalent** if they have the same value at a chosen date (called a **focal date**). Under a fixed interest-rate scheme, equivalence at one date implies equivalence at all dates.

Example. A business borrows \$100,000 today. Interest accrues at a nominal annual rate of 4% compounded monthly. The loan is to be repaid in three equal payments at the ends of months 4, 7, and 11, together with a final payment of \$20,000 at the end of month 13. Determine the amount of each of the three equal payments.

Solution. Let x denote the amount of each of the three equal payments. We select a focal date. Any date may be used; for convenience, take 7 months from the start of the loan, see Figure 4. At the focal date, the accumulated value of the payments must equal the accumulated value of the original loan.

- The payment at month 4 is carried forward 3 months.
- The payment at month 7 occurs at the focal date.
- The payment at month 11 is discounted 4 months.
- The \$20,000 payment at month 13 is discounted 6 months.
- The \$100,000 loan is accumulated 7 months.

Thus, using the monthly rate $i = 0.04/12$, we obtain:

$$x(1+i)^3 + x + x(1+i)^{-4} + 20,000(1+i)^{-6} = 100,000(1+i)^7$$

$$2.99681x + 19,604.63 = 102,356.80 \Rightarrow x \approx 27,613.42$$

Remark on the Choice of Focal Date: The focal date is arbitrary. For example, using the end of the loan (month 13) as the focal date gives:

$$x(1+i)^9 + x(1+i)^6 + x(1+i)^2 + 20,000 = 100,000(1+i)^{13} \Rightarrow x \approx 27,613.42.$$

This confirms that, under a consistent interest-rate model, equivalent payment streams have the same value at any focal date.

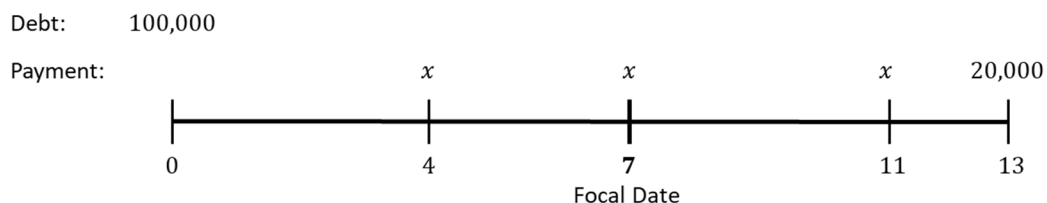


Figure 4. Debt and Payments Relative to a Focal Date

It may help to understand that x is a combination of interest payment and the downpayment of interest. The breakdown, with respect to the focal date to be 7 months after the start of the loan, is shown in the following table. The amount of \$19,999.99 is one penny off due to rounding.

Month	Balance before interest	Balance just before payment	Interest charged	Payment	Principal repaid	Balance after payment
0	100,000.00	—	—	—	—	100,000.00
4	100,000.00	101,340.01	1,340.01	27,613.42	26,273.41	73,726.59
7	73,726.59	74,466.32	739.73	27,613.42	26,873.69	46,852.90
11	46,852.90	47,480.74	627.84	27,613.42	26,985.58	19,867.32
13	19,867.32	19,999.99	132.67	19,999.99	19,867.32	0.00

3.8 Interest, Inflation and Taxes

At the beginning of a given period of time, assume that you have \$1 and the inflation rate is 3% for that period. An item that costs \$1 at the beginning of the period will cost \$1.03 at the end of the period. Your \$1 at the end of the period can only purchase $\frac{1}{1.03} \cong .97$ of what it could have at the beginning of the period. So, the effective simple interest rate is -3%. Another way to view this is ask “what is the future value of \$1 under -3% simple interest for one year?” The answer is

$$FV = P(1 + rt) = 1(1 - .03) = .97$$

noting that $P = 1, r = -.03$ and $t = 1$ in the above equation.

However, if you invested your \$1 at 5% interest at the beginning of the period, you would have \$1.05 which could be used to purchase $\frac{1.05}{1.03} \cong 1.0194$ of what you could have at the beginning of the period with \$1. So, the real rate of return (for the noted investment) is 1.94% in this scenario.

Look at it this way:

Let’s say that at the beginning of the year, a candy bar costs \$1. Also, at the beginning of the year, you invest \$1 at 5% simple interest. Over the course of the year, inflation is 3%. At the end of the year, you have \$1.05 and you want to know how many of the candy bars you can now buy (call this quantity x). Due to inflation, the candy bar now cost \$1.03. So, we have the task of solving the equation $1.05 = 1.03x$. This gives the result $x = \frac{1.05}{1.03} = 1.0194$ candy bars.

In general, if the interest rate for a given period is i and the inflation rate is r_{inf} , the **real rate of return** (i.e., interest adjusted for inflation) is

$$\frac{1 + i}{1 + r_{inf}} - 1$$

The above is known as the Fisher equation [5].

If there is a tax rate of i_t at the end of the year or given period, the effective interest becomes $i_t \cdot i$. This gives us the modified real rate of return (with taxes considered):

$$\frac{1 + (1 - i_t) \cdot i}{1 + r_{inf}} - 1$$

Returning to the previous problem with .05 simple interest and a .03 rate of inflation, and assuming a tax rate of 20%, we get the following real rate of return

$$\frac{1 + (1 - .2)(.05)}{1 + .03} - 1 = \frac{1.04}{1.03} - 1 \cong .0097 \text{ or } .97\%$$

3.9 Summary of Interest Formulas

Formula Name	Formula	Variables	Notes
Simple Interest			
Accumulated Value	$A = P(1 + rt)$	P = principal r = nominal annual rate t = time (years)	<i>Interest on original principal only; no compounding.</i>
Compound Interest			
Accumulated Value	$A = P \left(1 + \frac{r}{n}\right)^{nt}$	P = initial principal r = nominal annual rate n = compounding periods / year t = time (years)	<i>Standard form.</i>
Accumulated Value (alt.)	$A = P(1 + i)^N$	i = r/n (rate per period) N = nt (total periods)	<i>Equivalent; convenient in per-period terms.</i>
Continuous Interest			
Accumulated Value	$A = Pe^{rt}$	P = initial principal r = nominal annual rate t = time (years) e = Euler's number ≈ 2.71828	<i>Limiting case as compounding frequency $n \rightarrow \infty$.</i>
Effective Rate of Interest			
General Definition	$i_{\text{eff}} = \frac{A(t_1) - A(t_0)}{A(t_0)}$	A(t) = value at time t t ₀ = start of period, t ₁ = end of period	<i>Fractional growth of an investment over a period.</i>
Compound Interest (period k)	$i_{\text{eff}} = i$	i = rate per compounding period	<i>Constant across all periods.</i>
Simple Interest (period k)	$i_{\text{eff}} = \frac{r}{1 + r(k - 1)}$	r = annual rate k = period number (1, 2, 3, ...)	<i>Decreasing over time; no compounding benefit.</i>
Annual Percentage Yield (APY)	$\text{APY} = \left(1 + \frac{r}{n}\right)^n - 1$	r = nominal annual rate n = compounding periods / year	<i>Compares rates across compounding frequencies.</i>
Nominal Rate from APY	$r = n[(1 + \text{APY})^{1/n} - 1]$	APY = annual percentage yield	<i>Inverse of APY formula.</i>

Formula Name	Formula	Variables	Notes
		n = compounding periods / year	
Present Value			
Compound Interest	$PV = \frac{FV}{(1+i)^N}$ $= FV (1+i)^{-N}$	FV = future value i = rate per period N = total periods	<i>Amount to invest today to reach FV in N periods.</i>
Simple Interest	$PV = \frac{FV}{1+rt}$	FV = future value r = nominal annual rate t = years	
Continuous Interest	$PV = FV \cdot e^{-rt}$	FV = future value r = nominal annual rate t = years	
Discounting			
Effective Rate of Discount	$d = \frac{A(t_1) - A(t_0)}{A(t_1)}$	A(t) = value at time t	<i>Denominator is end-of-period value (vs. start for interest rate).</i>
Discount Rate — Compound	$d = \frac{i}{1+i}$	i = interest rate per compounding period	<i>Constant per period.</i>
Discount Rate — Simple	$d = \frac{rt}{1+rt}$	r = annual rate t = loan duration (years)	<i>Single-period simple interest discount.</i>
Accumulated Value (Compound Discount)	$A = \frac{P}{(1-d)^N}$	P = initial investment d = discount rate per period N = periods	<i>Growing a sum at a discount rate; interest paid upfront.</i>
Time Value of Money			
Future Equivalent Value	$FV = X (1+i)^n$	X = value at focal date i = periodic rate n = periods into the future	<i>Moves a payment forward in time.</i>
Past Equivalent Value	$PV = \frac{X}{(1+i)^n}$ $= X (1+i)^{-n}$	X = value at focal date i = periodic rate n = periods into the past	<i>Moves a payment backward in time.</i>

Formula Name	Formula	Variables	Notes
Interest, Inflation and Taxes			
Fisher Equation	$r_{\text{real}} = \frac{1+r}{1+f} - 1$	r = nominal interest rate f = inflation rate	<i>Real rate of return adjusted for inflation.</i>
Modified Real Rate (After-Tax)	$r_{\text{tax}} = \frac{1+r(1-\tau)}{1+f} - 1$	r = nominal rate f = inflation rate τ = tax rate	<i>After-tax, inflation-adjusted return.</i>

3.10 Supporting Programs

3.10.1 Interest Compounding Calculator

File name: interest.py

This calculator computes simple interest, compound interest or continuous interest, depending on user selection. The expected input is

- the original principal
- annual interest percentage
- interest type (compound, continuous or simple)
- compounding frequency in the case of compound interest
- time that interest accrues in years or fraction of years (e.g., 1.75 years is allowed).

The output is the final principal (which includes the original principal plus accrued interest).

3.11 Exercises

1. Determine the value of a simple interest investment of \$10,000 at 3% annual interest rate over a period of 2 years and 7 months. What would the value of the investment be if daily compounding were used instead?

Hint: For the second part of the problem, use the interest compounding calculator.

2. Country A makes a 100-year loan to Country B for 1 billion dollars at an annual interest rate of 6%. Compute the amount of interest paid back to Country A under the assumption of yearly compounded interest, daily compounded interest and continuous compounding. To be clear, the loan is paid back in full at the end of the 100-year period.

Hint: Use the interest compounding calculator for each scenario.

3. Mr. Wimble buys a diamond for investment purposes at \$55,000, holds the diamond for 5 years and 6 months and then sells the diamond for \$100,000. What is the effective interest rate per year (assuming simple interest)? For comparison, what is the effective interest rate if monthly compounding is assumed?

Hint: For the second question, use the compound interest formula with $P = 55,000$, $n = 12$ and $t = 5.5$, set the result equal to 100,000 solve for r , i.e., solve

$$55,000 \left(1 + \frac{r}{12}\right)^{12(5.5)} = 100,000$$

for r by taking the natural log on both sides and then apply some basic algebra. The answer is 10.9191% which, as expected, is smaller than the solution to the first question, i.e., 14.876%.

4. Given a nominal yearly interest rate of r , compute the effective annual interest rate under the assumption of continuous compounding between times t and $\hat{t}$. **Hint:** From the formula for effective interest rate, we have

$$\frac{V(\hat{t}) - V(t)}{V(t)} = \frac{Pe^{r(\hat{t}-t)} - P}{P} = e^{r(\hat{t}-t)} - 1$$

which is the effect interest rate for the entire period $\hat{t} - t$. To get the effective annual interest rate, just divide by $\hat{t} - t$.

5. What nominal annual interest rate r is required to quadruple an initial amount in 20 years, with monthly compounding? Same question with continuous compounding?

Hint and Answer: For the first question, solve the equation $4P = P \left(1 + \frac{r}{12}\right)^{240}$ to get $r = 6.9515\%$. For the second question, solve the equation $4P = Pe^{20r}$ to get $r = 6.9315\%$. In both cases, you need to take the natural log (i.e., log base e) on both sides and then do some algebra.

6. A rich uncle leaves \$5 million to 3 heirs (all nieces). Each niece is to receive an equal amount of money when she turns 21 years old, which will be 12 months, 27 months, and 43 months from the time the will is executed. In the meantime, the money is invested at a nominal annual interest rate of 6% (compounded monthly). How much money is each niece to receive?

Hint: Use the time that the will is executed as the focal date and equate present value of the investment with that of the payouts to get the following equation (where x is the amount to be paid to each heir): $x(1 + .005)^{-12} + x(1 + .005)^{-27} + x(1 + .005)^{-43} = 5,000,000$. Solve for x .

7. Given a debt of Q at time 0 which is resolved by n equal payments of amount x with periodic interest rate i , determine x .

Hint: Using the concept of equivalent payments/debts with the focal date being time 0, we get the equation $Q = x(1 + i)^{-1} + x(1 + i)^{-2} + \cdots + x(1 + i)^{-n}$. Use the formula for the sum of a geometric series and then solve for x . We will return to this problem in Section 4.3.1.

4 Annuities

An investment in knowledge pays the best interest. — Benjamin Franklin

4.1 Terminology

An **annuity** is a series of payments, usually (but not necessarily) made at equal intervals of time and in equal amounts. (The formulas in this book apply to the equal-interval, equal-payment case.) The time between successive payments of an annuity is called the payment interval (or period). The time from the beginning of the first payment interval to the end of the last payment interval is referred to as the **term** of an annuity.

When the dates of the first and the last payments of an annuity are fixed, the annuity is called an **annuity certain**. When the term of the annuity depends on some uncertain event (e.g., death of the annuity's holder), the annuity is called a **contingent annuity**.

When payment is made at the end of each payment interval, the annuity is called an **ordinary annuity** (or immediate annuity). When payment is made at the beginning of each payment interval, the annuity is called an **annuity-due**. A **deferred annuity** is an annuity whose first payment does not begin until after a specified deferral period has elapsed.

When the payment interval and the interest compounding period are equal — that is, payments and compounding occur at the same frequency — the annuity is called a **simple annuity**; otherwise, it is a **general annuity**. For example, a mortgage with monthly payments and monthly compounding is a simple annuity, whereas a mortgage with monthly payments but semi-annual compounding (as is common in Canada) is a general annuity.

The **accumulated (or future) value of an annuity** is the value of the recurring payments to an annuity at a certain date in the future, assuming a particular rate of return. The **discounted (or present) value of an annuity** is the current value of future payments of an annuity, given a specified rate of return, or discount rate. The higher the discount rate, the lower the present value of the annuity.

4.2 Accumulated Value of Ordinary, Simple, Certain Annuities

4.2.1 Concept and Formulas

Consider an annuity where an amount R is added (by the entity buying the annuity) at the end of each period, interest is compounded at the end of each period, the interest rate is i (per period) and there are n periods. The payment schedule for such an annuity is shown in Figure 5.

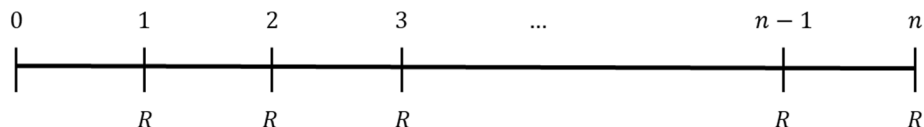


Figure 5. Payment Schedule for Accumulated Value of Ordinary, Simple, Certain Annuities

No interest is earned on the contribution at the end of period n since that is the end of the annuity's term. The contribution at the end of period $n - 1$ earns interest iR and so its

accumulated value at the end of the annuity's term is $R + iR = R(1 + i)$. The accumulated value for the contributions is summarized in Table 2.

It is important to read the table correctly. The "End of Period" column indicates **when each individual payment is deposited**, not a point in time at which the account balance is being measured. The "Accumulated Value" column shows what that individual payment grows to **by the end of the entire term (end of period n)** — this is the same reference date for every row in the table. In other words, the table is a snapshot of each payment's future worth, all evaluated at the same moment.

The payment deposited last (at the end of period n) has no time remaining to earn interest and so contributes exactly R to the final total. The payment deposited second-to-last (at the end of period $n - 1$) has one full period to grow, contributing $R(1 + i)$. Working backwards, the very first payment (deposited at the end of period 1) has the longest time to accumulate, i.e., $n - 1$ full periods, and therefore contributes $R(1 + i)^{n-1}$. The total accumulated value of the annuity at term is the sum of all these individual contributions.

Table 2. Accumulated Value of an Ordinary, Simple, Certain Annuity

End of Period	Accumulated Value of Given Investment
n	R
$n - 1$	$R + iR = R(1 + i)$
$n - 2$	$R(1 + i) + iR(1 + i) = R(1 + i)^2$
...	...
1	$R(1 + i)^{n-1}$

The value of the annuity at term is the sum (call it S) of the following geometric series:

$$\begin{aligned}
 S &= R + R(1 + i) + R(1 + i)^2 + \cdots + R(1 + i)^{n-1} \\
 &= R \frac{(1 + i)^n - 1}{(1 + i) - 1} = R \frac{(1 + i)^n - 1}{i}
 \end{aligned}$$

[Remark concerning notation: the quantity $\frac{(1+i)^n - 1}{i}$ is often represented by a shorthand notation in financial books and articles. The notation is usually some variation of $s_{n|i}$.]

Table 3. Accumulated Value: Ordinary, Simple, Certain Annuity

Formula Name	Formula	Variables	Notes
Accumulated value at term	$S = R \frac{(1+i)^n - 1}{i}$	S = accumulated value at term; R = payment per period; i = interest rate per period; n = number of periods	Geometric series of n end-of-period contributions. The factor $(1+i)^n - 1/i$ is often denoted $s_{\overline{n} i}$.
Payment per period (solve for R)	$R = \frac{iS}{(1+i)^n - 1}$	S = target accumulated value	Use when S , i and n are known.
Number of periods (solve for n)	$n = \frac{\ln\left(\frac{iS}{R} + 1\right)}{\ln(1+i)}$		Round up to the next integer for the annuity to meet or exceed S .

4.2.2 Examples

For example, Mr. Lebesgue agrees to an annuity savings plan from an investment company. Under the plan, Mr. Lebesgue will contribute \$300 at the end of each month for a period of three years. Interest is compounded at the end of each month at a nominal annual rate of 6%. How much is the annuity worth at term?

We have at $R = 300$, $n = 3(12) = 36$ and $i = \frac{.06}{12} = .005$. Using the accumulated value formula for an annuity gives the following result:

$$S = 300 \frac{1.005^{36} - 1}{.005} = \$11,800.83$$

Mr. Lebesgue reconsiders and tells that investment company that he really needs about \$15,000 to buy a car in three years and would like to know how much he needs to save each month to reach that goal under the same conditions as stated above.

In this case, we need so solve for R given $S = 15,000$, $n = 36$ and $i = .005$. We first solve the accumulation formula for R to get

$$R = \frac{iS}{(1+i)^n - 1}$$

We then substitute the specific values for the problem at hand into the above formula and get the following result

$$R = \frac{(.005)(15,000)}{1.005^{36} - 1} \cong \$381.33$$

In yet another variation of the problem, assume Mr. Lebesgue wants to limit his monthly payment to \$300 but is willing to wait longer to reach his goal of \$15,000. In this scenario, we need to solve the equation $S = R \frac{(1+i)^n - 1}{i}$ for n . With some rearrangement of the equation, we get

$$\frac{iS}{R} + 1 = (1+i)^n$$

Take the natural log on both sides and the equation, and then isolation n to get:

$$n = \frac{\ln\left(\frac{iS}{R} + 1\right)}{\ln(1 + i)}$$

For the problem at hand, we get $n = 44.74$ which we need to round-up to the next whole number, i.e., 45 payments of \$300 are required at the given interest rate for the annuity to surpass \$15,000 at term.

4.2.3 Blocking

Especially for longer-term annuities, the entity offering the annuity is not likely to provide a fixed interest rate for the entire term of the annuity. The investment entity may vary the rate based on something like the 30-year United States treasury bond interest rate. In such cases, the computation of the annuity's value needs to be blocked into several computations.

4.2.3.1 Example

An example will illustrate the point. Ms. Gold agrees to an annuity savings plan from a bank that stipulates a deposit of \$500 at the end of each month for 10 years. The interest rate is based on the average 30-year treasury bond rate for the past month. For the first 5 years (60 deposits), the nominal annual rate remained constant at 3% (compounded monthly), and for the last 5 years it remained constant at 3.5% (compounded monthly). How much was the annuity worth at term, assuming Ms. Gold has contributed the agreed \$500 per month?

We first compute value of the annuity at the end of the initial 5 years (call it S_1):

$$S_1 = 500 \frac{\left(1 + \frac{.03}{12}\right)^{60} - 1}{\frac{.03}{12}} = \$32,323.36$$

Next, compute the value of the contributions to the annuity during the second 5 years (call it S_2):

$$S_2 = 500 \frac{\left(1 + \frac{.035}{12}\right)^{60} - 1}{\frac{.035}{12}} = \$32,733.06$$

We are not done because it is necessary to consider the interest accrued to S_1 during the second 5 years of the annuity, which is $S_1 \left(1 + \frac{.035}{12}\right)^{60} = \$38,495.27$

So, the total value of the annuity at term is $\$32,733.06 + \$38,495.27 = \$71,228.33$, while Ms. Gold contributed a total of $120 \times \$500 = \$60,000$.

In a variation of the above problem, assume the term is changed to 15 years with the first 10 years being as before but for the last five years the nominal interest rate is 6% (compounded monthly). Further, Ms. Gold does not make further contributions beyond the first 10 years. In this scenario, we simply need to compound \$71,716.26 for another 5 years, i.e.,

$$\$71,228.33 \left(1 + \frac{.06}{12}\right)^{60} = \$96,076.34$$

4.2.3.2 Generalization

Blocking can be made even more general than what is suggested in the above example. Each block can have a different interest rate per period, a different payment per period and a different number of periods. So, each block is defined by three variables, i.e., interest rate, payment amount and number of periods. If either the interest rate or payment amount changes, we start a new block. So, these two attributes (i.e., interest rate and payment amount) determine the third variable (i.e., number of periods in a block).

The following algorithm can be used to determine the final value of an annuity with m blocks:

1. For each of the m blocks, compute the accrued value of the annuity for the interval when the block is active. This is done using the formula below for block k . The subscript k is put on i, R and n to emphasize these values can change for each block.

$$S_k = R_k \frac{(1 + i_k)^{n_k} - 1}{i_k}$$

2. For each S_k ($k = 1, 2, \dots, m - 1$), the value continues to accrue during subsequent blocks, based on the interest rate and number of periods for each subsequent block. So, the value at the end of the annuity for block k is

$$V_k = S_k(1 + i_{k+1})^{n_{k+1}}(1 + i_{k+2})^{n_{k+2}} \dots (1 + i_m)^{n_m}$$

3. Since there is no further accumulation of interest after the last block is finished, we have that $V_m = S_m$.
4. The total value of the annuity at term is $V = V_1 + V_2 + \dots + V_{m-1} + S_m$

Table 4. Blocking: Multiple Rate Blocks

Formula Name	Formula	Variables	Notes
Block accumulated value	$S_k = R_k \frac{(1 + i_k)^{n_k} - 1}{i_k}$	$k = \text{block index}; R_k, i_k, n_k = \text{payment, rate, periods for block } k$	Apply the ordinary accumulated-value formula separately to each block.
Block value at end of full term	$V_k = S_k(1 + i_{k+1})^{n_{k+1}} \dots (1 + i_m)^{n_m}$	$m = \text{total number of blocks}$	Compound S_k forward through all subsequent blocks. For the last block, $V_m = S_m$.
Total annuity value	$V = V_1 + V_2 + \dots + V_m$		Sum of all blocks' final values.

4.3 Discounted (Present) Value of Ordinary, Simple, Certain Annuities

4.3.1 Concept and Formulas

Consider a scenario in which someone purchases an annuity up front and then gets constant payments in the amount R at the end of each of n periods. At the end of period n , the value of the annuity is 0. The interest rate per period is i .

In order to cover the first payment (end of period #1), how much needs to be invested at beginning of period 1? If we call this quantity x_1 , then we need to solve $x_1 + x_1 i = R$ for x_1 . This gives us $x_1 = R(1 + i)^{-1}$. So, x_1 invested at the time 0 will be worth R at the end of the first period. This is basically a present value calculation for a future desired amount.

In general, we need to calculate the present value for each of the n payments of amount R and then sum the amounts, i.e.,

$$\begin{aligned} Q &= R(1 + i)^{-1} + R(1 + i)^{-2} + \cdots + R(1 + i)^{-n} \\ &= R(1 + i)^{-1} \{1 + (1 + i)^{-1} + [(1 + i)^{-1}]^2 + \cdots + [(1 + i)^{-1}]^{n-1}\} \end{aligned}$$

Noting that the above is the sum of a geometric series with $a = R(1 + i)^{-1}$ and $r = (1 + i)^{-1}$, we get the following equation

$$Q = R(1 + i)^{-1} \frac{1 - (1 + i)^{-n}}{1 - (1 + i)^{-1}}$$

Multiplying by $\frac{1+i}{1+i}$ and some simplification results in the following formula

$$Q = R \frac{1 - (1 + i)^{-n}}{i}$$

So, an initial investment in the amount of Q with interest rate i per period (compounded at the end of each period) is equivalent to n payments in the amount R at the beginning of each period. (Recall Exercise 7 from Section 3.11.)

Alternately, the above formula can be solved for R in terms of the other variables to get

$$R = \frac{iQ}{1 - (1 + i)^{-n}}$$

This is useful if one knows the interest rate, desired number of distributions and the initial investment, and wants to determine how much can be distributed at the end of each period.

We can also solve for n to determine how long an initial investment will last given the distribution amount per period and interest rate, i.e.,

$$n = \frac{\ln R - \ln(R - iQ)}{\ln(1 + i)}$$

where $\ln(x)$ is the natural log of x (i.e., log base e).

[Remark concerning notation: the quantity $\frac{1 - (1 + i)^{-n}}{i}$ is often represented by a shorthand notation in financial books and articles. The notation is usually some variation of $a_{n|i}$.]

Table 5. Discounted (Present) Value: Ordinary, Simple, Certain Annuity

Formula Name	Formula	Variables	Notes
Present (discounted) value	$Q = R \frac{1 - (1 + i)^{-n}}{i}$	Q = initial investment (purchase price); R = payout per period; i = interest rate per period; n = number of payouts	The factor $\frac{1 - (1 + i)^{-n}}{i}$ is often denoted $a_{(n i)}$.
Payout per period (solve for R)	$R = \frac{iQ}{1 - (1 + i)^{-n}}$		Use when Q , i and n are known.
Number of payouts (solve for n)	$n = \frac{\ln R - \ln(R - iQ)}{\ln(1 + i)}$		Gives the number of full periods the fund lasts; the last payment may be smaller than R .

4.3.2 Examples

What initial investment is needed to get a monthly payment of \$750 for 20 years given a nominal yearly interest rate of 5% (compounded monthly)?

We are given $R = \$750$, $n = 12 \times 20 = 240$ and $i = \frac{.05}{12}$. Using the formula for Q , we get

$$Q = 750 \frac{1 - \left(1 + \frac{.05}{12}\right)^{-240}}{\frac{.05}{12}} = \$113,643.98$$

As a second example, consider someone who has \$100,000 to invest in an annuity with a nominal annual interest rate of 4% (compounded monthly) and wants to have the annuity distributed monthly for 15 years. How much can be distributed each month for the given period?

In this scenario, we are given $Q = \$100,000$, $n = 12 \times 15 = 180$ and $i = .04/12$, and we need to determine R .

$$R = \frac{iQ}{1 - (1 + i)^{-n}} = \frac{\left(\frac{.04}{12}\right)(100,000)}{1 - \left(1 + \frac{.04}{12}\right)^{-180}} = \$739.69$$

In yet another example, we want to find the number of quarters (i.e., quarter-years) that an investment in the amount of \$27,976.08 will last, given equal quarterly payments in the amount of \$1000 with a quarterly interest rate of .013.

We are given $R = \$1,000$, $Q = \$27,976.08$ and $i = .013$, and want to find n (the number of quarters) the investment will last. Using the formula for n from the previous subsection, we get

$$n = \frac{\ln R - \ln(R - iQ)}{\ln(1 + i)} = \frac{\ln(1,000) - \ln(1,000 - (.013)(27,976.08))}{\ln(1 + .013)} = 35$$

So, the annuity can payout \$1,000 per quarter for 35 quarters or 8 years and 9 months.

(In most cases, the value of n will not be a whole number. In such cases, the final payment will be less than the previous recurring payments.)

4.3.3 Blocking on the Payout Side: Varying Interest Rates During Distribution

In Section 4.2.3, we saw how multiple interest-rate *blocks* affect the *accumulated* value of an annuity: each block is handled on its own, and the running balance is then compounded forward through all later blocks. The same structural idea applies on the *payout* side. If the rate environment changes during the distribution phase, the present value of the stream is obtained by discounting each block *backward* through all *earlier* blocks to time 0.

Typical scenarios include a retirement-income contract whose crediting rate resets every few years, a structured settlement funded by bonds laddered at different yields, and a pension valued under legislation that steps the discount rate up or down at policy-defined intervals.

Setup and notation

Suppose the payout stream is divided into m contiguous blocks. Within block k ($k = 1, 2, \dots, m$):

- R_k is the fixed payment per period,
- i_k is the interest rate per period, and
- n_k is the number of periods.

Let $N_k = n_1 + n_2 + \dots + n_k$ denote the total number of periods through the end of block k , with $N_0 = 0$.

Step 1 — Value each block at its own start.

Apply the ordinary present-value formula *within* the block:

$$Q_k^* = R_k \times \frac{[1 - (1 + i_k)^{-n_k}]}{i_k}$$

This is the value of block- k payments measured at time N_{k-1} (the *start* of block k).

Step 2 — Discount each block back to time 0.

Bring Q_k^* back to time 0 through all *preceding* blocks, using the rate that applied during each:

$$Q_k = Q_k^* \cdot (1 + i_{k-1})^{-n_{k-1}} \cdot (1 + i_{k-2})^{-n_{k-2}} \cdot \dots \cdot (1 + i_1)^{-n_1}$$

For $k = 1$, no extra discounting is required: $Q_1 = Q_1^*$.

Step 3 — Sum the block present values.

$$Q = Q_1 + Q_2 + \dots + Q_m$$

Notice the symmetry with Section 4.2.3: there, each block's accumulated value S_k was compounded *forward* through all *subsequent* blocks; here, each block's present value is discounted *backward* through all *preceding* blocks. The block index walks in the opposite direction, but the arithmetic is the mirror image.

Examples. A retiree purchases an annuity that will pay \$24,000 at the end of each year for 20 years. The insurer credits 4% for the first 8 years and 3% thereafter. Find the purchase price Q .

Block 1: $R_1 = \$24,000, i_1 = 0.04, n_1 = 8$

$$Q_1^* = 24,000 \cdot \frac{[1 - 1.04^{-8}]}{0.04} = 24,000 \cdot 6.732745 \approx \$161,585.88$$

Block 2: $R_2 = \$24,000, i_2 = 0.03, n_2 = 12$

$$Q_2^* = 24,000 \cdot \frac{[1 - 1.03^{-12}]}{0.03} = 24,000 \cdot 9.954004 \approx \$238,896.10$$

Discount block 2 back through block 1:

$$Q_2 = 238,896.10 \cdot 1.04^{-8} \approx \$174,559.04$$

Block 1 is already at time 0, so $Q_1 = \$161,585.88$. The purchase price is

$$Q \approx 161,585.88 + 174,559.04 = \$336,144.92$$

Remarks:

- If payments also vary by block (R_k depends on k), the same machinery applies — only the per-block payment changes.
- The discount path is what trips students up. A quick timeline diagram showing arrows sweeping from the *start* of each block back to time 0 is worth including in lecture.
- A companion reference entry appears below as Table 6 and is the mirror image of Table 4.

Table 6. Blocking on the Payout Side: Multiple Rate Blocks

Formula Name	Formula	Variables	Notes
Block present value (at start of block)	$Q_k^* = R_k \frac{[1 - (1 + i_k)^{-n_k}]}{i_k}$	k = block index; R_k, i_k, n_k = payment, rate, periods for block k	Ordinary present-value formula applied block by block.
Block present value at time 0	$Q_k = Q_k^* \cdot (1 + i_{k-1})^{-n_{k-1}} \cdot \dots \cdot (1 + i_1)^{-n_1}$	Discount factors use the rate that applied during each earlier block.	Discount Q_k^* back through all preceding blocks. For $k = 1, Q_1 = Q_1^*$.
Total present value	$Q = Q_1 + Q_2 + \dots + Q_m$	m = total number of blocks	Sum of the time-0 present values of all blocks.

4.4 Annuities Due

In an **annuity-due**, the periodic payments are due at the beginning of each payment interval. The term of an annuity-due starts at the time of the first payment and ends one payment period after the date of the last payment. The payment schedule for an annuity due is shown in Figure 6.



Figure 6. Payment Schedule for an Annuity Due

The accumulated (or future) value of an annuity-due at term is similar to the formula for S in Section 4.2.1, except that interest is earned on each contribution for one additional period. So, we need to multiply the formula for the accumulated value of an ordinary, simple, certain annuity by $(1 + i)$ to get the accumulated value for an annuity due:

$$(1+i)R \frac{(1+i)^n - 1}{i}$$

which can be written as

$$R \frac{(1+i)^{n+1} - (1+i)}{i}$$

The discounted (or present) value of an annuity-due is similar to the formula for Q in Section 4.3.1, except that all the payments are done at the beginning of each period. So, the discounted value of an annuity-due is

$$\begin{aligned} & R + R(1+i)^{-1} + R(1+i)^{-2} + \dots + R(1+i)^{-(n-1)} \\ &= R[1 + (1+i)^{-1} + (1+i)^{-2} + \dots + (1+i)^{-(n-1)}] \\ &= R \frac{1 - (1+i)^{-n}}{1 - (1+i)^{-1}} \end{aligned}$$

Note that the above is just $(1+i)$ times the formula for discounted (present) value of an ordinary, simple, certain annuity.

Table 7. Annuities Due

Formula Name	Formula	Variables	Notes
Accumulated value (annuity-due)	$S = R \frac{(1+i)^{n+1} - (1+i)}{i}$	<i>Payments made at the beginning of each period</i>	Equals the ordinary accumulated value multiplied by $(1+i)$.
Discounted value (annuity-due)	$Q = R \frac{1 - (1+i)^{-n}}{1 - (1+i)^{-1}}$		Equals the ordinary discounted value multiplied by $(1+i)$.

4.5 Deferred Annuities

As noted earlier, a deferred annuity is an ordinary annuity where the first payment is made after a given number of periods m . Take the example of two parents buying an annuity for their daughter's education. Assume the child is 11 years old when the annuity is purchased and there are no payments back to the annuity owners until the child turns 18. In this case, the annuity is deferred for 7 years (earning interest in the meantime) and the payments are made during the child's time in college (say 5 years). At the annuity's term (12 years from time of purchase), all the money has been distributed back to the parents who are presumably using the money to pay college bills for their daughter.

In general, assume

- the annuity is purchased at time 0
- there are to be n payments (to the owner of the annuity), each in the amount of R
- the payments start at the end of period m (i.e., at time $m+1$)
- the interest rate per period is i .

The scenario is summarized in Figure 7.

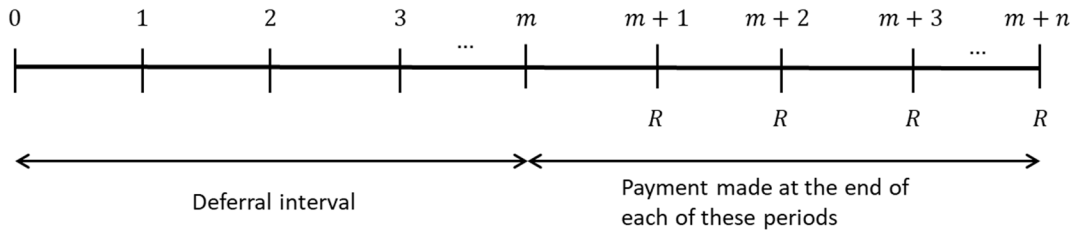


Figure 7. Deferred Annuity

To compute the overall present value (purchase price) of the annuity, we compute the present value of each payment. To cover the first payment (at time $m + 1$), how much needs to be invested at time 0? If we call that quantity x_1 , then we need to solve $x_1(1 + i)^{m+1} = R$ for x_1 . This gives us $x_1 = R(1 + i)^{-(m+1)}$. So, x_1 invested at the time 0 will be worth R at time $m + 1$. Using similar reasoning, the present value of the second payment is $R(1 + i)^{-(m+2)}$, and so on. Thus, the present value for the annuity is the sum

$$\begin{aligned} Q &= R(1 + i)^{-(m+1)} + R(1 + i)^{-(m+2)} + \dots + R(1 + i)^{-(m+n)} \\ &= R(1 + i)^{-(m+1)} [1 + (1 + i)^{-1} + (1 + i)^{-2} + \dots + (1 + i)^{-(n-1)}] \\ &= R(1 + i)^{-(m+1)} \frac{1 - (1 + i)^{-n}}{1 - (1 + i)^{-1}} \end{aligned}$$

If the interest rate is assumed to be different in the deferral period (call it j) from the rate during the payment periods (call it i), then we get a more general equation for Q . To be clear, it is assumed the interest rate changes from j to i **after** the first payment. This is almost the same as the previous equation, with the difference highlighted in bold.

$$Q = R(1 + \mathbf{j})^{-(m+1)} \frac{1 - (1 + i)^{-n}}{1 - (1 + i)^{-1}}$$

Example. Returning to the college expense savings example, we have the following inputs:

- $m + 1 = 7 \times 12 = 84$ and $m = 83$
- $n = 5 \times 12 = 60$
- $i = j = \frac{.05}{12} = .0041\overline{6}$ (assume interest is fixed over the entire term of the annuity)
- assume that \$2,500/month is needed during the child's college years, so $R = \$2,500$.

Assuming payments and compounding are done monthly, and using the formula for Q , we get

$$\begin{aligned} Q &= R(1 + i)^{-(m+1)} \frac{1 - (1 + i)^{-n}}{1 - (1 + i)^{-1}} \\ &= (2,500)(1.0041\overline{6})^{-84} \frac{1 - (1.0041\overline{6})^{-60}}{1 - (1.0041\overline{6})^{-1}} = \$93,811.97 \end{aligned}$$

In a variation of the above scenario, assume that during the deferral interval, the interest rate is 6% and 4% just after the payments start. Thus, $j = \frac{.06}{12} = .005$ and $i = \frac{.04}{12} = .00\overline{3}$ and using the second formula for Q , we get that $Q = \$89,583.59$.

Example: The Bobby Bonilla Deferred Annuity

In 2000, a professional baseball player, Bobby Bonilla, was owed \$5.9 million by his team (the New York Mets). Instead of receiving this amount immediately, he agreed to defer the payment under the following arrangement:

- No payments for 11 years
- Followed by 25 annual payments of \$1,193,248.20, beginning in 2011

This arrangement can be modeled as a deferred annuity.

Mathematical Interpretation

Let:

- $P = 5,900,000$ (present value in 2000)
- $i = 8\%$ annually
- Deferral period: $m = 11$ years
- Number of payments: $n = 25$

Step 1: Accumulate the value during deferral

$$P(1 + i)^m$$

Step 2: This accumulated amount funds an annuity:

$$R \cdot \frac{1 - (1 + i)^{-n}}{i}$$

Equating:

$$5,900,000(1.08)^{11} = 1,193,248.20 \cdot \frac{1 - (1.08)^{-25}}{0.08}$$

This shows that the future payments are exactly the annuity equivalent of the deferred lump sum.

4.6 Forborne Annuities

In a **forborne annuity**, the annuity continues to earn interest after the last payment into the annuity is made. We have already seen a forborne annuity in the variation of the example in Section 4.2.3.1.

This pattern is very common in practice. A parent may deposit into an education savings account for fifteen years and then let it accumulate for three more years until the child enrolls in a college or trade school; a worker may contribute to a retirement fund for thirty years and then leave it untouched for a few more years before beginning to draw down; a policyholder may pay premiums into a cash-value life insurance contract for a fixed term and let the fund grow until it is needed. In each case, the contributions form an ordinary annuity, and the *forbearance* period is pure compound interest applied to the lump sum that the annuity has produced.

This split is what makes the arithmetic tidy: first accumulate the payments to the end of the contribution phase, then let the resulting balance roll forward as a single lump sum. The two steps do not interact.

There are two steps in calculating the accumulated value of a forborne annuity:

- Calculate the accumulated value at the time of the last payment using the formula from Section 4.2.1, i.e., $R \frac{(1+i)^n - 1}{i}$, where i is the interest rate per period, R is the payment amount (in the annuity per period) and n is the number of payments.
- Calculate the interest earned while the annuity is held after the last payment into the annuity, using the formula for compound interest where the principal is given by $R \frac{(1+i)^n - 1}{i}$. Assuming the annuity is held for m periods after the last payment, the final accumulated value of the annuity is $R \frac{(1+i)^n - 1}{i} (1+i)^m$.

In compact actuarial notation, writing $s_{n|i}$ for the ordinary-annuity accumulation factor, the forborne accumulated value is

$$S_f = R \cdot s_{n|i} \cdot (1+i)^m$$

When $m = 0$ the forbearance vanishes and we recover the ordinary accumulated value $S = R \cdot s_{n|i}$ from Section 4.2.1.

Different rates during the forbearance period. If funds are rolled into a different vehicle after the last deposit — for example, from a contribution account paying rate i into a holding account paying rate j — replace $(1+i)^m$ by $(1+j)^m$ in the second step. This parallels the two-rate deferred-annuity formula in Section 4.5 and is useful when modelling a retirement fund that is moved from an employer plan into an annuity product after the contribution phase ends.

Forborne vs. deferred annuities. It is worth noting the symmetry with the deferred annuity in Section 4.5. A *deferred* annuity shifts the payment phase *later* in time: the first payment is received after a waiting period m . A *forborne* annuity shifts the *withdrawal* later: contributions finish at period n , but the balance is held for m more periods before being used. Both are obtained from the ordinary annuity by multiplying by $(1+i)^{\pm m}$ — forward for forborne, backward for deferred.

Example. Education savings fund

A parent deposits \$3,000 at the end of each year into an education savings account paying 5% per year, compounded annually. Deposits are made for 15 years. After the last deposit, the fund is left untouched for 3 more years until the child enrolls in a college or trade school. What is the accumulated value at the start of university?

Given. $R = \$3,000, i = 0.05, n = 15, m = 3$.

Step 1. Accumulated value at the time of the last deposit:

$$R \cdot s_{n|i} = 3,000 \cdot \frac{(1.05)^{15} - 1}{0.05} = 3,000 \cdot 21.578564 \approx \$64,735.69$$

Step 2. Compound the balance forward for the 3-year forbearance:

$$S_f = 64,735.69 \cdot (1.05)^3 = 64,735.69 \cdot 1.157625 \approx \$74,939.65$$

So, the fund is worth approximately \$74,940 when the child enters university. Of this, the deposits total \$45,000 ($15 \times \$3,000$) and the interest — both that earned during the contribution phase and the additional \$10,204 earned during the three-year forbearance — accounts for the remaining \$29,940.

Why the forbearance matters. Even though no new money is added during the last three years, the balance earns 5% compound interest on a principal that has already grown to nearly \$65,000. This

is why parents and retirement-savers who finish contributing *early* and simply leave the money invested can substantially outperform those who continue depositing the same annual amount but then withdraw as soon as their last deposit clears.

Table 8. Deferred and Forborne Annuities

Formula Name	Formula	Variables	Notes
Deferred present value (single rate)	$Q = R(1+i)^{-(m+1)} \frac{1 - (1+i)^{-n}}{1 - (1+i)^{-1}}$	$m = \text{deferral periods; payments begin at end of period } m+1; n = \text{number of payments; } i = \text{interest rate per period}$	Discounts each payment all the way back to time 0.
Deferred present value (two rates)	$Q = R(1+j)^{-(m+1)} \frac{1 - (1+i)^{-n}}{1 - (1+i)^{-1}}$	$j = \text{rate during deferral period; } i = \text{rate during payment phase}$	Use when the rate changes after the deferral ends.
Forborne accumulated value	$S_f = R \frac{(1+i)^n - 1}{i} \cdot (1+i)^m$	$n = \text{number of contributions; } m = \text{periods held after last contribution}$	Compounds the ordinary accumulated value forward by m more periods.

4.7 General Annuities

4.7.1 Concept and Interest Conversion Formulas

In the case of general annuities, the compounding and payment periods do not coincide (unlike the case for simple annuities). This problem can be solved by converting the general annuity into an equivalent ordinary simple annuity where the compounding and payments are on the same schedule.

We assume that compounding and payment times are equally spaced throughout the year. Thus, if compounding occurs s times per year, it occurs every $\frac{1}{s}$ year, and if payments are made t times per year, they occur every $\frac{1}{t}$ year.

In general, if compounding is done s times per year at a rate of i , and payments are made t times per year, we want to know the equivalent interest rate to i (call it j) if compounding is done t times per year. The problem is to solve for j in the following equation

$$(1+i)^s = (1+j)^t$$

Taking the $\frac{1}{t}$ power of both sides of the equation and then doing some rearrangement of terms gives us the following formula

$$j = (1+i)^{s/t} - 1$$

If interest is compounded continuously, then we need to solve $e^{r \cdot 1} = (1+j)^t$ in terms of j , where r is the nominal annual interest rate. Taking the $\frac{1}{t}$ power of both sides of the equation gives

$$j = e^{r/t} - 1$$

Alternately, one can solve for r to get

$$r = t \ln(1 + j)$$

For example, 6% interest compounded monthly (i.e., $t = 12$ and $j = .005$) is equivalent to $r = 12 \ln(1.005) = 0.05985$ or 5.985% compounded continuously.

Table 9. General Annuities: Interest Conversion

Formula Name	Formula	Variables	Notes
Equivalent periodic rate (discrete compounding)	$j = (1 + i)^{s/t} - 1$	i = rate per compounding period; s = compounding periods per year; t = payment periods per year; j = equivalent rate per payment period	Convert to an equivalent simple annuity, then apply Table A1 or A3 formulas.
Equivalent periodic rate (continuous compounding)	$j = e^{r/t} - 1$	r = nominal annual rate (continuously compounded)	Use when the stated annual rate is continuously compounded.
Continuous rate from periodic rate	$r = t \ln(1 + j)$		Inverse of the formula above.

4.7.2 Example – Accumulated Value of a General Annuity

Compute the accumulated value of a general annuity with term 11 years, where the interest is compounded 5 times per year at a nominal annual rate of 5%. Payments into the annuity are made 7 times per year in the amount of \$1,000 per payment.

Determine the interest rate per compounding period, i.e., $i = \frac{.05}{5} = .01$.

Determine the equivalent interest rate if compounding is done 7 times per year, i.e.,

$$j = (1 + .01)^{\frac{5}{7}} - 1 = 0.007132697$$

Use the formula for the accumulated value of a simple annuity (from Section 4.2.1) with per period interest rate 0.007132697, $n = 11 \times 7 = 77$ and $R = 1000$

$$S = R \frac{(1 + j)^n - 1}{j} = 1000 \frac{1.007132697^{77} - 1}{.007132697} = \$102,138.73$$

This can be verified using the block annuity calculator with an input csv file containing the following line:

.007132697, 1000, 77

Solve the same problem as above, except that the interest is compounded continuously. In this scenario, $j = e^{i/t} - 1 = e^{\frac{.05}{7}} - 1 = 0.007168428$ and $S = \$102,289.23$.

This can be verified using the block annuity calculator with an input csv file containing the following line:

.007168428, 1000, 77

4.7.3 Example – Discounted (Present) Value of a General Annuity

Compute the discounted value of a general annuity with term 13 years, where the interest is compounded 12 times per year at a nominal annual rate of 6%. Payments out of the annuity are made 26 times per year in the amount of \$500 per payment.

Determine the interest rate per compounding period, i.e., $i = \frac{.06}{12} = .005$.

Determine the equivalent interest rate if compounding is done 26 times per year, i.e.,

$$j = (1 + .005)^{\frac{12}{26}} - 1 = 0.0023046$$

Use the formula for the discounted value of a simple annuity (from Section 4.3.1) with per period interest rate 0.0023046, $n = 26 \times 13 = 338$ and $R = 500$ to get

$$Q = R \frac{1 - (1 + j)^{-n}}{j} = 500 \frac{1 - (1.0023046)^{-338}}{.0023046} = \$117,309.55$$

4.8 Perpetuities

A **perpetuity** is an annuity in which regular payments begin on a fixed date and continue indefinitely. It is sometimes referred to as a *perpetual annuity*.

The analysis of a perpetuity is straightforward. Suppose an initial investment of S earns interest at a rate of i per compounding period. If the interest earned each period is withdrawn, then the investment can support a constant payment of $R = iS$ at the end of every period forever, while preserving the original principal.

Equivalently, if a perpetuity pays R per period and the interest rate per period is i , then the required initial investment (i.e., the present value of the perpetuity) is $S = \frac{R}{i}$.

This relationship can be interpreted in two ways:

- As a funding problem: how much capital is required to generate a fixed payment indefinitely.
- As a valuation formula: the present value of a perpetual stream of payments.

If less than iS is withdrawn each period, then the principal will grow over time, allowing the payment to be increased in the future (for example, to account for inflation).

In practice, such idealized investment instruments are not directly available. However, endowment funds (e.g., for scholarships or charitable foundations) are conceptually similar. In these cases, the return is not guaranteed, and only a portion of the investment income is typically distributed, while the remainder is reinvested to maintain or grow the fund.

Example. Suppose a scholarship fund is designed to provide annual payments of \$5,000 indefinitely. If the fund earns 4% interest per year, how large must the initial endowment be?

Using the perpetuity formula,

$$S = \frac{R}{i} = \frac{5000}{0.04} = 125,000$$

Thus, an initial investment of \$125,000 is required to support the scholarship indefinitely.

Author's remark:

The formula $S = \frac{R}{i}$ can be viewed as the limiting case of the present value of an annuity as the number of payments becomes arbitrarily large. Recall that the present value of an annuity with n payments is

$$\frac{R(1 - (1 + i)^{-n})}{i}$$

Taking the limit as $n \rightarrow \infty$, we obtain

$$\lim_{n \rightarrow \infty} \frac{R(1 - (1 + i)^{-n})}{i} = \frac{R}{i}$$

since $(1 + i)^{-n} \rightarrow 0$ as $n \rightarrow \infty$.

The following table compares annuities and perpetuities.

Table 10. Comparison of Finite Annuities and Perpetuities

Concept	Finite Annuity	Perpetuity
Number of payments	n	Infinite
Payment per period	R	R
Interest rate per period	i	i
Future Value	$R \frac{[(1 + i)^n - 1]}{i}$	Not defined (diverges)
Present Value	$R \frac{[1 - (1 + i)^{-n}]}{i}$	$\frac{R}{i}$
Limiting behavior	Depends on n	Limit as $n \rightarrow \infty$ of annuity PV
Interpretation	Fixed number of payments	Payments continue indefinitely
Typical use	Loans, savings plans	Endowments, dividend models

4.9 Continuous Annuities

A **continuous annuity** is a type of an annuity where payments are made continuously but interest is compounded per some given period.

4.9.1 Discounted (Present) Value

To compute the present value formula for a continuous annuity, we start with a finite number of payments m within a given compounding period and then take the limit as m approaches infinity. To that end, start with an annuity that distributes n payments in the amount of R ($m = 1$ at this point). Interest is compounded at the end of each period at the rate i . Recalling from Section 4.3.1 that the current value of such an annuity is

$$Q = R(1 + i)^{-1} + R(1 + i)^{-2} + \cdots + R(1 + i)^{-n}$$

Next, assume that m payments in the amount of $\frac{R}{m}$ are made during each compounding period (see Figure 8).

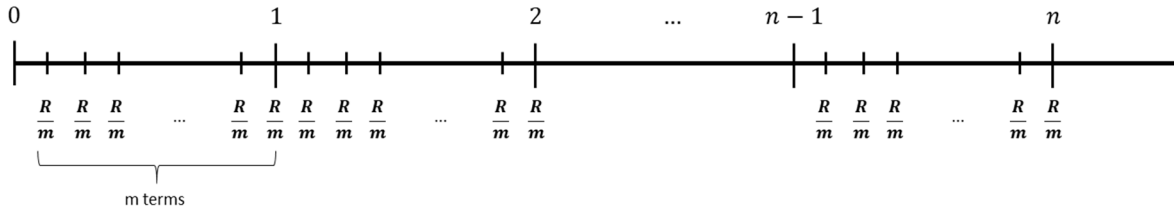


Figure 8. Multiple Payments per Compounding Period

We define the mini period rate j so that m compounding steps produce the same growth as one period of growth at rate i , i.e., $(1 + j)^m = 1 + i$, hence $1 + j = (1 + i)^{1/m}$. Noting that there are mn mini periods (i.e., payments), the present value of the annuity with the m payments per compounding period is given by

$$\begin{aligned} Q_m &= \frac{R}{m}(1 + j)^{-1} + \frac{R}{m}(1 + j)^{-2} + \cdots + \frac{R}{m}(1 + j)^{-mn} \\ &= \frac{R}{m} \left[(1 + i)^{-\frac{1}{m}} + (1 + i)^{-\frac{2}{m}} + \cdots + (1 + i)^{-\frac{mn}{m}} \right] \\ &= \frac{R}{m} \sum_{x=1}^{mn} (1 + i)^{-\frac{x}{m}} \end{aligned}$$

To get the formula for the present value of an annuity with a continuous number of payments in each compound period, we take the limit of the above expression as m goes to infinity, i.e.,

$$Q_\infty = \lim_{m \rightarrow \infty} Q_m = \lim_{m \rightarrow \infty} \frac{R}{m} \sum_{x=1}^{mn} (1 + i)^{-\frac{x}{m}} = R \lim_{m \rightarrow \infty} \sum_{x=1}^{mn} \left(\frac{1}{m} \right) (1 + i)^{-\frac{x}{m}} = R \int_0^n (1 + i)^{-x} dx$$

[**Aside:** Concerning the equivalence between the summation and the integral above, consider the

sum $\sum_{x=1}^{mn} \frac{1}{m} (1 + i)^{-x/m}$. To interpret this as a Riemann sum, define $t = \frac{x}{m}$.

The sum can be written as

$$\sum_{x=1}^{mn} (1 + i)^{-t} \Delta t, \quad \Delta t = \frac{1}{m}$$

Observe that the interval $[0, n]$ is divided into mn equal subintervals, where each subinterval has width

$$\Delta t = \frac{n - 0}{mn} = \frac{1}{m}$$

The values $t = \frac{x}{m}$ correspond to the right endpoints of the subintervals $\left[\frac{x-1}{m}, \frac{x}{m} \right]$.

Thus, the sum is a right Riemann sum for the function $f(t) = (1 + i)^{-t}$ on $[0, n]$.

Taking the limit as $m \rightarrow \infty$, we obtain

$$\lim_{m \rightarrow \infty} \sum_{x=1}^{mn} \frac{1}{m} (1+i)^{-t} = \int_0^n (1+i)^{-t} dt$$

Financial Interpretation: The factor $\frac{1}{m}$ represents a small time interval over which payment accrues. As $m \rightarrow \infty$, these intervals shrink to zero, and the discrete stream of payments becomes continuous, leading naturally to the integral representation.

End of Aside]

Next, solving the definite integral, we get

$$Q_{\infty} = R \int_0^n (1+i)^{-x} dx = R \left[\frac{-(1+i)^{-x}}{\ln(1+i)} \right]_{x=0}^{x=n} = R \frac{1 - (1+i)^{-n}}{\ln(1+i)}$$

If Q_{∞} , i and n are known, we can solve for R , i.e.,

$$R = Q_{\infty} \frac{\ln(1+i)}{1 - (1+i)^{-n}}$$

[Aside: To evaluate the integral $\int (1+i)^{-x} dx$, we use the identity

$$a^{u(x)} = e^{u(x) \ln(a)}$$

which allows us to rewrite the integrand as

$$(1+i)^{-x} = e^{-x \ln(1+i)}$$

Now apply the standard rule

$$\int e^{kx} dx = \frac{1}{k} e^{kx}, (k \neq 0)$$

Here, $k = -\ln(1+i)$, and so,

$$\int e^{-x \ln(1+i)} dx = \frac{1}{-\ln(1+i)} e^{-x \ln(1+i)}$$

Rewriting in terms of the original base gives

$$\int (1+i)^{-x} dx = \frac{-(1+i)^{-x}}{\ln(1+i)} + C$$

End of Aside]

As an example, consider two scenarios:

- Scenario A: Ordinary, simple, certain annuity with $n = 60$ equal payments of \$10,000 and an interest rate of .5% per period
- Scenario B: Annuity with continuous payments over $n = 60$ periods with an interest rate of .5% per period. The continuous payments equal \$10,000 in each period.

For Scenario A, the present value of the annuity is

$$Q = R \frac{1 - (1 + i)^{-n}}{i} = 10,000 \frac{1 - 1.005^{-60}}{.005} = \$517,255.61$$

For Scenario B, the present value of the annuity is

$$Q_{\infty} = R \frac{1 - (1 + i)^{-n}}{\ln(1 + i)} = 10,000 \frac{1 - 1.005^{-60}}{\ln(1.005)} = \$518,547.67$$

The present value of the annuity in Scenario B is larger than the present value of the annuity in Scenario A, which is to be expected since payments are effectively distributed sooner in Scenario B. Since $\ln(1 + i) < i$ for $i > 0$, the denominator in the continuous case is smaller, which explains why the present value is larger.

4.9.2 Accumulated Value

It is also possible to accumulate value with continuous investments into the annuity. We first consider the case of m mini investments into the annuity in the amount of $\frac{R}{m}$ during each compounding period. This is basically Figure 8 except that payments are accumulated forward rather than discounted. If the interest rate for each compounding period is i then (using the same calculation as in Section 4.9.1) the equivalent interest rate for the mini-period is $j = (1 + i)^{\frac{1}{m}} - 1$. The accumulated value of the annuity is given by

$$\begin{aligned} S_m &= \frac{R}{m} + \frac{R}{m}(1 + j) + \frac{R}{m}(1 + j)^2 + \cdots + \frac{R}{m}(1 + j)^{mn-1} \\ &= \frac{R}{m} \left[1 + (1 + i)^{\frac{1}{m}} + (1 + i)^{\frac{2}{m}} + \cdots + (1 + i)^{\frac{mn-1}{m}} \right] \\ &= \frac{R}{m} \sum_{x=0}^{mn-1} (1 + i)^{\frac{x}{m}} \end{aligned}$$

To get the accumulated value of an annuity with continuous input investments, we take the limit as m approaches infinity of S_m

$$S_{\infty} = \lim_{m \rightarrow \infty} S_m = \lim_{m \rightarrow \infty} \frac{R}{m} \sum_{x=0}^{mn-1} (1 + i)^{\frac{x}{m}} = R \lim_{m \rightarrow \infty} \sum_{x=0}^{mn-1} \left(\frac{1}{m} \right) (1 + i)^{\frac{x}{m}} = R \int_0^n (1 + i)^x dx$$

The second summation is the left Riemann sum of the function $(1 + i)^x$ from 0 to n , and thus taking the limit as $m \rightarrow \infty$ (and $\frac{1}{m} \rightarrow 0$) yields the stated integral. Solving the definite integral, we get

$$S_{\infty} = R \int_0^n (1 + i)^x dx = R \left[\frac{(1 + i)^x}{\ln(1 + i)} \right]_{x=0}^{x=n} = R \frac{(1 + i)^n - 1}{\ln(1 + i)}$$

As an example, compare the accumulated value of an ordinary, simple, certain annuity with 120 investments of \$5,000 at an interest rate of .6% per period to a continuous investment annuity with the same parameters.

For the ordinary, simple, certain annuity, the accumulated value is

$$S = R \frac{(1+i)^n - 1}{i} = 5,000 \frac{1.006^{120} - 1}{.006} = \$875,015.05$$

and for the continuous investment annuity, the accumulated value is

$$S_{\infty} = R \frac{(1+i)^n - 1}{\ln(1+i)} = 5,000 \frac{1.006^{120} - 1}{\ln(1.006)} = \$877,637.48$$

The accumulated value in the continuous case is slightly larger since money is deposited sooner, i.e., during each compounding period rather than at the end of each compounding period.

Table 11. Perpetuities and Continuous Annuities

Formula Name	Formula	Variables	Notes
Perpetuity payout	$R = iS$	$S = \text{principal}$ $R = \text{periodic payout}$ $i = \text{interest rate per period}$	Payout equals interest earned principal S is never depleted.
Continuous annuity — discounted (present) value	$Q_{\infty} = R \frac{1 - (1+i)^{-n}}{\ln(1+i)}$	$R = \text{payment rate per period (total payment per period, distributed continuously)}$ $i = \text{interest rate per period}$ $n = \text{number of periods}$	Payments made continuously within each compounding period. Replace denominator i with $\ln(1+i)$ relative to the ordinary formula.
Continuous annuity — payout rate (solve for R)	$R = Q_{\infty} \frac{\ln(1+i)}{1 - (1+i)^{-n}}$	$Q_{\infty} = \text{present value}$ $i = \text{interest rate per period}$ $n = \text{number of periods}$	Solve for R when Q_{∞} , when i, n are known.
Continuous annuity — accumulated value	$S_{\infty} = R \frac{(1+i)^n - 1}{\ln(1+i)}$	$R = \text{payment rate per period (total payment per period, distributed continuously)}$ $i = \text{interest rate per period}$ $n = \text{number of periods}$	Payments made continuously within each compounding period. Replace denominator i with $\ln(1+i)$ relative to the ordinary formula.

4.10 Supporting Programs

4.10.1 Block Annuity Calculator

File name: block_annuity.py

The program computes the accumulated value of:

- Simple and general annuities (Simple Annuity tab of the GUI).
- Block annuities with multiple interest-rate blocks (Block Annuity tab of the GUI, loaded from a CSV file).

Both the inputs and the outputs are handled entirely within the GUI. There are separate Calculate and Clear buttons on each tab.

Simple Annuity Tab

The tab takes five inputs: the nominal annual interest rate (as a percent, e.g., 5 rather than .05), the payment frequency, the compounding frequency, the payment per period, and the number of periods. Both frequencies are chosen from dropdown lists (Annual, Semi-annual, Quarterly, Monthly, Weekly, Daily).

When the payment frequency and compounding frequency are equal, the tab treats the input as a simple annuity and uses the per-period rate $i = \text{nominal_annual_rate}/p$, where p is the payment frequency. When they differ, the tab treats the input as a general annuity (see Section 4.7) and converts the per-compounding-period rate i_c to an equivalent per-payment-period rate via

$$j = (1 + i_c)^{c/p} - 1,$$

where c is the compounding frequency per year, p is the payment frequency per year, and $i_c = \text{nominal_annual_rate}/c$. The accumulated value is computed using j as the per-period rate. The Result panel reports the accumulated value and also shows j , n , and R , and tags the result as either “simple annuity” or “general annuity”. The Clear button resets all inputs to their defaults (both frequencies to Monthly).

Block Annuity Tab

This tab computes the total value of a savings annuity over several blocks, where:

- Each block has a fixed interest rate and fixed payment (i.e., investment) per period.
- The interest rate and payment amount may vary among blocks.
- The blocks may be of different sizes, i.e., different numbers of periods.
- The input is a CSV file whose rows each describe one block in the format:
interest_rate_per_period, saving_amount_per_period, number_of_periods

Click **Browse...** to open a file-picker dialog and select the CSV file; the selected path appears in the text field. Click **Calculate** to compute the results, which are displayed in the Results panel inside the GUI (not in the terminal). Click **Clear** to reset both the file path and the Results panel.

As an example, put the following lines in a CSV file and run the program:

```
0.0042, 500, 10
0.0033, 400, 25
0.0038, 300, 30
0.0045, 700, 15
0.0047, 200, 20
```

The Results panel will display the block-by-block summary, each block’s value at the end of its own term, each block’s value at the end of the full annuity term (including interest accrued during subsequent blocks), and the total value at the end of the annuity term. The output is shown below for the above example:

Accumulated value at end of each block:

```
Block 0: $ 5,095.57
Block 1: $ 10,406.20
Block 2: $ 9,513.95
Block 3: $ 10,837.29
Block 4: $ 4,183.74
```

Value of each block at end of full term:

Block 0: \$	7,283.76
Block 1: \$	13,698.88
Block 2: \$	11,177.33
Block 3: \$	11,902.79
Block 4: \$	4,183.74

Total: \$ 48,246.50

4.10.2 Annuity Payout Calculator

File name: simple_annuity_payout.py

This program does computations for ordinary, simple, certain annuities, with optional deferral of the payout phase. It is assumed that the compounding and the periodic payouts happen on the same schedule, e.g., at the end of each month.

Inputs

The required input is the per-period interest rate i during the payout phase (as a decimal, e.g., .05 rather than 5%). Three further inputs describe the annuity itself — the initial investment q , the number of payouts n , and the payout per period r — and the user supplies any two of the three, leaving the third at 0 for the program to compute. Two additional inputs allow for a deferral phase between the initial investment and the start of payouts:

- m = number of deferral periods before payouts begin. Set $m = 0$ for no deferral.
- j = per-period interest rate during the deferral phase (as a decimal). Leave $j = 0$ to default to i , i.e., to use the same rate in both phases.

Setting $m = 0$ recovers the non-deferred case exactly, so the same program handles both deferred and non-deferred ordinary, simple, certain annuities.

Behavior

The program solves for whichever of q , n , or r is left at 0, using the fund value at the start of the payout phase, $q' = q(1 + j)^m$:

If n is left at 0, the program computes the number of payouts using

$$n = \frac{\ln r - \ln(r - iq')}{\ln(1 + i)}$$

If q is left at 0, the program computes the initial investment using

$$q = r(1 + j)^{-m} \frac{1 - (1 + i)^{-n}}{i}$$

If r is left at 0, the program computes the payout per period using

$$r = \frac{iq(1 + j)^m}{1 - (1 + i)^{-n}}$$

Edge Cases

When solving for n :

- If the desired payout is less than or equal to the interest earned on the fund each period, i.e., $r \leq iq'$, the payouts can continue indefinitely. In this case the program outputs “infinite” for the number of payouts.
- If the desired payout exceeds the fund value at the end of the first payout period, i.e., $r > q'(1 + i)$, the payout is not possible and an error message is displayed in a pop-up dialog.

Invalid combinations of inputs (e.g., leaving two of q , n , r at 0, or providing a non-positive i) also produce a pop-up error message.

4.11 Exercises

1. Mr. Periwinkle agrees to purchase an annuity from a bank (this is intended to be a savings plan). He is to deposit \$750 at the end of each month for 5 years, with a guaranteed nominal annual interest rate of 3% (compounded monthly). How much is Mr. Periwinkle's annuity worth at term?

Hint: Use the "Simple Annuity" option in the `block_annuity` calculator. Alternately, use the "Block Annuity" with a single block. The input is a csv file with just one line, i.e., `.0025, 750, 60`. Note that `.0025` is the monthly rate based on a 3% nominal annual interest rate.

2. Continuing with the previous exercise, assume the term and interest rate is the same but Mr. Periwinkle asks the bank how much he would need to save each month to reach \$50,000 at the end of the term.

Hint and Answer: Use the formula for R in terms of i , S and n in Section 4.2.2. Be careful to calculate $i = \frac{.03}{12}$. The answer is \$773.44 (rounding up to the next cent). You can check the answer using the block annuity calculator.

3. Joan and John Song are saving for a down payment on a house. An investment company has offered them a savings annuity that pays a guaranteed nominal annual interest rate of 3% (compounded monthly) for the first 30 months, which may change (depending on market conditions) for the remaining 30 months of the annuity. However, it is guaranteed that the interest rate will drop at most .5% during the second 30 months. The couple would like to know how much the annuity will be worth at term if they invest \$1,000 per month, assuming the least optimistic interest rate for the second 30 months of the annuity, i.e., 2.5%.

Hint and Answer: Use the `block_annuity` calculator with two blocks to determine that the value of the annuity at term is \$64,041.91. Don't forget to convert 3% and 2.5% to monthly interest percentages (i.e., `.0025` and `.00208333`).

4. In the previous example, assume that the Song's really needed \$75,000. If the term (5 years) and interest rates are the same, how much do they need to save each month to reach their goal?

Hint: Clearly, there are multiple possible answers here. One approach is to increase the payment amount in the first block and leave the payment amount in the second block at \$1,000/month. Use the block annuity calculator and experiment with various payment amounts for the first 30 months. For example, a monthly payment of \$1,333/month for the first 30 months and \$1,000/month in the second 30 months yields a final accrued value of \$75,070.14.

5. Mr. Trubisky has just retired early after a lucrative career as a professional athlete. He has \$10,000,000 in a savings account and would like to invest half of that amount in an annuity. How long will the annuity last at 3% nominal annual interest rate (compounded monthly) if he takes a payment of \$50,000 per month?

Hint: Use the simple annuity payout calculator.

6. After doing the computation, Mr. Trubisky realizes that his withdraw amount will last about 9.6 years. Given that he is now 35 years old, he wants the payouts to continue for 50 years. With the interest rate conditions being the same, how much can Mr. Trubisky be paid each month if the annuity is to last 50 years? **Hint:** Use the simple annuity payout calculator.

Answer: \$16,098.87/month.

7. How can the block annuity calculator be used to calculate an annuity due where the payments are made at the beginning of each period and there is one accrual period after the last payment?

Hint: Use two blocks, with the second block consisting of a \$0 payment but at the same interest rate as the first block.

Explanation. The block annuity calculator computes each block's value using the ordinary-annuity formula, i.e., it assumes that payments are made at the end of each period. Block 1, with parameters (i, R, n) , therefore produces

$$S_1 = R \cdot \frac{(1+i)^n - 1}{i}$$

which is the accumulated value of an ordinary annuity at the end of period n . Adding Block 2 with $(i, 0, 1)$ does two things: its own accumulated value is zero (since the payment is zero), and it compounds Block 1's value forward by one additional period of interest. The total at the end of Block 2 is therefore

$$S_1 \cdot (1+i) = R \cdot \frac{(1+i)^n - 1}{i} \cdot (1+i),$$

which matches the closed-form expression for the accumulated value of an annuity due given in Section 4.3.3. Intuitively, the zero-payment block shifts each of the n ordinary-annuity payments forward by one period, which is algebraically identical to moving every payment from the end of a period to the beginning.

8. Test out your methodology from Exercise 7 on the following annuity due: \$150 payment (at the beginning of each monthly payment period), for four years and at a nominal annual interest percentage of 6% (compounded monthly). Check your answer by using the annuity due formula in Section 4.4.

Hint: Use the block annuity option of the block annuity calculator with input file `block_4_11_8.csv`. **Answer:** \$8,155.25.

9. Milton invests \$75,000 in a deferred annuity at a nominal annual interest percentage of 4% (compounded monthly). The annuity is deferred for 5 years, after which 90 monthly payments are made to Milton. Compute the amount of the regular payments.

Hint: use the deferred annuity calculator.

10. As a variation of the previous exercise, assume Milton wants to get payments of \$1,175.50 per month for 90 payments with the first payment 5 years from now. How much money does he need to invest up front in the deferred annuity?

Hint: use the deferred annuity calculator.

11. Modify the Python program for the deferred annuity calculator to support the case where the interest rate during the time of deferral is different from the interest rate during the payment periods.

Hint: You need to add another input for the interest rate during the deferral period, and then use that rate in the calculation of interest during the deferral period.

12. Compute the accumulated value of a general annuity with term 13 years, where the interest is compounded 7 times per year at a nominal annual rate of 7% and payments into the annuity are made 23 times per year in the amount of \$500 per payment.

Hint: The interest rate per compounding period is $i = \frac{.07}{7} = .01$. If compounding is done at the same rate as the payments, the equivalent rate is $j = (1 + .01)^{\frac{7}{23}} - 1 = .003032952$. Next, use the formula for the accumulated value of a simple annuity with per period interest rate j , $n = 13 \times 23 = 299$ and $R = 500$. Check your answer with the block annuity calculator using a csv file with the following line: .003032952,500,299.

13. Compare the present value of an ordinary, simple, certain annuity with 360 equal payments of \$2,000 at an interest rate of .005 per period with a continuous annuity of the same term, payout, and interest rate.

Hint: See the example in Section 4.9.1.

Answer: \$333,583.23 for the ordinary, simple, certain annuity and \$334,416.49 for the continuous annuity.

5 Repayment of Loans

If you think nobody cares about you, try missing a couple of payments. — Earl Wilson

5.1 Amortization of Loans

5.1.1 General Formula and Equivalence to Annuities

Amortization refers to the distribution of payments into multiple equal installments where each payment consists of both principal and interest. Amortization is mainly used in loan repayments (e.g., home mortgages and car loans) and in sinking funds (i.e., money set aside over a period of time to fund a future capital expense, or repayment of a long-term debt). Repayments are divided into equal amounts for the duration of the loan. However, the amount of interest and principal pay-back in each installment varies, as demonstrated in the example that follows below. This process is known as Equated Monthly Installment (EMI).

(**Note:** In what follows, we say “payment” although technically it is a “repayment” of money that had been previously loaned, since “payment” is the term more commonly used in the literature.)

Make the following assignments:

P = principal amount borrowed

A = equal periodic amortization payment

i = the interest rate per compounding period

n = total number of payment periods.

It is assumed that the compounding and payment periods coincide.

When doing the calculation, it is important to note that the loanee (person who receives the loan) must pay back the amount of interest due at the end of each payment period.

At the end of the first payment period, the amount still owed to the lender (“the balance”) is the original principal plus the interest due for the first period minus the amount of principal paid back by the loanee to the lender, i.e.,

$$B_1 = P + iP - A = (1 + i)P - A$$

At the end of the second payment period, the balance is

$$B_2 = (1 + i)B_1 - A = (1 + i)^2P - A(1 + i) - A$$

At the end of the third payment period, the balance is

$$\begin{aligned} B_3 &= (1 + i)B_2 - A \\ &= (1 + i)^3P - A(1 + i)^2 - A(1 + i) - A \end{aligned}$$

In general, after payment period k , the balance is

$$B_k = (1 + i)^kP - A[(1 + i)^{k-1} + \cdots + (1 + i) + 1]$$

At the end of n payment periods (for $n \geq 1$), the balance of the mortgage should be zero, i.e.,

$$0 = B_n = (1 + i)^nP - A[(1 + i)^{n-1} + \cdots + (1 + i) + 1]$$

which implies

$$(1+i)^n P = A[(1+i)^{n-1} + \cdots + (1+i) + 1]$$

Solving for A and noting that the terms in the square brackets constitute a geometric series whose sum we know how to calculate from Section 2.1.2, we get

$$A = \frac{iP(1+i)^n}{(1+i)^n - 1}$$

which can be rewritten as

$$A = \frac{iP}{1 - (1+i)^{-n}}$$

The above formula should look familiar as it is equivalent to the formula for regular payments from an ordinary, simple, certain annuity (see Section 4.3.1). A mortgage can be thought of as an annuity (purchased by a lending institution such as a bank) from an individual. The discounted (present) value of the annuity (Q) is the amount of the mortgage (P), and the annuity payments (R) are essentially the periodic amortization payments (A).

Table 12. Amortization of Loans: General Formulas

Formula Name	Formula	Variables	Notes
Balance after period k (recursive)	$B_k = (1+i)B_{k-1} - A$	B_{k-1} = balance at end of period k A = periodic payment i = rate per period	Each period's balance grows by interest and drops by the payment.
Balance after period k (closed form)	$B_k = (1+i)^k P - A \frac{(1+i)^k - 1}{i}$	P = principal borrowed n = total number of periods	Retrospective form expressed as a geometric series.
Periodic amortization payment	$A = \frac{iP(1+i)^n}{(1+i)^n - 1} = \frac{iP}{1 - (1+i)^{-n}}$	Obtained by setting $B_n = 0$ and solving for A .	Equivalent to the payment formula for an ordinary, simple, certain annuity (Section 4.3.1).

5.1.2 Breakdown of Interest versus Principal Repayment

As noted, each payment is a combination of principal payback and interest payment where the sum of the two is equal to A for each payment. However, the ratio of principal payback to interest payment changes over time. At the end of each payment period, the interest owed is i times the amount of outstanding balance at the beginning of the period. Thus, the interest owed at the end of period k is given by the following formula:

$$I_k = iB_{k-1} = i(1+i)^{k-1}P - iA[(1+i)^{k-2} + \cdots + (1+i) + 1]$$

and of course, the amount of principal repayment at the end of period k (call it R_k) is given by

$$R_k = A - I_k$$

The interest owed at the end of period k can also be represented recursively, i.e.,

$$\begin{aligned} B_{k-1} &= (1 + i)B_{k-2} - A \\ \Rightarrow iB_{k-1} &= (1 + i)iB_{k-2} - iA \\ \Rightarrow I_k &= (1 + i)I_{k-1} - iA \end{aligned}$$

which is convenient for programming purposes.

As an example, consider a 3-year loan on a car for \$10,000 at a 3% annual interest rate where the loanee pays back the loan on a monthly basis. In terms of the above formulas, $P = 10,000$, $i = \frac{.03}{12} = .0025$ and $n = 12 * 3 = 36$. Using the formula for equal periodic amortization payment yields $A = \$290.81$. A breakdown (interest versus principal) for the first four repayments is given in Table 13. Given the relatively short term of the loan, the principal payments outweigh the interest payments from the start. For home mortgages (e.g., 30 years), the interest payments are greater than principal payback at the beginning of the loan.

Table 13. Interest-Principal Repayment Breakdown Example

Period	Interest Owed Formula	Interest to be paid for given period	Principal to be repaid for given period
1	$I_1 = iP$	\$25.00	\$265.81
2	$I_2 = (1 + i)I_1 - iA$	\$24.34	\$266.48
3	$I_3 = (1 + i)I_2 - iA$	\$23.67	\$267.14
4	$I_4 = (1 + i)I_3 - iA$	\$23.00	\$267.81

A summary of the formulas in this subsection is provided in the following table.

Table 14. Interest vs. Principal Breakdown per Payment

Formula Name	Formula	Variables	Notes
Interest owed in period k	$I_k = iB_{k-1}$	$I_k = \text{interest portion of payment } k$	Interest is charged on the previous period's balance.
Closed-form interest formula	$I_k = i(1+i)^{k-1}P - A[(1+i)^{k-1} - 1]$	$k \geq 1$	Expresses interest directly in terms of P and A (no need for B_{k-1}).
Principal repaid in period k	$R_k = A - I_k$	$R_k = \text{principal portion of payment } k$	Each payment of A splits into principal and interest.
Recursive interest formula	$I_k = (1+i)I_{k-1} - iA$	<i>Uses interest from the previous period.</i>	Convenient for iterative / programmatic calculation.

5.1.3 Rounding Mortgage Payments

It may be desirable to round upward (or downward) the periodic mortgage payments to the next dollar, 10-dollar, or 100-dollar value (or any other convenient round figure), and then adjust the final payment downward (or upward). The algorithm for doing this is as follows:

1. Determine the periodic payment A .
2. Round A up to the next \$1, \$10, or \$100 value, depending on the granularity desired. Call this new value $\hat{A}$.
3. Determine the penultimate balance, assuming the amount $\hat{A}$ was used for all past payments as well as the penultimate payment and we have

$$B_{n-1} = (1+i)^{n-1}P - \hat{A}[(1+i)^{n-2} + \cdots + (1+i) + 1] = (1+i)^{n-1}P - \hat{A} \frac{(1+i)^{n-1} - 1}{i}$$

4. Calculate the adjusted last payment which is

$$B_{n-1} + iB_{n-1} = (1+i)B_{n-1} = (1+i)^n P - \hat{A} \frac{(1+i)^n - (1+i)}{i}$$

Actually, one can go directly to the formula in Step 4 and skip Step 3, or do Step 3 and then just multiple by $(1+i)$.

As an example, consider a \$200,000 mortgage at a 6% nominal annual interest with payments due each month. The term of the mortgage is short (just 10 years). The 6% nominal annual interest means the monthly interest rate is $\frac{.06}{12} = .005$.

The monthly payment is $A = \$2,220.41$ which we round up to $\hat{A} = \$2,221$. Using the formula from Step 4 in the above algorithm gives us the amount of the final payment:

$$(1+i)^n P - \hat{A} \frac{(1+i)^n - (1+i)}{i} = (1.005)^{120} (200,000) - (2,221) \frac{(1.005)^{120} - 1.005}{.005} = \$2,124.32$$

Alternately, we could round the payments to the nearest multiple of \$10 (\$2,220 in this case) and then calculate the final payment, i.e., \$2,287.20 which as one would expect, is slightly higher than the other 119 payments of \$2,220.

We could even round to the nearest multiple of \$100 (i.e., \$2,200) and compute the final payment, i.e., \$5,544.78.

In terms of rounding down, keep in mind that the loanee is expected to pay at least the amount of interest owed each period, and so we cannot go beyond that limit.

In terms of rounding up, it is possible to round up to the point where some of the last several payments are not needed since all the principal is already paid off. The number of payments actually required can be found by solving $B_k = 0$ for k , giving

$$k = \frac{\ln\left(\frac{\hat{A}}{\hat{A} - iP}\right)}{\ln(1+i)}$$

and rounding up to the next integer.

Table 15. Rounded Mortgage Payments

Formula Name	Formula	Variables	Notes
Penultimate balance	$B_{n-1} = (1+i)^{n-1}P - \hat{A} \frac{(1+i)^{n-1} - 1}{i}$	$n = \text{total number of periods}$	Balance assuming $\hat{A}$ was paid in all prior periods.
Adjusted final payment	$(1+i)B_{n-1} = (1+i)^n P - \hat{A} \frac{(1+i)^n - (1+i)}{i}$	Equal to B_{k-1} plus one period of interest (evaluated at $k = n$).	Absorbs the rounding difference into the last payment.

5.1.4 Early Repayment via Additional Principal Payments

Borrowers often accelerate mortgage repayment by contributing additional amounts toward principal each period. This reduces the outstanding balance more rapidly, decreases total interest paid, and shortens the loan term.

To cast the problem more formally, we make the following assignments.

- P = original principal
- A = required periodic payment
- C = additional principal payment made each period
- i = interest rate per period

- n = original number of periods
- m = number of periods (at beginning of loan) that amount A is paid each period
- k = number of additional periods (beyond m) to payoff the loan when amount $A + C$ is paid per each period, starting with period $m + 1$

Suppose that after m periods of regular payments in the amount A , the loanee begins making additional principal payments of C per period. The remaining balance at time m is

$$B_m = (1 + i)^m P - A \frac{(1 + i)^m - 1}{i}$$

From that point forward, the loan can be viewed as a new loan with principal B_m and periodic payment $A + C$. The number of remaining periods k required to repay the loan is given by

$$k = \frac{\ln\left(\frac{A + C}{A + C - iB_m}\right)}{\ln(1 + i)}$$

Thus, the total number of periods required to repay the loan is $m + k$.

Example 1 – Additional Monthly Principal Payment

Mr. Marino has a \$300,000 mortgage for 30 years at 4% nominal annual interest with monthly payments.

- Monthly payment $A \approx \$1,432.25$.
- After 5 years, balance $\approx \$271,000$.

If he pays an extra \$1,000 per month, total payment becomes \$2,432.25.

With this adjustment, we have

$$B_{60} = \left(1 + \frac{.04}{12}\right)^{60} (300,000) - (1,432.25) \frac{\left(1 + \frac{.04}{12}\right)^{60} - 1}{i} = \$271,342.27$$

$$k = \frac{\ln\left(\frac{2432.25}{2432.25 - \left(\frac{.04}{12}\right)(271,342.27)}\right)}{\ln\left(1 + \frac{.04}{12}\right)} = 139.5$$

So, with the new repayment arrangement, it would take $m + k = 60 + 139.5 = 199.5$ months to repay the loan (down from a total of 300 months).

Example 2 – Targeting a 25-Year Payoff

To repay the mortgage in 25 years (300 months), after 5 years there are 240 months remaining. The required payment is approximately \$1,644, implying an additional payment of about \$212 per month.

Comparison of Additional Monthly Principal Payments (after 5 years):

Additional Payment (\$)	Total Monthly Payment (\$)	Remaining Term (months)	Total Loan Length (years)
0	1432.25	300	30.0
200	1632.25	243	25.25
500	1932.25	190	20.83
1000	2432.25	139.5	16.67

Required Additional Payments to Achieve Target Payoff Periods:

Target Total Loan Length (years)	Remaining Term After Year 5 (months)	Required Monthly Payment (\$)	Additional Payment (\$)
30	300	1432	0
27	264	1547	115
25	240	1644	212
20	180	2007	575

Remarks

1. Additional payments reduce interest significantly.
2. Effects are strongest early in the loan.
3. Assumes extra payments go fully to principal.
4. Refinancing is a separate consideration.

5.2 Balloon Loans

5.2.1 Concept and Formulas

In Section 3.1, an example of a balloon loan was given. In the example, payment of all principal was deferred until the last payment of the loan. All the intermediate payments only entailed interest payment on the initial principal P . In the more general case, an amount B (less than or equal to the original principal) is made at the last payment. The loanee does, in some scenarios, repay some of the principal along the way (i.e., $P - B$).

In the following calculations, the terms A , P , B_n and i are as defined in Section 5.1.1. It is assumed that n equal payments are made in the amount A and then a final payment in the amount of B (at the same time as the n^{th} payment).

Relying on the derivation from Section 5.1.1, at the end of n periods (for $n \geq 1$, the total amount owed to the loaner should be B (the balloon payment). Using this fact, we solve for A in the equations below:

$$B = B_n = (1+i)^n P - A[(1+i)^{n-1} + \dots + (1+i) + 1]$$

$$B = (1+i)^n P - A \frac{(1 - (1+i)^n)}{-i}$$

$$A \frac{((1+i)^n - 1)}{i} = (1+i)^n P - B$$

$$A = \frac{iP(1+i)^n - iB}{(1+i)^n - 1}$$

If A , i , B and n are known, the above equation can be solved for P :

$$P = \frac{A[(1+i)^n - 1] + iB}{i(1+i)^n}$$

which can also be written as

$$P = A \frac{1 - (1+i)^{-n}}{i} + B(1+i)^{-n}$$

If $B = 0$, the formula for A is the same as for a non-balloon mortgage. If $B = P$ (all the principal is paid at the end of the term), then

$$A = \frac{iP(1+i)^n - iP}{(1+i)^n - 1} = iP$$

which is to be expected since one is only paying interest during the course of the loan, with the principal repayment deferred until the end of the loan.

There are variants on the assumptions for balloon loans which lead to slightly different formulas for A . For example, one could assume that n payments in the amount of A are made, followed by payment $n + 1$ in the amount of B . In this case, $B = B_n + iB_n = B_n(1+i)$ which when solved for A leads to

$$A = \frac{iP(1+i)^n}{(1+i)^n - 1} - \frac{iB}{(1+i)^{n+1} - (1+i)}$$

5.2.2 Example – Balloon Loan as a General Annuity

A car loan stipulates payments of \$500 at the end of each quarter for 5 years, with an additional payment of \$7,000 at the end of 5 years. What is the present value of the loan at a 6% nominal annual interest (compounded monthly)?

The monthly rate is $i = \frac{.06}{12} = .005$ which can be converted to an equivalent quarterly rate as

follows: $j = (1+i)^{s/t} - 1 = 1.005^{\frac{12}{4}} - 1 = 0.015075125$

We are essentially being asked to work backwards and find the cost of the car, given payments ($A = 500$), interest rate ($j = 0.015075125$), the balloon amount ($B = 7,000$) and the number of payment ($n = 20$). Using the formula for P from Section 5.2.1 yields

$$P = \frac{A[(1+j)^n - 1] + jB}{j(1+j)^n} = \frac{500[(1.015075125)^{20} - 1] + (.015075125)(7,000)}{(.015075125)(1.015075125)^{20}} = \$13,767.57$$

Table 16. Balloon Loans

Formula Name	Formula	Variables	Notes
Periodic payment with balloon at period n	$A = \frac{iP(1+i)^n - iB}{(1+i)^n - 1}$	B = balloon payment at period n	Reduces to the amortization formula when $B = 0$, and to iP when $B = P$.
Principal borrowed given A, B, i, n	$P = A \frac{1 - (1+i)^{-n}}{i} + B(1+i)^{-n}$	Present value of n payments plus present value of B .	Useful for working out the amount that can be financed.
Periodic payment with balloon at period $n + 1$	$A = \frac{iP(1+i)^n}{(1+i)^n - 1} - \frac{iB}{(1+i)^{n+1} - (1+i)}$	n regular payments are followed by a balloon one period later.	Variant in which the balloon is due one period after the final regular payment.

5.3 Computation of Remaining Balance on a Loan

In Section 5.1.1, we used a retrospective method to determine the balance on a loan at the end of a given period. In the formula below, B_k is the accumulated value of the debt at time k minus the accumulated value of the payments at time k . This difference is the balance owed at time k . As before, A is the equal periodic amortization payment.

$$B_k = (1+i)^k P - A[(1+i)^{k-1} + \cdots + (1+i) + 1]$$

applying the formula for the sum of a geometric series to the above, we get the more compact form

$$B_k = (1+i)^k P - A \frac{(1+i)^k - 1}{i}$$

Another approach (the prospective method) is to sum the present (discounted) value of the remaining $n - k$ payments at time k , i.e.,

$$B_k = A(1+i)^{-1} + A(1+i)^{-2} + \cdots + A(1+i)^{-(n-k)}$$

$$B_k = A(1+i)^{-1} [1 + (1+i)^{-1} + \cdots + (1+i)^{-(n-k-1)}]$$

$$B_k = A(1+i)^{-1} \frac{1 - (1+i)^{-(n-k)}}{1 - (1+i)^{-1}}$$

$$B_k = A \frac{1 - (1+i)^{-(n-k)}}{i}$$

In the above, we used the fact that $1 - (1 + i)^{-1} = \frac{i}{1+i}$.

The first formula works even if the final payment is irregular (e.g., a balloon payment) while the second formula does not work in that case. However, the second formula works even when the original amount of the loan is unknown.

The two formulas are algebraically equivalent, as is shown below:

At the end of the loan (focal date), the dated value of the debt minus the dated value of the payments should equal zero, i.e.,

$$0 = (1 + i)^n P - A[(1 + i)^{n-1} + \cdots + (1 + i) + 1]$$

which is equivalent to

$$(1 + i)^n P = A \frac{(1 + i)^n - 1}{i}$$

The above can be rewritten as

$$P = A \frac{1 - (1 + i)^{-n}}{i}$$

Subtracting $A \frac{1 - (1 + i)^{-k}}{i}$ from both sides, we get

$$P - A \frac{1 - (1 + i)^{-k}}{i} = A \frac{1 - (1 + i)^{-n}}{i} - A \frac{1 - (1 + i)^{-k}}{i} = A \frac{(1 + i)^{-k} - (1 + i)^{-n}}{i}$$

Multiplying by $(1 + i)^k$ yields

$$P(1 + i)^k - A \frac{(1 + i)^k - 1}{i} = A \frac{1 - (1 + i)^{-(n-k)}}{i}$$

Thus, the two formulas for B_k are algebraically equivalent.

Table 17. Remaining Balance on a Loan

Formula Name	Formula	Variables	Notes
Retrospective method	$B_k = (1 + i)^k P - A \frac{(1 + i)^k - 1}{i}$	$k = \text{current period (already paid)}$	Works even when the final payment is irregular (e.g., a balloon).
Prospective method	$B_k = A \frac{1 - (1 + i)^{-(n-k)}}{i}$	$n - k = \text{number of remaining payments}$	Works even when the original principal P is unknown; assumes regular payments.

5.4 Refinancing a Loan

When interest rates drop, it is not unusual for people with existing mortgages to consider refinancing. Refinancing entails the termination of an existing mortgage (perhaps with a penalty to pay) and the initiation of a new mortgage, which also involves various costs—for example, legal fees, an application fee, a title search, local taxes, and an appraisal fee. A typical goal for the borrower is to lower monthly payments and to reduce the amount of interest paid over the remainder of the mortgage from the time of refinancing.

For example, the Butterfields have a \$300,000, 30-year mortgage that is repaid monthly at 0.005 per month (a 6% nominal annual rate). Exactly five years into that mortgage, they can obtain a new 30-year loan at a 4% nominal annual rate, i.e., $\frac{.04}{12} \approx 0.003333 \dots$ per month (often written $0.00\overline{3}$ for one-third of one percent per month). Assume the total cost of terminating the existing mortgage and taking out the new one is \$6,000. Does it make sense for the Butterfields to refinance?

The balance on the initial loan is \$279,163.07. The periodic payment for the initial loan is \$1,798.65, and the periodic payment for the new loan is \$1,332.77. The remaining payments on the initial loan, if they do not refinance, equal \$539,595.47, of which \$260,432.40 is interest. The payments in the refinancing scenario, including the costs to terminate the initial loan and to start the new one, equal \$485,796.19. The interest paid in that scenario is \$200,633.12. It will take 13 periods to recover the cost of refinancing. Assuming the Butterfields do not sell the house before they recover the \$6,000 in refinancing costs, refinancing makes sense.

A natural follow-up question is: at what interest rate does refinancing still make sense? Table 18 lists total payments (for the new loan) and the payment per period (here, per month) for several interest rates, beginning with the rate on the existing mortgage. Total payments for the new loan stay below the remaining payments on the existing loan up to a nominal annual rate a little above 4.90% but below 4.96%.

The first row of Table 18 is the status quo: keep the existing loan and do not refinance. The “total payments” figure in that row is the amount still to be paid under the original loan terms.

Table 18 was generated by the “Loan Refinance Toolkit”, see Section 5.5.4 for a description of this program.

Table 18. Interest Rate Analysis for Refinancing

Interest Rate per Period	Nominal Annual Interest Rate	Total Payments	Payment per Period
0.005	6.00%	\$539,595.47	\$1,798.65
0.0033333	4.00%	\$485,796.19	\$1,332.77
0.0033833	4.06%	\$489,279.02	\$1,342.44
0.0034333	4.12%	\$492,774.81	\$1,352.15
0.0034833	4.18%	\$496,283.50	\$1,361.90
0.0035333	4.24%	\$499,805.05	\$1,371.68
0.0035833	4.30%	\$503,339.39	\$1,381.50
0.0036333	4.36%	\$506,886.47	\$1,391.35
0.0036833	4.42%	\$510,446.22	\$1,401.24
0.0037333	4.48%	\$514,018.59	\$1,411.16
0.0037833	4.54%	\$517,603.52	\$1,421.12
0.0038333	4.60%	\$521,200.95	\$1,431.11
0.0038833	4.66%	\$524,810.82	\$1,441.14
0.0039333	4.72%	\$528,433.07	\$1,451.20
0.0039833	4.78%	\$532,067.64	\$1,461.30
0.0040333	4.84%	\$535,714.46	\$1,471.43
0.0040833	4.90%	\$539,373.48	\$1,481.59
0.0041333	4.96%	\$543,044.63	\$1,491.79

5.5 Supporting Programs

5.5.1 Annuity Payout Calculator

As described in Section 4.10.2

5.5.2 Amortization Calculator

File name: amortizations_app.py

This program is an interactive amortization calculator with two groups of capabilities:

- **Core loan calculations (standard amortizing loan)**
 - **Compute the periodic payment** given:
 - loan amount P
 - **annual interest rate entered as a percentage** (e.g., enter 6.5 for 6.5%)
 - number of payments per year
 - length of the loan in years (can be non-integer)
 - **Compute the maximum affordable loan** given the same rate and term inputs plus a **maximum affordable payment per period**.

- **Generate a per-period amortization schedule** (payment, interest, principal, remaining balance).
 - Output is written to a CSV file named **breakdown_N.csv** in the current directory, where N is auto-incremented to avoid overwriting prior runs.
- **Additional principal payments (“pay extra each period”)**
 This section analyzes the impact of adding an extra principal amount C after an initial phase of regular payments.
 - **Compute payoff acceleration from an extra payment C :**
 Given m initial periods paid at the regular payment A , then paying $A + C$ thereafter, the program reports the **new payoff time** and **interest saved** (relative to the original schedule).
 - **Compute the extra payment needed to hit a target payoff time:**
 Given a target total number of payoff periods T (with $T > m$), the program computes the **required extra payment C** to retire the loan by period T , and reports the associated **interest savings**.
 - **Generate a full schedule with extra payments:**
 Writes a detailed schedule (including a “Phase” column identifying regular/extra/final-adjusted periods) to **breakdown_extra_N.csv**, again auto-numbered to avoid overwrites.

Notes:

- The program automatically adjusts the **final payment** in schedules so the balance reaches exactly zero (rather than going slightly negative due to rounding).
- In the extra-payment section, entering $m = 0$ means the extra payment begins immediately (starting with period 1).
- This is also an executable for Windows. The file name is AmortizationCalculator.exe. It is located in subdirectory dist under https://github.com/sfratini33/art-of-managing-things-external/tree/master/fin_math/code.

5.5.3 Balloon Loan Calculator

File name: balloon_loan.ipynb

This calculator supports balloon-loan computations by solving for whichever of these two quantities you leave as 0:

- **Compute the periodic payment amount** (regular payment) given:
 - loan amount P
 - **interest rate per period as a decimal** (e.g., 0.005 for 0.5% per period)
 - total number of payments n
 - balloon payment amount B
- **Compute the present value (loan amount P)** given:
 - periodic payment amount

- **interest rate per period as a decimal**
- total number of payments n
- balloon payment amount B

How to use:

- To compute the **periodic payment**, enter the loan amount and leave the periodic payment field at **0**, then click “**Compute unknown variable.**”
- To compute the **loan amount (present value)**, enter the periodic payment and leave the loan amount field at **0**, then click “**Compute unknown variable.**”
- The program validates that **exactly one** of (loan amount, periodic payment) is left at zero and displays an error message otherwise.

5.5.4 Loan Refinance Toolkit

File name: loan_refinance_toolkit.ipynb

There is also an executable for Windows, with filename LoanRefinanceToolkit.exe and in subdirectory dist under https://github.com/sfratini33/art-of-managing-things-external/tree/master/fin_math/code.

Note: This program **replaces** the earlier stand-alone programs loan_refinance.ipynb and loan_refinance_int_rate_analyzer.ipynb (the same material is now combined in one notebook).

The **Loan Refinance Toolkit** is a small graphical application (built with Tkinter) that you run from the notebook. It uses a tabbed window with two tools:

1. **Single Refinance** — analyzes one refinance scenario and shows whether refinancing reduces total cash outflows (including upfront costs) and how long it takes for payment savings to cover those costs, if the new payment is lower.
2. **Breakeven Rate Table** — sweeps refinanced interest rates and writes a table to a CSV file for comparing refinance totals to staying on the existing loan.

Assumptions

- The payment period for the original loan (for example, a month) is the same as the period for the new loan; payment and compounding coincide.
- All interest rates are **per-period decimals** (for example, 0.005 for 0.5% per period).
- The refinanced loan need not have the same term as the remaining periods on the original loan (for example, you may refinance a 30-year loan after 10 years into a 15-year loan).

Shared inputs (both tabs)

These describe the existing loan and the cost of switching:

- Amount of the initial (existing) loan
- Interest rate per period for the initial loan (decimal)
- Number of periods in the initial loan
- Number of periods already completed when refinancing begins

- Penalties on termination of the initial loan plus financing costs for the new loan

Single Refinance tab — additional inputs

- Interest rate per period for the refinanced loan
- Number of periods for the refinanced loan

Use **Analyze single refinance** to display balances, periodic payments, remaining payments if you do *not* refinance (with interest broken out), total payments in the refinance scenario (including termination and new-loan costs, with interest broken out), and either how many periods it takes to recover refinancing costs from lower payments or a message that the new payment is not lower than the old one.

Breakeven Rate Table tab — additional inputs

- Number of periods for the refinanced loan
- Starting per-period rate for the sweep (i_{low})
- Step size for the rate sweep (Δi)
- Payments per year (used only to show a **nominal annual** percentage alongside each per-period rate in the output)

The analyzer builds scenarios by increasing the refinanced rate until the refinance scenario's total payments (including costs) are at least as large as the remaining payments on the original loan if you do not refinance. It writes **breakeven_int.csv** in the current working directory (full path is shown in the window). A short preview appears in the application; a fuller table is suitable for a spreadsheet or for a table in the book (for example, Table 18).

Example (Single Refinance)

Enter the following values in the shared fields and in the single-refinance section (plain numbers, no commas or percent signs), matching the field labels in the window:

Concept	Value
Amount of initial loan	300000
Interest rate per period for initial loan	.005
Number of periods for initial loan	360
Periods completed when refinancing starts	60
Refinancing costs (termination + new loan)	6000
Interest rate per period for refinanced loan	0.0033333
Number of periods for refinanced loan	360

Output:

The balance at refinance point is \$279,163.07

The periodic payment for the existing loan is \$1,798.65

The periodic payment for the new loan is \$1,332.77

Remaining payments if not refinanced equal \$539,595.47, of which \$260,432.40 are interest.

Refinance total payments (including costs) equal \$485,796.19; interest portion (excluding costs) is \$200,633.12

It will take 13 periods to recover the refinancing cost based on payment savings.

5.6 Exercises

- Complete Table 13 and generate the entire repayment schedule.
Hint: Use the “Compute Interest – Principal breakdown” feature in the amortization calculator.
- Assume that a potential home buying couple has determined that they can afford a maximum of \$1,500 per month for a home mortgage. Further, they have found the best 15-year mortgage rate to be 3%, the best 20-year mortgage rate to be 3.5% and the best 30-year mortgage rate to be 3.75%. For all three scenarios, assume that payments on the mortgage are made monthly. What is the maximum mortgage (initial principal) that they can afford?
Hint: Solve the equation $A = \frac{iP(1+i)^n}{(1+i)^n - 1}$ for P and use the derived formula to determine the value for P in each of the three cases. Check your calculations using the “Max affordable loan given interest rate ...” feature in the amortization calculator.
Answer: In terms of being able to afford the most expensive house, the 30-year mortgage is the best option. It allows the couple to purchase a \$323,893.22 house.
- For a \$300,000, 30-year mortgage at a nominal annual interest rate of 4% with monthly payments back to the lender, determine the amount of the monthly payments. Adjust the monthly payment to the nearest multiple of \$10 and determine the final payment in this case.
Hint: Use the formula from Step 4 in Section 5.1.3.
Answer: From the amortization calculator, we determine the monthly payment to be \$1432.25. So, the adjusted monthly payment (to the nearest \$10) is \$1,430 and the final payment (based on the modified monthly payment) is \$2,988.76. If we just wanted to eliminate the cents (i.e., pay \$1432 per month), the final payment would be \$1602.66. If we adjust the monthly payment to the nearest multiple of \$100 (i.e., \$1,400), then the final payment is a surprising \$23,780.24.
- Continuing with the previous problem, how many payments would it take if the monthly payment is increased to \$1,500?
Hint: Use the Additional Principal Payments option in the Amortization Calculator with the additional principal payment set to \$67.75 and m to so 0 (since the initial principal payments start from the very beginning of the loan).
Answer: 330.13 payments or about 2.5 years early.
- For a car loan of \$30,000 at a nominal annual rate of 2% (compounded monthly) for 3 years, and a balloon payment of \$15,000 at the same time as the last monthly payment, how much is each monthly payment?
Hint: Use the balloon formula, i.e., $A = \frac{iP(1+i)^n - iB}{(1+i)^n - 1}$, and check your answer with the balloon loan calculator.
Answer: \$454.81.
- Continuing with the previous exercise, how does the answer change if the loanee makes the balloon payment one period after the last regular payment?
Hint: Use the alternate formula at the end of Section 5.2.1.
Answer: \$455.48 which is slightly more than the answer to Exercise 5, as expected since the balloon payment is made a month later.

7. Jedediah (goes by “Jed”) has a 30-year mortgage in the amount of \$5 million at a nominal annual interest rate of 3% (compounded monthly and with monthly payments back to the bank). How much is the balance on the mortgage at the end of 73 months?
Hint: Use either of the formulas for the remaining balance from Section 5.3. Check your answer with the Loan Refinance Toolkit (with $\frac{.03}{12} = .0025$ for the interest rate per period). Use the amount 0 in the penalties and costs field. The interest rate and number of periods for the refinanced loan are not relevant, but to avoid an error, use .0025 and 360.
8. Continuing with the previous exercise, Jed has the opportunity (73 months into the original mortgage) to refinance at a 1% nominal annual interest rate (compounding and payments are monthly). The new mortgage would be for 30 years. There is no cost to terminate the original mortgage (as written in the mortgage agreement if he uses the same bank to refinance, which he is doing) but the new mortgage costs \$55,000. Should Jed refinance, assuming he intends not to sell the house in the next ten years?
Hint: Use the Loan Refinance Toolkit.
9. Continuing with the previous exercise, what is roughly the highest interest rate (above 1% and less than 3%) at which it still makes sense for Jed to refinance in terms of having a lower total (remaining) payments?
Hint: Use the Loan Refinance Toolkit - Breakeven Rate Table tab.

6 Bonds

Don't look for the needle in the haystack. Just buy the haystack! — John C. Bogle

6.1 Terminology

A **bond** is a fixed income instrument that represents a loan made by an investor to a borrower (usually a corporation or government entity). When an entity needs to raise money for a long period of time (e.g., a city looking to build a sports stadium), they issue bonds (also known as debentures) which are sold to investors. Bonds are a form of debt.

A bond is essentially a contract between the issuer (lender or bond issuer) and the investor (borrower or bondholder) that specifies:

- The **face value**, or the denomination, of the bond, which is stated on the front of the bond. This is usually (but not always) the amount of money to be paid to the bondholder when the bond matures.
- The **date of maturity** (also known as the redemption date) is when the loan will be repaid. In some cases, it is sometime possible to redeem a bond after the maturity date but typically, no interest is accumulated after the maturity date. For example, United States Series EE savings bonds can be held beyond the maturity but no interest accrues after the maturity.
- The rate of interest that the bond issuer pays the bondholder is known as the **coupon rate**. It is critical to keep in mind that the coupon rate is on the face value of the bond and not on the purchase price of the bond, i.e., the coupon rate is applied to the face value, not the market price.
 - There is also something called a **zero-coupon bond** which does not have a coupon rate and does not pay regular interest to the holder but rather is sold at a discount and renders a profit at maturity when the bond is redeemed for its full face value, with no periodic interest payments.
- The price at which the bond issuer originally sells a bond is known as the **issue price**. Since bonds can be resold, there could be additional purchase prices. It is important to note that the coupon rate does not change when a bond is resold. After issuance, bonds trade in the **secondary market**, where price fluctuates.
- The amount of money that will be paid on the date of maturity is known as the **redemption value**. If the redemption value is the same as the face value, the bond is said to be **redeemed at par**.
- The **yield** of a bond is the rate of return the bond generates. This is not to be confused with a bond's coupon rate which is based on the face value of a bond.
 - An example may help distinguish the two concepts. If a \$100 bond with a 6% coupon rate sells for \$100, then the current yield is also 6%. However, because of market price fluctuation, it may be possible to purchase this bond for the price of \$80. So, if a \$100 (face value) bond is purchased for \$80, the current yield is 7.5% since the \$6 annual coupon payment is now 7.5% of the \$80 purchase price. If the \$100 (face value) bond was purchased for \$120, then the yield will be just 5%.

Note (there are three different “yields”): In bond discussions, people use the word *yield* in several ways, and it is easy to mix them up.

- **Coupon rate (stated rate):** the annual coupon as a percent of **face value** (or, equivalently, the periodic coupon divided by face value, annualized in the market’s convention). It does **not** depend on the purchase price.
- **Current yield:** the annual coupon cash flow as a percentage of the **purchase price** (often quoted as an annual figure). This ignores any capital gain or loss you realize at maturity when the bond redeems at its redemption value.
- **Yield to maturity (YTM):** the Internal Rate of Return (IRR) implied by the bond’s **price**, its coupon schedule, and the redemption payment at maturity—i.e., the discount rate y that makes the present value of all future cash flows equal the price. In this book, when we write “yield” in pricing formulas, we mean **YTM** unless we explicitly say “current yield.” YTM accounts for both coupon income and capital gain/loss, whereas current yield ignores price changes.

An example may help distinguish the concepts. Recall the example where a \$100 bond with a 6% coupon rate is purchased for \$80 with 2.5 years (5 semi-annual periods) remaining. In this case, the coupon rate is 6% and the current yield is $\frac{6}{80} = .075$ or 7.5%. To find the YTM, we solve for the semi-annual rate y that satisfies:

$$80 = \frac{3}{(1+y)^1} + \frac{3}{(1+y)^2} + \frac{3}{(1+y)^3} + \frac{3}{(1+y)^4} + \frac{3}{(1+y)^5} + \frac{100}{(1+y)^5}$$

where the \$3 semi-annual coupons and \$100 redemption value represent all future cash flows. Solving iteratively (or using a financial calculator or spreadsheet), we find $y \approx 5.0\%$ per semi-annual period, or approximately 10% annualized YTM.

A **callable bond** (also known as a redeemable bond) allows the issuer to redeem the bond before it reaches the stated maturity date. In other words, on the call date, the bond issuer has the option (not the obligation) to buy back the bond from the bond holder at a defined call price. The issuer of a callable bond may choose to call their bonds if market interest rates move lower, and then re-issue another series of bonds at a lower rate. Callable bonds compensate investors for the possibility of the bond being called by offering a higher interest rate than comparable non-callable bonds.

For callable bonds, yield-to-maturity may overstate an investor’s realistic return because the issuer may redeem early when it is advantageous (often when market yields decline). Practitioners also use yield-to-call and yield-to-worst to summarize callable-bond returns under early-redemption scenarios. Unless call features are explicitly modeled, this book’s yield discussion should be interpreted as the standard fixed-maturity cash-flow case.

Savings bonds (as offered by the United States Treasury) can typically be redeemed before or after maturity, subject to holding period rules and penalties. The investor (bond holder) will receive the face value of the bond (typically the purchase price) plus accrued interest.

Marketable bonds (e.g., corporate bonds or government bonds) cannot be redeemed early by the holder, but may be sold in the secondary market; some may be callable by the issuer. If the holder of a marketable bond needs to liquidate a bond before the maturity date, the holder must sell the

bond on the open bond market at a price that depends on current interest rates and other market factors (e.g., the financial stability of the bond issuer).

The above concepts are summarized in Table 19 and in Figure 9.

Table 19. Bond terminology summary

Term	Definition	Key Insight
Face Value (Par Value)	Amount paid to the bondholder at maturity	Basis for coupon payments
Coupon Rate	Interest rate applied to face value	Fixed; does NOT depend on price
Coupon Payment	Periodic interest payment	= (Coupon Rate × Face Value)
Maturity Date	Date the redemption amount is repaid	Bond ends (typically)
Redemption Value	Amount paid at maturity	Usually = face value
Issue Price	Price when bond is first sold	May differ from face value
Market Price	Price in secondary market	Fluctuates with interest rates
Current Yield	Coupon ÷ Market Price	Ignores capital gain/loss
Yield to Maturity (YTM)	IRR of bond cash flows	True total return measure
Zero-Coupon Bond	No coupons; sold at discount	Return comes at maturity
Callable Bond	Issuer can redeem early	Adds reinvestment risk
Marketable Bond	Can be traded	Liquidity feature
Savings Bond	Non-marketable government bond	May have special redemption rules

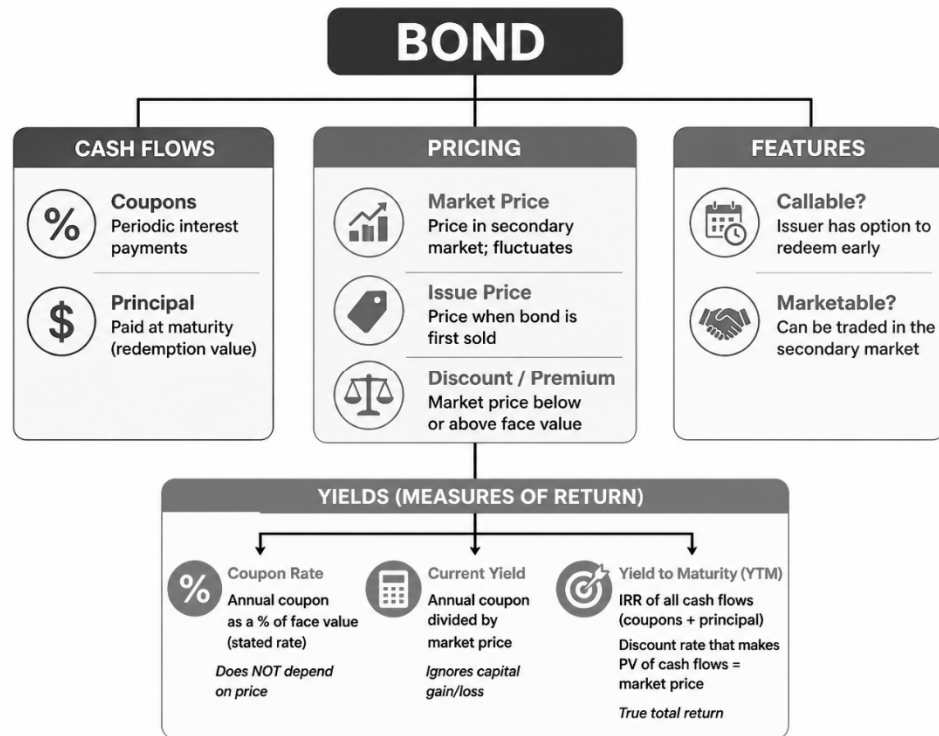


Figure 9. Summary of concepts related to bonds

6.2 Similarity between Bonds and Balloon Loans

Consider a bond that has the following characteristics:

- coupon rate j (per compounding period)
- face value F
- coupon payment A (computed by applying the coupon rate against the face value of the bond, i.e., $A = j \times F$)
- a final payout (redemption value) of B
- a purchase price P
- y is the discount rate per period

To determine the price of the bond, we need to determine the present value (at time of purchase) for each interest payout and the final payout.

This is the same problem as computing the principal for a balloon loan given the per period interest rate, payout per period and the balloon payout at the end of the loan. The only difference is that the entity providing the loan is the bond purchaser and the entity taking the loan is the bond issuer. Simply restating the formula for the principal of a balloon loan, we also have the formula for the price of a bond (with the conditions stated above):

$$P = A \frac{1 - (1 + y)^{-n}}{y} + B(1 + y)^{-n}$$

The first part of the right-hand side of the above equation is the present value (at the time of bond purchase) of the individual interest payments and the second term is the present value of the final bond payout.

Remarks:

- The bond purchaser is sold the bond under the guarantee that the yield will be y and the purchase price P is computed under this assumption. The coupon rate is only used to compute the coupon payment A .
- It is assumed that the yield rate applies on the same schedule as the coupon schedule (e.g., semi-annually or quarterly). If not, then the yield rate needs to be converted to an equivalent rate on the same schedule as the coupon rate using the conversion formula, i.e., $j = (1 + y)^{s/t} - 1$ where the yield rate is compounded s times per year and the coupon is distributed t times per year.
- As long as you know A, y, n and B , the above formula works whether time 0 is the initial bond issue date, or the purchase date (provided the purchase date is immediately after a coupon payment).

The payout timeline for the type of bond described above is shown Figure 10.

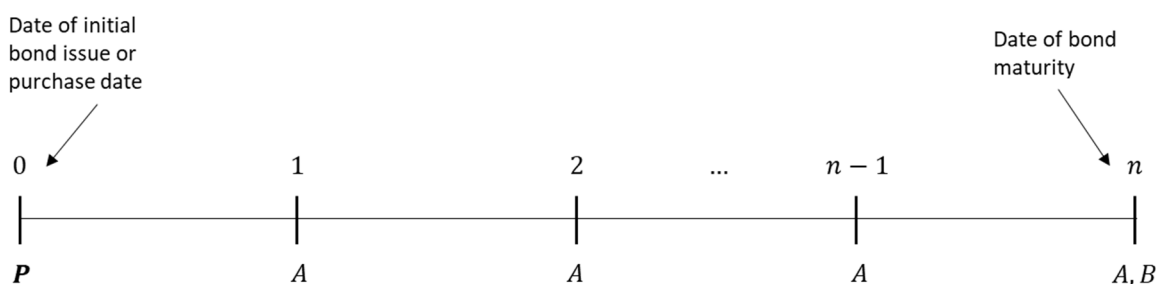


Figure 10. Payouts timeline for a bond

6.2.1 School Bond Example

Consider the scenario where a municipality wants to raise \$1,000,000 for a new school building. The head financial planner for the town wants to offer 5-year bonds at a semi-annual yield of 1.5%, with a coupon rate of 2% paid semi-annually and with a redemption value of \$5,000 (which is equal to the face value in this case). How many bonds does the municipality need to sell?

The coupon payment A is $(.02)(5,000) = 100$. We are given that $B = 5,000$, $y = 1.5\%$ and $n = 10$. Using the balloon loan calculator with these values yields the required issue price of the bond, i.e., \$5,230.55. Dividing the issue price of the bond into \$1,000,000, we get 191.18 and so the town needs to sell at least 192 bonds to fully cover the cost of their planned school building.

Pedagogical Context: In this example, the coupon rate (2%) differs from the market yield (1.5%), resulting in a premium bond. This could reflect realistic scenarios such as:

1. **Timing Delay** — The municipality issued bonds at a 2% coupon when that matched market demand. However, if there was a delay before actual sale and market yields subsequently

fell to 1.5%, the bonds now trade at a premium because their higher coupon is more attractive than current market rates.

2. Secondary Market Trading — After issuance, the coupon remains fixed at 2%, but if market conditions shift and investors are willing to accept lower yields (1.5%), existing bondholders can sell their bonds at a premium to new buyers. This example demonstrates how bond prices and yields adjust in secondary market trading based on changing market conditions.

6.2.2 Determine Purchase Price of Bond to Generate Various Yields

A purchase price other than the bond face value will lead to a different yield from the coupon rate, as we saw in the example in the previous section. As a second example, consider a bond with the following characteristics:

- Redemption value (equal to face value in this example) of the bond is \$1,000
- The coupon rate is compounded semi-annually at a nominal annual rate of 6%
- Length of the bond is 5 years.

Determine the purchase prices that will give yields of 4%, 6% and 8% (compounded semi-annually in each case). In all three cases, the semi-annual coupon payout is $A = \frac{.06}{2}(1,000) = 30$, $j = .03$ and $n = 10$.

In the case that a 6% yield is desired, $y = j = .03$ and the price should be set at the redemption value of \$1,000. As noted previously, when the redemption value and face value are equal, a bond is said to be purchased at par.

In the case a 4% yield, we have $y = .02$. It should be emphasized the A does not change. Using the balloon loan calculator, the required price to give the desired yield is \$1,089.83. Since the bond is being purchased for more than its redemption value, it is said to be purchased at a **premium**. Such a bond would be listed as $\frac{1,089.83}{1,000} \times 100 \cong \108.99 (it is typical to list a bond's price per \$100 of face value).

In the case of an 8% yield, we have $y = .04$. Using the balloon loan calculator, the required price to produce the desired yield is \$918.89. Since the bond is being purchased for less than its redemption value, it is said to be purchased at a **discount**. The price of this bond would typically be listed as \$91.89 (i.e., the price per \$100 of face value).

6.2.3 Bond Prices and Yields

Extending the previous example to include semi-annual yields between 1% and 5%, in .5% increments, we get the following table relating yields to purchase price:

Table 20. Bond Purchase Price versus Yield

Semi-annual Yield	Bond Price	Purchased
1%	\$ 1,189.43	at a premium
1.5%	\$ 1,138.33	at a premium
2%	\$ 1,089.83	at a premium
2.5%	\$ 1,043.76	at a premium
$i = y = 3\%$	\$ 1,000.00	at par
3.5%	\$ 958.42	at a discount
4%	\$ 918.89	at a discount
4.5%	\$ 881.31	at a discount
5%	\$ 845.57	at a discount

In general, as the yield increases, the bond price drops and vice versa. For example, assume Abe purchases a bond with a 6% coupon rate (compounded semi-annually), a face value (with equal redemption value) of \$1,000 and a term of 5 years. Abe decides to sell his bond after 2.5 years. Excluding brokerage fees, for how much should Abe expected to sell his bond?

First, note that Abe is basically selling a 2.5 year bond at this point with a redemption value of \$1,000. We need to compare this to other bonds with a 2.5 year term. Assume that such bonds now (at the time Abe wants to sell) have an average coupon rate of 7% (semi-annual compounding). Thus, the price estimate should be based on a bond with $y = .035$, $A = (.03)(1,000) = 30$ and $n = 5$. The key here is the A remains at \$30 (the bond issuer does not change their conditions when the bond gets sold). Using the balloon loan calculator, an estimate for price of the resold bond is \$977.42. This is different from the 3.5% row in Table 20, because the calculations in Table 20 are based on $n = 10$ periods.

6.3 Characterizing the Purchase of a Bond

There are several ways of characterizing whether a bond is purchased at a premium, at par or at a discount.

If we use the annuity notation mentioned in Section 4.3.1 (i.e., $a_{n|i} = \frac{1-(1+i)^{-n}}{i}$), the price formula for a bond (from Section 6.2) can be written as

$$P = Aa_{n|y} + B(1 - ya_{n|y})$$

and with some rearrangement, we get

$$P - B = (A - yB)a_{n|y}$$

If $P > B$ (i.e., purchase price exceeds redemption value), the bond is purchased at a premium. From the formula above, the amount of the premium is $P - B$. The above formula also provides

another way to characterize a premium bond purchase, i.e., when $A - yB > 0$. In other words, if the coupon exceeds the periodic yield (i.e., effective interest paid on the redemption value of the bond for a period), then the bond is purchased at a premium. If the redemption value equals the face value, then $A = jB$ (where j is the periodic coupon rate). Substituting $A = jB$ into $A - yB > 0$, we get $(j - y)B > 0$ which implies $j - y > 0$ (i.e., a bond is purchased at a premium if the coupon rate is greater than the yield).

Similarly, if $P < B$ (equivalently, $A - yB < 0$ or $j - y < 0$), then the bond is purchased at a discount.

Of course, if a bond is purchased at its redemption value (assuming this is the same as the face value) then $P = B$, $A = yB$ and $j = y$.

6.3.1 Schedule for a Bond Bought at a Premium

If a bond is purchased at a premium, the redemption amount (not the purchase amount) is returned to the bond holder on the redemption date and in effect, the bond holder will incur a loss. Depending on the applicable tax laws for the bond holder, there are at least two cases:

- This loss can be applied against the bond holder's income at the time of redemption, i.e., for a given tax year.
- Alternatively, the bond holder could allocate (i.e., distribute) the loss over each bond interest period (over several tax years with the exception of short term bonds that do not span more than one year).

Ms. Erdnussbutter purchased a \$100,000 bond (redemption and face value being the same) with a nominal annual coupon rate of 8% (with semi-annual coupon payments) and at a nominal annual yield of 6% (semi-annual compounding). The remaining duration of the bond is 10 years.

The coupon rate is $j = .08/2 = .04$ per period (i.e., half-year), the yield rate is $y = .06/2 = .03$ per period, and $n = 20$. Since the $j > y$, we can conclude the bond has been purchased at a premium. Using the bond schedule creator, we get the schedule shown in Table 21. The purchase price is shown as the book value in row 0, i.e., \$114,877.47.

Depending on her circumstances, Ms. Erdnussbutter can either declare a single loss of \$14,877.47 in the final year of the bond's term, or she could take a series of smaller losses in each of the years she holds the bond. The smaller losses are listed in the "Book Value Adjustment" column in Table 21. Note that the sum of the book value adjustments equals the total amount of loss, i.e., \$14,877.47. In either case, she needs to pay taxes on the coupon payments received each year.

For each period, the book value adjustment (loss in this case) is subtracted from the coupon payment and thus reduces the taxable income. For example, if the bond scheduling technique is used, the taxable amount on the bond from period 1 is \$3,446.32 rather than \$4,000.

The computations are as follows:

- (interest on book value for period k) = (periodic yield) **times** (book value from period $k - 1$)
- (book value adjustment for period k) = (coupon) **minus** (interest on book value for period k).

Table 21. Schedule for Bond Purchased at a Premium

Period (k)	Coupon	Interest on Book Value	Book Value Adjustment	Book Value
0	-	-	-	\$114,877.47
1	\$4,000.00	\$3,446.32	\$553.68	\$114,323.80
2	\$4,000.00	\$3,429.71	\$570.29	\$113,753.51
3	\$4,000.00	\$3,412.61	\$587.39	\$113,166.12
4	\$4,000.00	\$3,394.98	\$605.02	\$112,561.10
5	\$4,000.00	\$3,376.83	\$623.17	\$111,937.94
6	\$4,000.00	\$3,358.14	\$641.86	\$111,296.07
7	\$4,000.00	\$3,338.88	\$661.12	\$110,634.96
8	\$4,000.00	\$3,319.05	\$680.95	\$109,954.00
9	\$4,000.00	\$3,298.62	\$701.38	\$109,252.62
10	\$4,000.00	\$3,277.58	\$722.42	\$108,530.20
11	\$4,000.00	\$3,255.91	\$744.09	\$107,786.11
12	\$4,000.00	\$3,233.58	\$766.42	\$107,019.69
13	\$4,000.00	\$3,210.59	\$789.41	\$106,230.28
14	\$4,000.00	\$3,186.91	\$813.09	\$105,417.19
15	\$4,000.00	\$3,162.52	\$837.48	\$104,579.71
16	\$4,000.00	\$3,137.39	\$862.61	\$103,717.10
17	\$4,000.00	\$3,111.51	\$888.49	\$102,828.61
18	\$4,000.00	\$3,084.86	\$915.14	\$101,913.47
19	\$4,000.00	\$3,057.40	\$942.60	\$100,970.87
20	\$4,000.00	\$3,029.13	\$970.87	\$100,000.00
Totals	\$80,000.00	\$65,122.53	\$14,877.47	-

Remarks:

- Table 21 and Table 22 can be seen as effective-interest amortization schedules showing how the book value transitions from purchase price to redemption value. For each period:
 - interest on book value = yield rate $\times$ prior book value, and
 - book value adjustment = coupon – interest, which amortizes the premium or discount and ensures the return matches the yield y .
- There is the opposite issue when a bond is purchased at a discount. In this case, the bondholder may want to distribute the gains over the course of the bond's life. We provide an example of that case in the following subsection.

6.3.2 Schedule for a Bond Bought at a Discount

In this example, Mr. Everest buys a \$50,000 bond. The term of the bond is 5 years, the coupon rate is 4% (with payments semi-annually) and the yield is 5% (with semi-annual compounding).

The coupon rate is $j = .04/2 = .02$, the yield rate is $y = .05/2 = .025$, and $n = 10$. Since the $j < y$, we can conclude the bond has been purchased at a discount. Using the bond schedule creator, we get the schedule shown in Table 22. Note that the amounts in parentheses are meant to be negative (this is a convention in accounting).

Mr. Everest can choose to pay tax on the realized gain of \$2,188.02 in the final year of the bond's terms, or distribute the gains over the life of the loan (by the amounts shown in the book value adjustment column). In either case, he needs to pay taxes on the coupon payments received each year. The formulas for the calculations are the same as those used to construct Table 21.

[Author's remark: Showing the book value adjustments as negative in Table 22 seems odd to me since the amounts are being listed as profits on one's tax return. I have a similar comment on Table 21. I guess the idea is for the Interest on Book Value plus the Book Value Adjustment to equal the Coupon.]

Table 22. Schedule for Bond Purchased at a Discount

Period	Coupon	Interest on Book Value	Book Value Adjustment	Book Value
0	-	-	-	\$47,811.98
1	\$1,000.00	\$1,195.30	(\$195.30)	\$48,007.28
2	\$1,000.00	\$1,200.18	(\$200.18)	\$48,207.47
3	\$1,000.00	\$1,205.19	(\$205.19)	\$48,412.65
4	\$1,000.00	\$1,210.32	(\$210.32)	\$48,622.97
5	\$1,000.00	\$1,215.57	(\$215.57)	\$48,838.54
6	\$1,000.00	\$1,220.96	(\$220.96)	\$49,059.51
7	\$1,000.00	\$1,226.49	(\$226.49)	\$49,285.99
8	\$1,000.00	\$1,232.15	(\$232.15)	\$49,518.14
9	\$1,000.00	\$1,237.95	(\$237.95)	\$49,756.10
10	\$1,000.00	\$1,243.90	(\$243.90)	\$50,000.00
Totals	\$10,000.00	\$12,188.02	(\$2,188.02)	-

6.4 Correction if Bond Not Purchased on a Coupon Date

The discussion in the previous sections has been under the assumption that the bonds were purchased on the date of a coupon payout (with the payout going to the bond seller) or at the time of the initial issue of the bond. However, it is far more likely that a bond is sold between coupon payout dates. In such cases, a correction is needed to the calculations.

6.4.1 Dirty Price of a Bond

Figure 11 depicts a situation where a bond is purchased between two coupon payout dates (or possibly between the issue date of the bond and the first coupon payout date). The bond purchase price at time 0 is P . The coupon payouts are in the amount A . At maturity, the bond holder receives the final coupon payout plus the redemption value of the bond B . P is the price of the bond if it had been purchased at time 0 with the given yield.

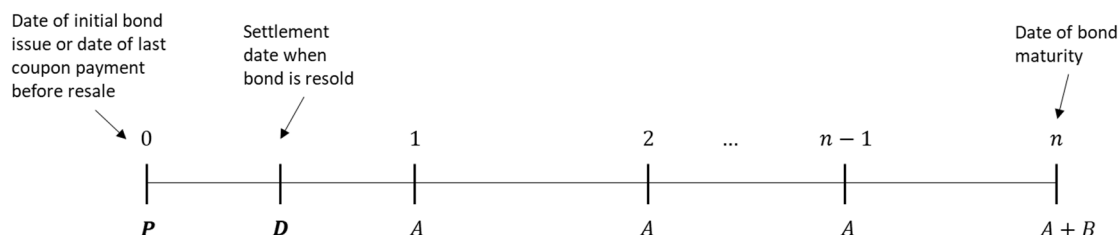


Figure 11. Bond Purchases between Coupon Dates

The price P of the bond at time 0 can be calculated with the bond price formula, i.e., $A \frac{1-(1+y)^{-n}}{y} + B(1+y)^{-n}$, since this time coincides with the initial bond issue date or a coupon payment date. Using the concepts from Section 3.7, the equivalent value of P at the time of settlement can be computed (this is shown as D in the figure and in the calculations that follow).

If simple interest (at rate y) is assumed over the fraction of the coupon period between time 0 and the settlement date (call this fraction k), then

$$D = P(1 + ky)$$

An alternate approach is to accumulate the bond value using compound interest over the fractional period, giving

$$D = P(1 + y)^k$$

These two approaches reflect slightly different viewpoints: one is the present bond value to term using the yield, while the other models the increase in value through accrued coupon interest over the fraction of the coupon period. The quantity D is known as the full or **dirty price** of the bond. In the examples and tables that follow, we will use the simple-interest version $D = P(1 + ky)$. For the examples considered here, the compound-interest formula $D = P(1 + y)^k$ produces nearly identical numerical results.

The term “dirty price” is used because it includes accrued interest, which pollutes the quoted bond price with a predictable, day-by-day buildup between coupon payments. Traders therefore quote the clean price (excluding accrued interest) so the quoted price reflects market moves rather than this mechanical accrual.

In bond markets, the fraction of a coupon period that has elapsed at settlement—what we denote by k —is not uniquely defined: it depends on a day-count convention chosen for that bond (common examples include 30/360, Actual/Actual, Actual/360, and others). In this book, unless stated otherwise, we compute k using a simple elapsed-days / days-in-coupon-period ratio as an intuitive approximation. If you use Excel bond functions (such as YIELD/PRICE) or institutional pricing systems, select the *basis* that matches the bond’s stated convention, because small differences in k can change quoted clean/dirty prices slightly.

6.4.2 Dirty Bond Price Example

As an example, consider a 30-year US Treasury bond with an annual yield of 1.61%, an annual coupon rate of 1.25% (interest paid semi-annually) and redemption value (same as face value in this case) of \$10,000. The bond was purchased by a brokerage house. At what price should the bond be sold to achieve the advertised yield if the settlement date is 5 June 2020 (i.e., when the bond is resold to a customer of the brokerage house) and the maturity day is 15 May 2050?

For this worked example, we assume semi-annual coupon dates on **May 15** and **November 15**, and we take k to be the fraction of the current coupon period elapsed from **May 15 to June 5**, computed as **21 actual days divided by 184 actual days** in the May–November coupon window (a standard Treasury-style day-count approximation for semi-annual Treasuries). So, $k = \frac{21}{184} \cong$

.11413. The other input variables are as follows:

- yield rate per period = $.0161/2 = .00805$
- total number of payments = 60 (note that no payments are missed as the broker resells the bond before the first coupon payout)
- redemption value = \$10,000
- coupon rate per period = $.0125/2 = .00625$
- coupon payout per period = $.00625 \times 10,000 = \$62.50$

Using the bond price formula, we get $P = \$9,146.12$ (this can be checked using the balloon loan calculator as described previously). Finally, we get $D = P(1 + ky) \cong \$9,154.52$. Using the alternate formula, we get $D = P(1 + y)^k = \$9,154.49$.

We can use the Microsoft Excel PRICE function to compute the clean and dirty price. The following expression in a cell gives the result \$91.47355667

=PRICE(DATE(2020,6,5),DATE(2050,5,15),0.0125,0.0161,100,2,1)

Interpretation: Excel uses cost/price per \$100 of face/redemption value, which we use in the following explanation. The value $P = \$9146.12$ (that we got from the bond price formula) is the value of the bond as of the previous coupon date, May 15. Accumulating this value forward to the settlement date (using the alternate formula) gives the dirty price of approximately \$9154.49. The market clean price is then found by subtracting accrued coupon interest (i.e., $\frac{21}{184} \times .6250 \cong 0.071331522$), giving approximately \$91.4736 which is the value returned by Excel's PRICE function.

(If the bond was sold later in the coupon period (e.g., 10 November 2020) then k would be $\frac{179}{184} \cong .972826$ and $D \cong \$9217.75$. Using the alternate formula, we get $D = P(1 + y)^k = \$9,217.74$.)

6.4.3 Clean Price of a Bond

Before we get to the definition of the clean price for a bond, let's further analyze how the dirty price changes over time. For the example in the Section 6.4.2, Table 23 shows the dirty price for the bond starting at time 0 and going through the end of period 4 (the other 55 periods are not shown in the table). Each period is divided into 8 equal time intervals. (There is nothing special about 8. The point is to get a few representative points for the graph in Figure 12.) For example, the dirty price of the bond at time $2\frac{115}{184}$ is \$9,214.59 (shown in bold in the table). The column on the extreme right (Diff) shows the difference between the dirty bond price at the end of one interval

and the beginning of the next (just before and after a coupon payout). For example, the difference between the price at the end of period 0 (just before the first coupon payout) and the beginning of period 1 (just after the first coupon payout) is \$9,219.75 - \$9,157.24 = \$62.51 (see the shaded cells table below). The difference should be exactly the coupon payout amount A but there is a small rounding error in the calculations.

The fact that the dirty price drops by exactly A at the end of a period can be proven as shown below. Note that k is 1 just before the end of a period and 0 at the beginning of a period. Further, we used the simple interest version of the dirty bond price, but you get the same result if you use the compound interest version.

Take the dirty price of a bond just before the end of a period with m remaining coupon payments and subtract the dirty price of the bond at the beginning of the next period (at which point there are $m - 1$ coupon payments remaining) to get the following (with terms in bold cancelling each other):

$$\begin{aligned}
 & (1 + 1 \cdot y) \left[A \frac{1 - (1 + y)^{-m}}{y} + B(1 + y)^{-m} \right] - (1 + 0 \cdot y) \left[A \frac{1 - (1 + y)^{-(m-1)}}{y} + B(1 + y)^{-(m-1)} \right] \\
 &= A \frac{(1 + y) - (1 + y)^{-(m-1)}}{y} + \mathbf{B(1 + y)^{-(m-1)}} - A \frac{1 - (1 + y)^{-(m-1)}}{y} - \mathbf{B(1 + y)^{-(m-1)}} \\
 &= \frac{Ay}{y} + A \frac{1 - (1 + y)^{-(m-1)}}{y} - A \frac{1 - (1 + y)^{-(m-1)}}{y} = A
 \end{aligned}$$

Table 23. Dirty Price Changes over Time for a Bond

Period	0	23/184	46/184	69/184	92/184	115/184	138/184	161/184	<1	Diff
0	9,146.12	9,155.32	9,164.53	9,173.73	9,182.93	9,192.14	9,201.34	9,210.54	9,219.75	
1	9,157.24	9,166.45	9,175.67	9,184.88	9,194.10	9,203.31	9,212.53	9,221.74	9,230.96	62.51
2	9,168.46	9,177.69	9,186.91	9,196.14	9,205.36	9,214.59	9,223.81	9,233.04	9,242.27	62.50
3	9,179.76	9,189.00	9,198.23	9,207.47	9,216.71	9,225.95	9,235.18	9,244.42	9,253.66	62.51
4	9,191.16	9,200.41	9,209.66	9,218.91	9,228.15	9,237.40	9,246.65	9,255.90	9,265.15	62.50

The points in Table 23 are plotted on the graph shown in Figure 12. Each upward line represents one period (one row from the table). The drop-off from one period to another is actually discontinuous, i.e., the downward lines should not be in the figure (limitation of the plotting tool used).

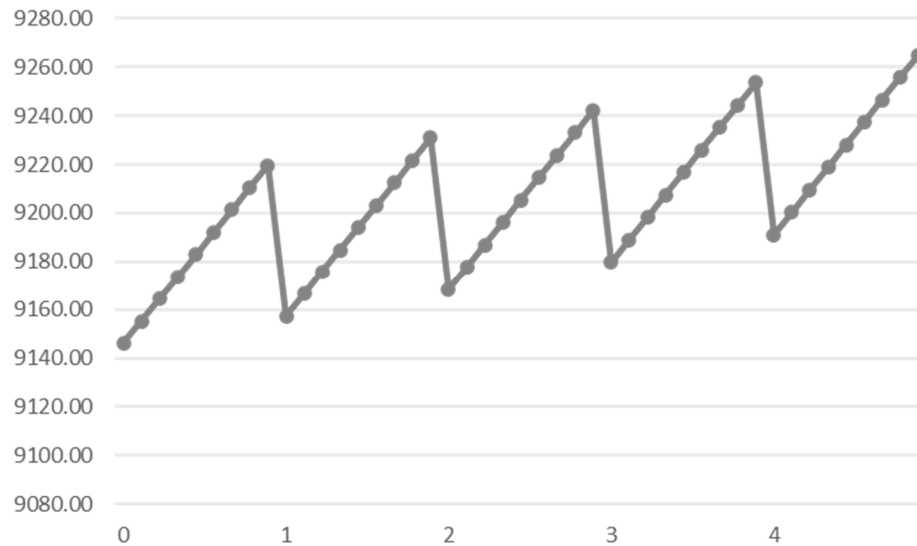


Figure 12. Graph of Dirty Bond Prices over Time

The reason for the increase in the price over a period is that the dirty price reflects the accrued interest on the coupon before it is paid out at the end of the period. Look at it this way, if the price did not reflect the accrued interest on the coupon, everyone would buy bonds just before the coupon dates and immediately get the coupon payment for the previous period.

The **dirty price** (or full price) is the total cash you pay at settlement: it equals the clean price plus accrued interest earned by the seller since the last coupon date. Economically, the dirty price is the value of the bond's remaining cash flows as of settlement, which lines up with paying clean price plus accrued interest. Equivalently, the **clean price** (also called the flat price) is the dirty price with that accrued interest stripped out, i.e., $\text{clean} = \text{dirty} - \text{accrued interest}$. In many markets (especially in the US), bonds are quoted at the clean price so the quoted figure does not jump on coupon dates; you add accrued interest to that quote to get the actual purchase amount (the dirty price). So bonds are typically advertised at clean prices and settled at dirty prices.

In terms of equations, we have the following definitions:

- Accrued interest: $I_k = kA$ where k is the fraction of a given coupon period that has transpired at settlement time (i.e., time of purchase).
- Clean price: $C = D - I_k$
- Equivalently: $D = C + kA$.

Table 24 shows the clean prices corresponding to the dirty prices in Table 23.

Table 24. Clean Price Changes over Time for a Bond

Period	0	23/184	46/184	69/184	92/184	115/184	138/184	161/184	<1
0	9,146.12	9,147.51	9,148.90	9,150.29	9,151.68	9,153.07	9,154.46	9,155.86	9,157.25
1	9,157.24	9,158.64	9,160.04	9,161.45	9,162.85	9,164.25	9,165.65	9,167.05	9,168.46
2	9,168.46	9,169.87	9,171.29	9,172.70	9,174.11	9,175.53	9,176.94	9,178.35	9,179.77
3	9,179.76	9,181.18	9,182.61	9,184.03	9,185.46	9,186.88	9,188.31	9,189.73	9,191.16
4	9,191.16	9,192.60	9,194.03	9,195.47	9,196.90	9,198.34	9,199.78	9,201.21	9,202.65

The points in Table 24 are plotted on the graph shown in Figure 13. This would be a straight line but for some rounding errors in the computations.

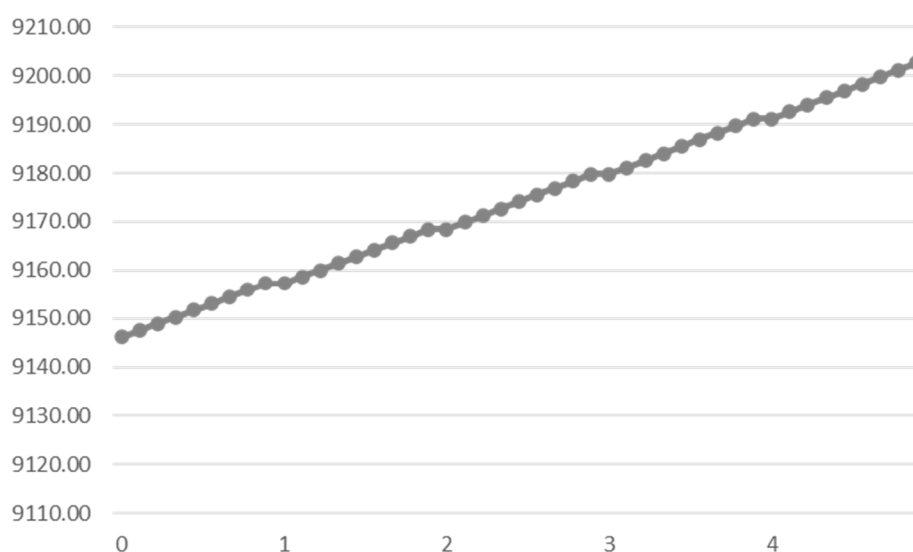


Figure 13. Graph of Clean Bond Prices over Time

6.5 Computing the Yield given the Bond Price

So far, all the work in this section assumes that the yield rate is given (along with the coupon rate, redemption value, and term of the bond) and the bond price is derived. In fact, it is possible to determine the yield when given the bond price, coupon rate and term of the bond, but there is no closed formula for the yield in terms of the other characteristics of a bond. A numerical approximation method is required to solve the problem.

Fortunately, Microsoft Excel offers a yield function that computes the yield for a bond given:

- Settlement – the date when the security is sold to the buyer (this could be the issue date of the bond or a later date). Need to use the Excel DATE function to enter the date.
- Maturity – the date when the security expires. Again, one needs to use the Excel DATE function to enter the date.
- Coupon rate per year

- Purchase price per \$100 of face value.
- Redemption value per \$100 face value.
- Frequency – the number of coupon payments per year. For annual payments, frequency = 1; for semiannual, frequency = 2; for quarterly, frequency = 4. No other options are supported.
- Basis (optional) - the type of day count basis to use.

Consider the example from Section 6.3.2 concerning Mr. Everest but this time assume we are given the purchase price \$47,811.98 and asked to find the yield. In order to use the Excel function, we need actual dates that span (in this case) 5 years. We can arbitrarily choose January 1, 2000 as the settlement date and January 1, 2005 as the maturity date. The nominal annual coupon rate is 4%. The purchase price per \$100 of face value is $47,811.98 \times \frac{100}{50,000} = 95.62396$. The redemption value is equal to the \$100 per \$100 of face value. The frequency is 2.

If the following is entered in an Excel cell, the result is 0.050000019 (not exact but pretty good):

= YIELD(DATE(2000,1,1), DATE(2005,1,1),0.04,95.62396,100,2)

Note: Given a bond price, solving for yield y is generally a nonlinear problem: there is typically no closed-form algebraic formula for y . Functions like Excel YIELD compute y by a numerical root-finding procedure (internally solving for the discount rate that equates the present value of coupons plus redemption to the quoted price). Implementations in Python follow the same idea (bisection/Newton-style solvers), not algebraic inversion.

6.6 Supporting Programs

This section describes two companion programs used throughout Section 6. Both programs ask for interest rates **as decimals per period** (for example, enter 0.05 for **5% per period**), not as whole-number percents (do **not** enter 5 intending 5%).

6.6.1 Balloon Loan Calculator

The balloon loan calculator is described in Section 5.5.3. For bond pricing in Section 6, it is convenient to view a bond by analogy to a balloon loan: the bond's coupons and redemption cash flows are discounted at the purchase yield in the same way that loan payments and a balloon are discounted in the balloon-loan setup.

Table 25. Balloon Loan – Bond Comparison

Balloon Loan Concept	Bond Concept
Amount of loan P	Purchase price of the bond P
Interest rate per period i	Yield rate per period y
Number of payments n	Number of coupon periods n
(not used in the same way)	Coupon rate per period j
Amount of each periodic payment A — computed using the balloon-loan payment formula	Coupon cash flow per period — computed using $A = jB$ (coupon rate $\times$ redemption/face)
Balloon payment B	Redemption value B (this chapter typically assumes redemption equals face value unless stated otherwise)

How to enter bond inputs into the balloon loan calculator

- Interest rate per period = bond yield per period y
- Number of payments = bond number of coupon periods n
- Amount of each periodic payment = coupon cash flow per period $A = jB$
- Balloon payment = redemption value B

Note on interpretation: A typical amortizing balloon loan's scheduled payments often include interest and principal over time (depending on the loan structure), whereas the standard bond setup in this chapter pays coupon cash flows through time and pays redemption at the end. The mapping above is for pricing/discounting the cash-flow stream, not a claim that bonds and amortizing loans are identical in accounting treatment.

6.6.2 Bond Schedule Creator

Program files:

- `bond_schedule_creator.py` (standalone program)
- `bond_schedule_creator.ipynb` (Jupyter launcher notebook that runs the same program)

The program builds an effective-interest style bond schedule like Table 21 and Table 22: for each period it shows the coupon cash flow (constant while the bond is outstanding), interest on book value ($\text{yield} \times \text{prior book value}$), book value adjustment, and book value. Negative adjustments are displayed in parentheses (discount schedules).

Inputs (all per period, as decimals):

- redemption value (face value)
- number of payment periods
- coupon rate per period
- yield rate per period

Output: writes `bond_schedule.csv`. When running `bond_schedule_creator.py`, the CSV is written beside that script. When running from Jupyter, the CSV is typically written to the notebook's working directory (usually the folder containing the notebook).

The `bond_schedule.csv` will overwrite prior output unless the reader renames/moves it.

6.7 Exercises

1. Compute the price of a 10-year zero coupon bond that effectively yields 3.5% per year (under the assumption of yearly compounding). The redemption value of the bond is \$15,000.

Answer: Use the formula for the bond price from Section 6.2 with $A = 0$ (since there is no coupon) and $B = 15,000$ to get the result $P = 15,000(1 + .035)^{-10} = \$10,633.78$.

Note: If no compounding is assumed, then we need to compute the present value of 15000 under the assumption of simple interest, i.e., $P = \frac{15,000}{1+10(.035)} = \$11,111.11$.

2. Compute the price of a 5-year bond with semi-annual coupon payments at a nominal annual rate of 5%. The redemption and face value are the same, i.e., \$50,000. The nominal annual yield rate is 4% (compounded semi-annually).

Hints: Use the balloon loan calculator in the manner described in Sections 6.2 and 6.6.1. The coupon is $\frac{.05}{2}(50,000) = 1,250$.

Answer: \$52,245.65

3. Is the bond from Exercise 2 purchased at a premium, at a discount or at par?

Answer: The coupon rate is higher than the yield. So, the bond is purchased at a premium.

4. Using the information from Exercise 2, create a table that shows the bond price for the following nominal annual yields (again assuming semi-annual compounding): 2%, 3%, 4%, 5%, 6%, 7% and 8%.

Hints: This is similar to the example in Section 6.2.3. Don't forget to convert the nominal annual yields to semi-annual yields. Also, keep in mind that the coupon is the same for all cases in this exercise.

Answer:

Table 26. Answer to Exercise 4

Semi-annual yield	Bond price	Purchased
1	\$57,103.48	at a premium
1.5	\$54,611.09	at a premium
2	\$52,245.65	at a premium
2.5	\$50,000.00	at par
3	\$47,867.45	at a discount
3.5	\$45,841.70	at a discount
4	\$43,916.83	at a discount

5. For the bond described in Exercise 2, create bond schedules for two cases, i.e., when the yield is 3% and 7% (semi-annual compounding in both cases).

Hint: Use the bond schedule creator.

6. Compute the dirty (full) price for a 20-year US Treasury bond that is to have an annual yield of 1% (compounded semi-annually), an annual coupon rate of 1.25% (interest paid semi-annually) and redemption value (same as face value in this case) of \$25,000. The settlement (purchase) date is 15 June 2020 and the redemption (maturity) date is 31 August 2030.

Hints and answer: Clearly, the bond is being resold since the settlement date is less than 20 years prior to the redemption date. *The coupon dates are August 31st and March 3rd (note the issue with February having only 28 days) and so there are 21 coupon payments remaining, i.e., $n = 21$. Use the balloon calculator to compute the price of the bond at the time of last coupon payment before the settlement date which is \$25,621.50 (this is P in the formula for the dirty price from Section 6.4.1). Use the online application at <https://www.timeanddate.com/date/duration.html> to count the number of days from 3 March 2020 to 15 June 2020 (105 days), and between 3 March 2020 to 31 August 2020 (182 day) which means that $k = \frac{105}{182} \cong .5769$. Applying the dirty price formula, we get $D = P(1 + ky) \cong 25,621.50(1 + (.5769)(.005)) \cong \$25,695.41$. If we use the alternate formula for the dirty price, we get $D = P(1 + y)^k \cong \$25,695.33$.*

7. Using the simple interest version of the dirty price formula, show that the clean price can be represented by the formula $C = P + k(yP - iB)$ with P, k, y, i and B as defined in Section 6.4.1. This also explains why the clean price can be slightly below P : here $yP < iB$, so the term $k(yP - iB)$ is negative.

Hint: Assume that the face value is redemption value and thus $A = iB$.

8. For the bond described in Exercise 6, compute the clean price.

Answer: $C = D - kA \cong 25,695.41 - .5769(156.25) \cong \$25,605.27$.

7 Capital Budgeting

The index fund is a most unlikely hero for the typical investor. — John C. Bogle

This section covers some financial concepts that help in decision making with regard to business investments, and the incremental cash flows that results from such investments. The reader is assumed to be familiar with cash flow and how it is different from profit. For a refresher on cash flow and profit, see the YouTube video “What is a cash flow statement?” [8] and the Investopedia article on cash flow [9]. For a more extensive discussion of cash flow analysis and capital budgeting, see Chapters 12 and 13 of The Basics of Finance [6].

[Author’s remark: While I have not provided any Python programs in support of this section, there is an Excel spreadsheet that covers many of the computations related to the examples. It is recommended that you have this spreadsheet in hand when reading this section. For a summary of the various sheets within the spreadsheet, see Section 7.7.]

7.1 Net Present Value (NPV)

7.1.1 Concept with an Example

Net Present Value (NPV) refers to the present value of all cash flows related to a given project and associated investments for a given entity. NPV is used as a method for making decisions concerning capital budgeting. If the NPV for a project is negative, then it is recommended not to proceed with the project. If the NPV for a project is positive, then it is recommended to proceed with the project. The NPV can also be used to compare several projects (see the example in Section 7.1.6). As we shall see, NPV is but one measure used to decide whether to go ahead with a project or not.

Calculation of the NPV depends on the stipulation of a given rate of return (known as the cost of capital). The general idea is that a proposed capital investment project should provide a higher expected return than an alternative investment of similar risk. For example, if a company can make a 5% rate of return with, in their opinion, a moderate risk investment in the bond market, then an internal project, with similar risk, would be expected to make more than 5% per year.

Wikipedia provides the following definition of **Cost of Capital** [7]:

In economics and accounting, the cost of capital is the cost of a company's funds (both debt and equity), or, from an investor's point of view “the required rate of return on a portfolio *[of a]* company's existing securities.” It is used to evaluate new projects of a company. It is the minimum return that investors expect for providing capital to the company, thus setting a benchmark that a new project has to meet.

So, the cost of capital is the minimum return a company must earn on an investment to justify its risk and create value. It is used to evaluate projects, compare investment opportunities, and assess financial decisions. Investors and analysts also use it to estimate stock values and expected returns. The cost of capital and NPV are closely related, but they are not the same:

- The cost of capital is the required rate of return (or discount rate) used to evaluate an investment.
- NPV (Net Present Value) is the value created after discounting future cash flows using the cost of capital. NPV measures how much value an investment adds after accounting for the cost of capital.

Interpretation of NPV:

- $NPV > 0$: the project earns more than the required return and creates value.
- $NPV = 0$: the project exactly earns the required return.
- $NPV < 0$: the project earns less than the required return.

In practical terms, a positive NPV means the project is expected to outperform an alternative investment of similar risk.

As an example, consider a small business (Acme Furniture) that makes custom furniture and does antique furniture restoration. The business is currently run by Alice and Bob in a converted garage. On a regular basis, Alice and Bob are turning down business because of the lack of resources. To address this issue, they are considering an expansion of the garage, the purchase of additional equipment, the hiring of additional staff, additional parking spaces and a small showroom for customers. The upgrade to their business will require an investment of \$225,000. On the other hand, Alice and Bob could invest the \$225,000 in three other ways, with the following semi-annual rate of returns

- 2% (conservative investment, e.g., bank certificates of deposit)
- 4% (moderate risk, e.g., government and corporate bonds)
- 6% (higher risk, e.g., stocks and higher risk bonds).

Alice and Bob know their business well and are able to make (in their opinion) reasonably accurate projections of cash flows for the business expansion (as a result of the planned expansion). The problem at hand is to compare the projected cash flows against the three alternative investment options.

Table 27 shows Alice and Bob's projected cash flows concerning Acme Furniture after the planned expansion.

- The top row is time in half-year increments, up to the end of year number 5.
- The second row contains the cash inflows to Acme Furniture, e.g., sales of furniture.
- The third row contains the cash outflow, e.g., the initial investment, supplies, wages.
- The bottom row shows net cash flows for each 6-month period. In this example, there is only a net outflow at time 0. However, in general, any of the periods could be a net inflow or outflow of cash.

(Note: In the examples that follow, we usually only show the net cash flow for each period and omit the details concerning cash inflows and outflows.)

Table 27. Initial Investment and Cash Flow in Half-year Increments

Year	0	.5	1	1.5	2	2.5
Cash Inflow		30,000	35,000	35,000	40,000	35,000
Cash Outflow	(225,000)	(10,000)	(13,000)	(11,000)	(14,000)	(7,000)
Net Cash Flow	(225,000)	20,000	22,000	24,000	26,000	28,000

Year	3	3.5	4	4.5	5
Cash Inflow	40,000	45,000	50,000	50,000	50,000
Cash Outflow	(10,000)	(13,000)	(16,000)	(14,000)	(12,000)
Net Cash Flow	30,000	32,000	34,000	36,000	38,000

Recall from Section 3.7 that we can compare the value of money realized at different times by computing the various values at some common focal date. In other words, NPV converts all projected future cash flows into equivalent dollars at the focal date (typically the present time). This allows cash flows occurring at different times to be compared on a consistent basis. For the problem at hand, we will use the time of the initial investment (i.e., time 0) as the focal date and compute the value of the various net cash flows for each period based on their value at the focal date. In general, we make the following assignment

- let C_0 be the initial investment
- let C_k be the net cash flow for period k which ends at time t_k (measured in time periods, e.g., half-years, quarters or months)
- i is the interest rate per period, i.e., the cost of capital
- n is the number of cash flows.

With the above assignments, the general formula for net present value is given by

$$NPV = -C_0 + \sum_{k=1}^n C_k(1+i)^{-t_k}$$

The formula allows for the cash flows to be unequally spaced in terms of the number of periods. For example, we could have cash flow estimates at periods number 3, 5, 11 and 13. In this case, $n = 4$, $t_1 = 3$, $t_2 = 5$, $t_3 = 11$ and $t_4 = 13$. However, it is more usual to have a net cash flow estimate for each period, in which case, $t_k = k$ and the formula becomes

$$NPV = -C_0 + \sum_{k=1}^n C_k(1+i)^{-k}$$

Getting back to the problem at hand, we can use the Microsoft Excel function NPV to compute the NPV for each of the 3 cost of capital scenarios, i.e., 2%, 4% and 6% semi-annual rates.

(The Excel NPV function is easy to use. The first parameter is the cost of capital (discount rate), and the remaining parameters are the projected net cash flows for future periods. One must be careful, however, since Excel assumes that the listed cash flows begin one period after the initial investment. Consequently, the initial investment at time 0 is usually added separately to the result of the NPV function.)

The results are summarized in Table 28 (the parenthesis indicate a negative NPV). For 2% and 4%, Acme Furniture will have a positive NPV, and thus, go forward with the expansion in these two cases. However, at 5% and 6%, the NPV is negative and it may not be a good idea to go forward with the expansion in this case.

Table 28. Investment Analysis for Acme Furniture

Semi-annual Cost of Capital	NPV
2%	\$32,561.90
4%	\$4,980.62
5%	(\$7,261.21)
6%	(\$18,593.62)

Notice that the NPV decreases as the cost of capital increases. This occurs because larger discount rates reduce the present value of future cash flows more aggressively.

For the sake of argument, let's say that Alice and Bob think that the risk in expanding their business is similar in risk to the 6% cost of capital alternative. **[This is a key point, i.e., the cost of capital needs to be at the same estimated risk as the proposed investment!]** With the given information, they should not go forward with their business expansion. However, it may be that consideration of a longer time frame for the cash flows will change the decision. For example, assume Alice and Bob are able to project the cash flow for a 6th year, as shown in Table 29.

Table 29. Initial Investment, and Cash Flow in Half-year Increments - Revised

0	.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
(225,000)	20,000	22,000	24,000	26,000	28,000	30,000	32,000	34,000	36,000	38,000	41,000	44,000

The two extra semi-annual cash flow projections change the investment analysis, as shown in Table 30. With the extra year of cash flow taken into consideration, it now makes sense to go forward with the business expansion, even for the case of a 7% return on the alternative investment. It is not until about 8% where the alternative investment makes more sense. **[This is a second key point, i.e., the time frame used in the cash flow analysis is also critical. However, the farther one projects into the future, the more uncertain the estimates become.]**

Table 30. Investment Analysis for Acme Furniture - Revised

Semi-annual Cost of Capital	NPV
2%	\$100,230.39
4%	\$59,095.71
6%	\$24,871.32
7%	\$9,918.07
8%	(\$3,787.51)

7.1.2 Negative Cash Flows

The previous example showed positive cash flows for each period but this is not always the case, and it is quite possible to have negative net cash flows for some periods. For example, the capital expenditure could be negative in some periods because of start-up issues, seasonal businesses, or expected charges (e.g., insurance payments) at various points in time.

As an example, consider a vacation resort that has a seasonal business, with Fall being a time for repairs and a reduced number of customers. The resort owners are considering a \$500,000 investment to improve the resort. Table 31 shows the estimated project cash flows for each season (quarter) in the 5 years following the potential investment.

Table 31. Project Quarterly Cash Flows for a Resort Business

Year	Winter	Spring	Summer	Fall
1	40,000	41,000	42,000	(3000)
2	41,000	42,000	43,000	(2000)
3	41,000	42,000	43,000	(2000)
4	43,000	44,000	45,000	(1500)
5	43,000	44,000	45,000	(1500)

Table 32 shows an analysis of the potential investment at various costs of capital (keeping in mind that the cost of capital refers to alternative investments). The project's cash flows outweigh the alternative investments up to and including a quarterly rate of 2.25%. At 2.5%, it **may** make more sense not to invest in the resort but rather in the alternative investment. On the other hand, the 2.5% alternative investment may be very risky (perhaps junk bonds and small growth stocks) and the resort improvement investment may be seen as much more certain than the 2.5% alternative investment. The point is that the final decision by the resort owners is not black and white, and the NPV analysis is just one tool to help with their decision.

Table 32. Investment Analysis for Resort Investment

Quarterly Cost of Capital	NPV
.5%	\$98,166
.75%	\$83,553
1%	\$69,445
1.25%	\$55,823
1.5%	\$42,668
1.75%	\$29,959
2%	\$17,680
2.25%	\$5,813
2.5%	\$(5,658)
2.75%	\$(16,748)

7.1.3 Multiple Investments

The examples thus far have only had one investment and that occurred at the beginning of the project in question. This does not always need to be the case and in fact, it is quite possible that a project will require several investments at various times. In the case of several investments, one can separate the cash flow into what are called operating cash flow and investment cash flow (these are not to be confused with cash inflow and outflow). These two types of cash flow are added together for each period to get a net cash flow. In general, investment cash flows are negative. Operating cash flows can be positive or negative (as we saw in example from the previous section).

Table 33 shows a project that requires an initial investment of \$70,000 (e.g., purchase of some equipment) and another investment of \$10,000 in year 4 of the project (e.g., maintenance on the equipment or perhaps purchase of another piece of equipment).

Table 33. Operating and Investment Cash Flows

Time in Years	0	1	2	3	4	5	6	7
Operating Cash Flow		\$ 7,500	\$ 10,000	\$ 12,500	\$ 15,000	\$ 15,000	\$ 17,500	\$ 20,000
Investment Cash Flow	\$(70,000)				\$(10,000)			
Net Cash Flow	\$(70,000)	\$ 7,500	\$ 10,000	\$ 12,500	\$ 5,000	\$ 15,000	\$ 17,500	\$ 20,000

The operating and investment cash flows are added to get a net cash flow for each period, and we proceed as before to compute the NPV for various rates of return, as shown in Table 34. The investments in the project make financial sense up to a little over a 5% rate for the cost of capital.

Table 34. NPV Analysis in the Case of Multiple Investments

Yearly Cost of Capital	NPV
0.02	\$9,900
0.03	\$6,446
0.04	\$3,201
0.05	\$150
0.06	\$(2,722)
0.07	\$(5,427)

7.1.4 Range of Cash Flow Estimates

In the previous examples, it was assumed that the decision-making parties (regarding capital expenditure) had a good idea of potential cash flows as a result of their investment. Another approach (particularly when the decision makers are not so sure of the potential cash flows) is to stipulate a range of values. For example, consider the case of Mr. Throckmorton who is starting a new restaurant. Unfortunately, Mr. Throckmorton has only a rough idea of cash flow based on market analysis and consequently, he can only stipulate a range of projected cash flows for each period. It is clear, however, that the restaurant will require an initial investment of \$625,000. Table 35 shows the yearly (projected) cash flows for the restaurant under pessimistic and optimistic scenarios.

Table 35. Optimistic-Pessimistic Cash Flows for New Restaurant

Year	0	1	2	3	4	5	6	7	8	9	10
Pessimistic	(625,000)	50,000	55,000	60,000	65,000	70,000	75,000	80,000	85,000	90,000	95,000
Optimistic	(625,000)	60,000	66,000	72,000	78,000	84,000	90,000	96,000	102,000	108,000	114,000

Table 36 lists the NPV results for the optimistic and pessimistic cases. For the pessimistic case, the NPV is only positive up to 2.5%. The optimistic case has a positive cash flow up to 5.5%. Mr. Throckmorton could, for example, make his decision based on a compromise between the optimistic and pessimistic scenarios, and use 4% as the cut-off point, i.e., if he can get more than 4% with an alternative investment of similar risk to the restaurant business then he should **not** go ahead with the restaurant business.

Table 36. NPV for Optimistic and Pessimistic Scenarios

Cost of Capital per Year	Pessimistic	Optimistic
0.02	\$18,904.75	\$147,685.70
0.025	\$619.18	\$125,743.02
0.03	(\$16,945.91)	\$104,664.91
0.035	(\$33,824.42)	\$84,410.69
0.04	(\$50,048.45)	\$64,941.86
0.045	(\$65,648.38)	\$46,221.94
0.05	(\$80,653.01)	\$28,216.38
0.055	(\$95,089.61)	\$10,892.46
0.06	(\$108,984.04)	(\$5,780.85)
0.065	(\$122,360.81)	(\$21,832.97)

This example illustrates an important practical issue with NPV analysis: the results are only as reliable as the projected cash flows and estimated cost of capital. In many real-world situations, these values can only be approximated.

7.1.5 Initial Investment is Loaned

All the previous examples in this section had the tacit assumption that the parties involved had the money required for the initial investment. It is likely, however, that some or all of the money may need to be borrowed. This clearly affects the NPV analysis since the loan repayments need to be subtracted from the cash flow estimates.

Consider the Acme Furniture example in the case where Alice and Bob need to take out a 5-year small business loan in the amount of \$50,000 at a nominal interest rate of 3% (repaid semi-annually). Under this scenario, Alice and Bob would need to make payments of \$5,421.71 twice per year for five years. These payments need to be subtracted from the cash flow estimates as shown in Table 37.

Notes:

- In some locales, interest payments on debt are tax deductible. This would reduce the impact on cash flow, but we didn't consider this issue in our analysis of the project.
- While investment is entirely within the cash outflow category, operating cash flow entails both cash inflow and outflow.

Table 37. Acme Furniture - Cash Flow Estimates with Loan Repayments

Time in years	0	.5	1	1.5	2	2.5	3	3.5
Operating Cash Flow		20,000	22,000	24,000	26,000	28,000	30,000	32,000
Investment Cash Flow	(225,000)	(5,421.71)	(5,421.71)	(5,421.71)	(5,421.71)	(5,421.71)	(5,421.71)	(5,421.71)
Net Cash Flow	(225,000)	14,578	16,578	18,578	20,578	22,578	24,578	26,578

3.5	4	4.5	5	5.5	6
32,000	34,000	36,000	38,000	41,000	44,000
(5,421.71)	(5,421.71)	(5,421.71)	(5,421.71)	0	0
26,578	28,578	30,578	32,578	41,000	44,000

Table 38 shows the NPVs for various rates of return for alternative investments. In this scenario, there is positive NPV up to and including 4%. Recall from the associated scenario without the loan that it still made business sense to go ahead with the improvements to Acme Furniture up to something over 7% but less than 8%. This is to be expected since the loan decreases net cash inflow for each period.

Table 38. Investment Analysis for Acme Furniture with Loan Repayments

Semi-annual Cost of Capital	NPV
2%	\$51,529.41
3%	\$32,453.34
4%	\$15,120.78
5%	(\$653.52)
6%	(\$15,032.93)

In the above example, the loan is fully repaid by the end of year 5, while the cash flow projection continues through year 6. This means the analysis includes one additional year of projected cash flows after the loan payments have ended. As expected, borrowing reduces the NPV because the loan repayments lower the net cash flow during the repayment period. However, once the loan is paid off, the projected cash flows are no longer reduced by loan payments. Alice and Bob may wish to extend their cash flow analysis even further beyond the end of the loan term in order to better assess the long-term value of the business expansion.

7.1.6 Comparison of Projects

It is possible to use NPV analysis to decide where to make an investment among several projects of similar risk. In this case, the same cost of capital can be used to evaluate each project. If the projects have substantially different risk levels, then different costs of capital may be more appropriate for each project. Table 39 shows projected yearly cash flows for Projects A and B, based on an initial investment of \$155,000.

Table 39. Cash Flows for Projects A and B based on Same Initial Investment

Time in Years	0	1	2	3	4	5	6	7	8
Project A	(155,000)	20,000	20,000	30,000	40,000	30,000	30,000	25,000	5,000
Project B	(155,000)	50,000	10,000	20,000	20,000	20,000	30,000	25,000	25,000

Table 40 shows the computed NPVs for Projects A and B at various costs of capital. For lower rates (from .02 to .035), Project A has a higher NPV and would thus be a slightly better choice for investment. For rates between .04 and .06, Project B has a higher NPV and would be a better choice in this case. For rates higher than .06, the NPV is negative for both projects. In this case, it is better to put the capital into the alternative investment.

Table 40. NPV Comparison for Projects A and B

Cost of Capital	NPV Project A	NPV Project B
0.02	\$28,330.62	\$28,244.78
0.025	\$24,550.62	\$24,474.07
0.03	\$20,912.86	\$20,855.77
0.035	\$17,411.50	\$17,383.12
0.04	\$14,040.96	\$14,049.69
0.045	\$10,795.93	\$10,849.36
0.05	\$7,671.35	\$7,776.33
0.055	\$4,662.36	\$4,825.07
0.06	\$1,764.37	\$1,990.33
0.065	(\$1,027.06)	(\$732.89)
0.07	(\$3,716.12)	(\$3,349.36)

Author's remark: The analysis above assumes that Projects A and B have similar risk levels. If one project is substantially riskier than the other, then different costs of capital may be needed for a meaningful comparison.

7.2 Internal Rate of Return (IRR)

In the examples from the previous section, we evaluated the NPV of a series of cash flows against a rate of return for an alternative (and assumed to be external) investment such as bank notes, stocks or bonds. However, we did not pinpoint the exact rate of return where the NPV is 0, i.e., the point where there is an exact balance between the internal investment and the external alternative (i.e., the cost of capital). This balance point is known as the **internal rate of return**. Equivalently, the IRR is the discount rate that makes the project's NPV equal to zero.

From Wikipedia [11] we have the following definition:

The internal rate of return (IRR) is a measure of an investment's expected future rate of return. As the IRR is an estimate of a future annual rate of return, IRR should not be confused with the actual achieved investment return of an historical investment. The term internal refers to the fact that the calculation excludes external factors, such as the risk-free rate, inflation, the cost of capital, or various financial risks.

Technically, to find the IRR, one solves the equation for i :

$$NPV = -C_0 + \sum_{k=1}^n C_k(1+i)^{-k} = 0$$

which is equivalent to

$$C_0 = \sum_{k=1}^n C_k(1+i)^{-k}$$

In words, we are looking for the value (or values) of i such that the initial investment equals the present value of the subsequent cash flows resulting from the initial investment.

The decision rules concerning the IRR are as follows:

- If the IRR is greater than the cost of capital, go forward with the project.
- If the IRR is less than the cost of capital, do not go forward with the project.
- In the unlikely event that the IRR exactly equals the cost of capital, the analysis is indifferent as to whether the project should go forward or not.

7.2.1 Acme Furniture Example Revisited

Consider Table 29 regarding the Acme Furniture example from Section 7.1.1. From the analysis in Table 30, we see that the IRR is somewhere between 7% and 8%. Using the Microsoft Excel IRR function with the information from Table 29, we get the more precise value of 7.71% for the IRR. The Excel IRR function is very easy to use (just input the initial investment and the subsequent cash flows). For more details, do a web search on "Excel IRR function".

So, if the cost of capital is less than 7.71%, Acme Furniture should invest in the project. For cost of capital rates higher than 7.71%, the NPV for the project becomes negative.

7.2.2 Multiple IRR Values for a Given Cash Flow

It is possible for a given cash flow to have more than one IRR, i.e., for the NPV equation to have several values of i that give an NPV value of 0. For example, the cash flow in Table 41 has two IRRs.

Table 41. Cash Flow with Two IRRs

Time	0	1	2	3	4	5
Net Cash Flow	\$ (10,000)	\$ 10,000	\$ 11,000	\$ 13,000	\$ 10,000	\$ (37,000)

The existence of two IRRs can be seen by graphing the equation for the NPV as a function of i (divided by 1000 to simplify the equation), i.e.,

$$-10 + 10(1 + i)^{-1} + 11(1 + i)^{-2} + 13(1 + i)^{-3} + 10(1 + i)^{-4} - 37(1 + i)^{-5}$$

The graph of the NPV as a function of i is shown in Figure 14. In the graph, the y-axis is the NPV and the x-axis represent the cost of capital i . As one can see, the graph crosses the x-axis around .05 and then again around .83. The Microsoft Excel IRR function has an optional parameter called “guess” that allows one to start the numerical algorithm for root determination around a given value. This is exactly what is needed in the case of several IRRs. If we use a guess of .05, the IRR function in Excel returns the more precise value of .0507, and if we guess .8, the Excel IRR function returns the value of .8243. Thus, the two IRRs are 5.07% and 82.43%. Between these two values, the NPV is positive and it makes sense to go ahead with the initial investment of \$10,000. For values of i less than 5.07% or greater than 82.43%, the NPV is negative and so, the initial investment is not advised in these cases (at least not based on IRR analysis).

The possibility of multiple IRRs is one of the shortcomings with this approach to capital budgeting. As we shall see in Section 7.3, there is a modification to the IRR concept that eliminates the possibility of multiple solutions to the equation $NPV = 0$.

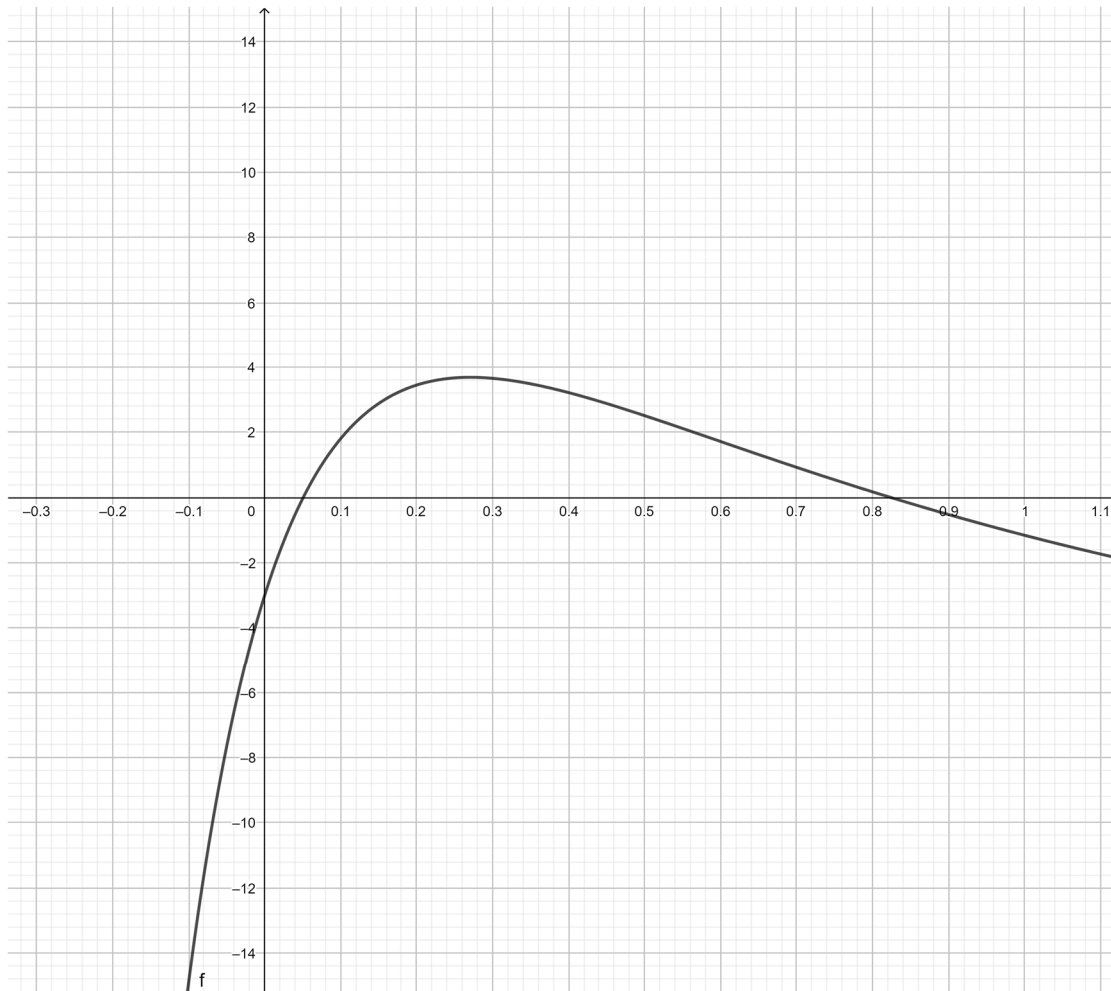


Figure 14. Graph of NPV as a Function of the Rate of Return

7.3 Comparing NPV to IRR

When comparing mutually exclusive projects, the project with the highest IRR may not be the project with the highest NPV. In such situations, the NPV approach is generally preferred because it measures the actual increase in value created by the investment.

The conflict between the IRR and NPV approaches is often explained in terms of their different reinvestment assumptions.

- The NPV approach implicitly assumes that intermediate net cash flows can be reinvested at the cost of capital (or required rate of return).
- The IRR approach implicitly assumes that intermediate net cash flows can be reinvested at the project's own internal rate of return. Issues with the IRR approach are well explained in an article from McKinsey & Company [12].

If the IRR is very large, this assumption may be unrealistic. For example, a project with an IRR of 40% implicitly assumes that interim cash flows can continue earning 40% elsewhere. In practice, opportunities with such high returns are often unavailable. As a result, the IRR approach can exaggerate the attractiveness of certain projects, especially projects with large early cash flows.

A common situation in which IRR and NPV disagree occurs when comparing mutually exclusive projects of different scale or timing.

Author's Remark: The commonly stated reinvestment assumptions associated with NPV and IRR are somewhat misleading if interpreted literally. Neither the NPV nor IRR formula explicitly reinvests interim cash flows during the computation. Rather, the assumptions arise from economic interpretations of the methods, particularly when interpreting IRR as a compound rate of return over the life of a project. The primary advantage of the NPV method is that it directly measures the increase in value created by an investment using a market-based required rate of return.

Example: Suppose a company must choose between the following two projects, and the cost of capital is 10%.

Year	Project A	Project B
0	-1000	-1000
1	1500	0
2	0	1700

Project A has:

- IRR $\approx$ 50%
- NPV $\approx$ \$364

Project B has:

- IRR $\approx$ 30%
- NPV $\approx$ \$405

Although Project A has the higher IRR, Project B creates more value because it has the higher NPV. The IRR method favors Project A because of its extremely high percentage return over a short period, while the NPV method recognizes that Project B ultimately generates more wealth in present-value terms.

For this reason, when mutually exclusive projects conflict, financial theory generally recommends choosing the project with the higher NPV rather than the higher IRR.

7.4 Modified IRR

The **Modified Internal Rate of Return** (MIRR) is a measure of an investment's attractiveness. It is used in capital budgeting to rank alternative investments of equal size. The MIRR is a modification of the IRR concept and as such aims to resolve some problems with the IRR.

Conceptually, the MIRR simplifies a complex series of cash flows into a single investment lifecycle. It calculates the compound annual growth rate necessary to turn the total present value of a project's costs into the total future value of its returns.

The MIRR is calculated as follows for a given investment:

- Determine the future value of all periods with positive cash flows using the cost of reinvestment (call it i). The calculation is done from the time of the inflow until the end of the project. The compute value is referred to as the Terminal Value (TV) of the cash flows. In the equations that follow, we call this value x .

- Determine the present value (at time zero) of all the periods with negative cash flows (including the initial investment), using the cost of financing (call it j). In the equations that follows, we call this value y .
- Determine the MIRR, i.e., the rate of return at which the value y accumulates to the value x by the end of the project. Assume the project transpires over n time periods. In the equations that follows, the MIRR is represented by the variable z .

More formally, we have the following equation

$$-y(1+z)^n = x$$

which can be solved for z as follows

$$(1+z)^n = -\frac{x}{y}$$

$$z = \sqrt[n]{-\left(\frac{x}{y}\right)} - 1$$

where

$$x = \sum_{k=1}^u C_k(1+i)^{n-t_k}, y = \sum_{h=1}^v C_h(1+j)^{-t_h} \text{ and } C_m \text{ is the cash flow for period } t_m.$$

The MIRR is used in a manner similar to the IRR. If the MIRR for a project is greater than the cost of capital, then go forward with the project; otherwise, do not go forward with the project. When comparing several projects, the project with the highest MIRR is preferred. Further, the MIRR can be used to rank projects of different sizes.

Unlike the IRR, there is only one MIRR for a given project. Further, and as noted in the definitions above, the MIRR computation allows one to use a reinvestment rate of return i that is different from the MIRR itself.

As an example, consider the investment and cash flow data for a potential project in Table 42.

Table 42. Input for MIRR Example

Year	0	1	2	3	4	5
Net Cash Flow for year	(25,000)	(5,000)	5,000	20,000	20,000	20,000

Assume that the positive net cash flows in years 2, 3 and 4 are reinvested at rate $i = .04$. The cash inflow in year 5 is at the end of the project and so, there is no reinvestment (at least not during the time duration of the project). We now have sufficient information to calculate x .

$$x = 5,000(1.04)^3 + 20,000(1.04)^2 + 20,000(1.04)^1 + 20,000 = 68,056.32$$

Next, we compute the negative net cash flow at $t = 0$, assuming a financing rate $j = .05$

$$y = -(25,000 + 5,000(1.05)^{-1}) = -29,761.90476$$

The MIRR is

$$\sqrt[5]{-\left(\frac{68,056.32}{-29,761.90476}\right)} - 1 \cong 0.1799$$

This can be cross checked with the Microsoft Excel MIRR function, which does in fact return the result of 17.99%. A description of the Excel MIRR function can be found by doing a web search on “Modified IRR in Excel.”

Using the Excel IRR function, we find that the IRR for this problem is 23.99%.

For an additional example, see the Wikipedia article on MIRR [13].

7.5 Payback and Discounted Payback Period

Payback Period and **Discounted Payback Period** (DPP) are decision techniques used to evaluate investment projects. The payback period approach entails a calculation of the time duration required to recover the cost of an initial investment. The shorter the time, the better the investment. The discounted payback period approach is similar except that one uses the discounted values of the cash flows.

From the Investopedia article on Payback Period [14]:

The payback period refers to the amount of time it takes to recover the cost of an investment. Simply put, the payback period is the length of time an investment reaches a break-even point.

From the Investopedia article on DDP [15]:

The discounted payback period is a capital budgeting procedure used to determine the profitability of a project. A discounted payback period gives the number of years it takes to break even from undertaking the initial expenditure, by discounting future cash flows and recognizing the time value of money. The metric is used to evaluate the feasibility and profitability of a given project.

The DPP concept is very simple as can be illustrated via an example. Consider Projects A and B from the example in Section 7.1.3, with an assumed cost of capital of 2.5%. Table 43 shows the discounted cash flows for Project A and the cumulative sum of the discounted cash flows. As can be seen, it is not until the end of year 6 that Project A has recovered the initial investment. (The discounting is done with respect to time 0.)

Table 43. DPP for Project A

Years	0	1	2	3	4	5	6	7	8
Cash Flows	(155,000)	20,000	20,000	30,000	40,000	30,000	30,000	25,000	5,000
Discounted Cash Flows	(155,000)	19,512	19,036	27,858	36,238	26,516	25,869	21,032	4,104
Accumulated Discounted Cash Flows	(155,000)	(135,488)	(116,452)	(88,594)	(52,356)	(25,840)	29	21,061	25,164

Table 44 shows similar calculations for Project B, which takes an additional year to break even.

Table 44. DPP for Project B

Years	0	1	2	3	4	5	6	7	8
Cash Flows	(155,000)	50,000	10,000	20,000	20,000	20,000	30,000	25,000	25,000
Discounted Cash Flows	(155,000)	48,780	9,518	18,572	18,119	17,677	25,869	21,032	20,519
Accumulated Discounted Cash Flows	(155,000)	(106,220)	(96,701)	(78,129)	(60,010)	(42,333)	(16,464)	4,567	25,086

Based on DPP analysis, the decision should be to go forward with Project A since it achieves payback of the initial investment in 6 years whereas payback for Project B is not until the 7th year.

7.6 Profitability Index

The **Profitability Index** (PI) is the ratio of payoff to investment for a proposed project. It is useful for ranking projects because it allows one to quantify the amount of value created per unit of investment. More precisely, the profitability index is defined as the NPV of the positive net cash flows for a project divided by the NPV of the negative net cash flows. In terms of a formula, we have

$$PI = \frac{\sum_{k=1}^n PNCF_k (1+i)^{-t_k}}{-\sum_{j=1}^m NNCF_j (1+i)^{-t_j}}$$

where $PNCF_k$ stands for k^{th} positive net cash flow at time t_k and $NNCF_j$ stands for j^{th} negative net cash flow at time t_j . Notice that the sum of the negative net cash flows is converted to a positive number so that the PI is a positive number.

[Author's remarks: This looks worse than it is. It's just a matter of computing the present value for the positive and negative net cash flows separately, and then dividing to get the PI.

Also, some (perhaps most) sources use an alternative definition of PI where the denominator only includes the initial investment and the numerator is the sum of the present values of all other cash flows. See, for example, the article on PI in Investopedia [16] which uses the alternate definition of PI.]

As an example, we will compute the PI for Projects A and B from Section 7.1.3, under the assumption of a 2.5% cost of capital:

$$PI \text{ for Project A} = \frac{180,164}{155,000} = 1.1624$$

$$PI \text{ for Project B} = \frac{180,086}{155,000} = 1.1618$$

The general rule is to select the project with the highest PI. So, in this case, we should go forward with Project A (albeit by a small margin).

7.7 Computational Support via Microsoft Excel

There are no supporting Python programs for this section. However, there is an Excel spreadsheet that covers many of the concepts in this section. The spreadsheet is in a file named “Capital Budgeting.xlsx”. The spreadsheet is located in the same Github directory as the Python programs.

A brief summary of the sheets within the spreadsheet is listed below:

- Acme Furniture – this sheet shows the NPV calculations for the Acme Furniture example under the 5 year and 6 year scenarios. The calculation of the IRR is also provided (using the Excel IRR function).
- Resort – this sheet provides an NPV analysis for the resort example in Section 7.1.2 concerning negative cash flows.
- Multiple Inv – this sheet supports the multiple investments example from Section 7.1.3.
- Restaurant – this sheet provides an NPV analysis for pessimistic and optimistic scenarios regarding a restaurant (from Section 7.1.4).
- Acme Furniture – scroll down on this sheet and there is a separate calculation for the loan scenario in Section 7.1.5.
- Project Comparison – this sheet provides an analysis of Projects A and B in Section 7.1.6.
- Multiple IRRs – this sheet supports the multiple IRR example in Section 7.2.2.
- MIRR – this supports the MIRR example in Section 7.3.
- DDP and PI – this sheet supports DDP and PI examples that span Sections 7.5 and 7.6.
- Ex 1 – this sheet provides the answer to Exercise 1 in Section 7.8.
- Ex 2 – this sheet provides the answer to Exercise 2 in Section 7.8.

7.8 Exercises

1. Given an initial investment of \$70,000 and projected yearly cash flows of \$5,000, \$7,500, \$10,000, \$12,500, \$15,000, \$17,500 and \$20,000, compute the NPV for cost of capital of 2, 3, 4, 5, 6 and 7 percent.

Answer: See the sheet entitled “Ex 1” in the Excel file “Capital Budgeting.xlsx”.

2. For the example in Section 7.1.5, assume the term of the loan (in the amount of \$50,000) is 10 years at 3% per year with semi-annual payments. Further, assume the project returns no cash inflows after year 6. Compute the NPV for the (semi-annual) cost of capital ranging from 2% to 8% in increments of .5%.

Hint: Use the Amortization Calculator to determine the semi-annual payments.

Answer: See the sheet entitled “Ex 2” in the Excel file “Capital Budgeting.xlsx”.

3. For the Acme Furniture company loan example, what is the maximum amount that Alice and Bob can borrow under the same loan terms — 3% nominal annual interest, repaid semiannually over 5 years — and still have a positive NPV at a 2% semiannual cost of capital?

Hint: First determine the largest semiannual loan payment that can be subtracted from the project cash flows while keeping the NPV approximately equal to zero at a 2% semiannual cost of capital. Then use the loan payment formula, or the amortization calculator, to determine the loan amount corresponding to that semiannual payment.

Answer: For a semiannual loan payment of about \$11,158.30, the NPV is approximately zero at a 2% semiannual cost of capital. Under the stated loan terms — 3% nominal annual interest, paid semiannually over 5 years — this payment corresponds to a loan amount of about \$102,904. Therefore, to maintain a positive NPV, Alice and Bob should borrow slightly less than \$102,904.

See the sheet entitled “Ex 3” in the Excel file *Capital Budgeting.xlsx*.

4. Use the Microsoft Excel IRR function to compute the IRRs for Projects A and B in the example from Section 7.1.6.

Answer: 6.31% for Project A and 6.36% for Project B.

5. Use the Microsoft Excel MIRR function to compute the MIRR for Project A and B in the example from Section 7.1.6. Assume a reinvestment rate of .015 and a financing rate of .025. Do the calculation again with a reinvestment rate of .08 and a financing rate of .06.

Answer: In the first scenario, 3.96% for Project A and 3.95% for Project B. In the second scenario, 7.15% for Project A and 7.20% for Project B.

7.9 Summary and Recommendations: Choosing Among Capital Budgeting Methods

The preceding sections of this chapter introduced five primary techniques for evaluating capital investments: Net Present Value (NPV), Internal Rate of Return (IRR), Modified Internal Rate of Return (MIRR), Payback Period (including the Discounted Payback Period, DPP), and the Profitability Index (PI). Each method offers useful information, but none is perfect for every situation. In practice, financial decision-makers rarely rely on a single metric. Instead, they combine methods to gain a more complete picture of a project’s value, risk, liquidity, and efficiency—especially when projects differ in size, timing of cash flows, or risk level.

The choice of method (or combination of methods) depends on the decision context: whether you are screening a single project, ranking mutually exclusive projects, operating under capital rationing (limited funds), or facing high uncertainty about future cash flows. The following table summarizes the strengths, limitations, and most appropriate uses of each technique.

Table 45. Comparison of Capital Budgeting Methods

Method	Primary Output	Key Advantages	Key Limitations	Best Used For
NPV	Dollar value added (present value)	Directly measures increase in wealth; handles uneven or negative cash flows; consistent reinvestment assumption (cost of capital); no multiple-solution problem	Requires accurate cost-of-capital estimate; absolute dollar figures can be hard to interpret for projects of different scales	Primary decision rule for accept/reject decisions and ranking mutually exclusive projects; any situation where value maximization is the goal
IRR	Breakeven discount rate (percentage)	Intuitive percentage metric that is easy to communicate and compare with the cost of capital; quick screening tool (accept if IRR > cost of capital)	Can produce multiple solutions with non-conventional cash flows; implicit (and often unrealistic) reinvestment assumption at the IRR itself; can conflict with NPV when comparing projects of different size or timing	Communicating expected returns; preliminary screening; supplement to NPV
MIRR	Adjusted percentage rate of return	Eliminates multiple-rate problem; uses explicit, realistic reinvestment and financing rates	Requires two additional rate inputs (reinvestment and financing rates); slightly more complex calculation	Projects with non-conventional cash flows; when IRR assumptions appear unrealistic; ranking projects of similar size
Payback / DPP	Time (years) to recover investment	Simple to calculate and understand; highlights liquidity and risk (shorter payback = lower risk of not recovering money); DPP incorporates time value of money	Ignores all cash flows after payback period; arbitrary cutoff; DPP still does not measure total profitability	Quick liquidity screen; high-uncertainty environments; preliminary filter before applying NPV/IRR; industries where rapid recovery is critical (e.g., technology)
PI	Ratio of present value of inflows to outflows	Measures bang-for-the-buck (value created per dollar invested); useful when capital is limited	Does not directly show absolute value created; same data requirements as NPV	Capital rationing (ranking many small projects when funds are scarce); comparing projects of different initial investment sizes

7.9.1 Practical Guidelines

1. **Use NPV as the primary decision rule.** Financial theory and the majority of corporate finance textbooks regard NPV as the most reliable single metric because it directly answers the question: “How much value (in today’s dollars) does this project create?” When NPV and IRR conflict (as shown in Section 7.3), choose the project with the higher NPV. NPV correctly accounts for the scale of the investment and the realistic reinvestment rate (the firm’s cost of capital).
2. **Supplement NPV with IRR or MIRR for communication and intuition.** A percentage rate is often easier for non-financial stakeholders to grasp than a dollar amount. Report the IRR (or, better, MIRR) alongside NPV so everyone can see both the absolute value created and the breakeven rate of return relative to the cost of capital. Use MIRR whenever cash-flow

signs change more than once or when the simple IRR reinvestment assumption seems unrealistic.

3. **Employ Payback/DPP as a screening or risk filter.** Even a project with a strong NPV may be rejected if the payback period is excessively long (e.g., the firm needs cash quickly or faces high technological obsolescence). DPP is preferred over simple payback because it respects the time value of money.
4. **Use the Profitability Index when capital is constrained or when comparing projects of different sizes.** When projects require different initial investment levels (i.e., different scales), the Profitability Index is especially helpful because it measures value created per dollar invested. When a company cannot fund every positive-NPV project, rank them by PI and select the combination that maximizes total NPV within the available budget.
5. **Combine methods for robust decision-making.** A common real-world approach is:
 - First, apply the Payback/DPP screen to eliminate projects that tie up capital too long.
 - Then compute NPV (the decisive metric) and IRR/MIRR (for intuition and benchmarking against the cost of capital).
 - If capital is limited, calculate PI to finalize the ranking.
6. **Always remember the limitations of the numbers.** As noted throughout this section, the reliability of any technique depends on the quality of the cash-flow forecasts and the chosen cost of capital. Sensitivity analysis (varying cash flows, discount rates, or project life) and scenario analysis (optimistic/pessimistic cases) should accompany every major decision. Non-financial factors—strategic fit, environmental impact, regulatory risk, or employee morale—may also tip the scale even when the numbers look favorable.

All of the methods discussed above can be implemented quickly in the Excel spreadsheet “Capital Budgeting.xlsx” provided with this section (see Section 7.7). The spreadsheet already contains the examples used in the text, plus ready-made templates for sensitivity tables and charts that make it easy to compare methods side-by-side.

In summary, treat NPV as your anchor, use IRR/MIRR and PI to add insight and handle special situations, and keep Payback/DPP as a practical safety check. By applying the appropriate combination of these tools, you will make capital-budgeting decisions that are both financially sound and easy for stakeholders to understand.

8 Stocks

October: This is one of the particularly dangerous months to invest in stocks. Other dangerous months are July, January, September, April, November, May, March, June, December, August, and February." — Mark Twain

All of the financial instruments we have studied so far — loans, annuities, and bonds — share a common mathematical structure: they generate contractually specified cash flows at known times, and their present value can be computed by discounting those flows at an appropriate interest rate. A **stock** (also called a share or equity) is fundamentally different. A stock represents a fractional ownership stake in a company. There is no maturity date, no guaranteed redemption value, and no contractual obligation to pay dividends. Yet the same core idea — present value as the sum of discounted future cash flows — still applies. The challenge is that the future cash flows are uncertain, and we must make assumptions about them.

This chapter develops a mathematical framework for stock valuation based on dividends. We begin with the simplest case — constant dividends, which reduces to a perpetuity — and then extend to the more realistic case in which dividends grow at a constant rate. Throughout, the mathematics leans heavily on results already established: the geometric series from Section 2, the present-value formulas from Section 3, and perpetuities from Section 4.8.

8.1 Terminology

Before deriving valuation formulas, it is useful to fix some terms and compare stocks with bonds, since the contrast sharpens our understanding.

Table 46. Stock terminology

Feature	Bond	Stock
Cash flows	Fixed coupons (contractual)	Dividends (discretionary)
Maturity	Fixed date	None (perpetual in principle)
Redemption value	Face value F at maturity	None guaranteed
Priority on assets	Senior (paid before shareholders)	Junior (residual claimant)
Upside	Capped at coupon + face value	Unlimited (capital gains)

We use the following notation throughout this chapter:

- P — the current price of one share of stock.
- D — a dividend payment per share (when constant).
- D_0 — the most recently paid dividend (time 0).
- D_1 — the dividend expected one period from now.
- g — the expected constant growth rate of dividends per period (as a decimal).
- r — the required rate of return per period (as a decimal); also called the discount rate or cost of equity.
- n — the number of periods.

Author's remark: The required rate of return r reflects what an investor demands in order to hold the stock, given its risk. In bond pricing we used the yield y in the same role. Determining r in practice involves considerations beyond this text (such as the Capital Asset Pricing Model); here we treat it as a given input.

8.2 The Constant Dividend Model

Suppose a stock is expected to pay a dividend D at the end of every period, forever, and the required rate of return is r . The present value of these dividends is

$$P = \frac{D}{(1+r)} + \frac{D}{(1+r)^2} + \frac{D}{(1+r)^3} + \dots$$

This is an infinite geometric series with first term $a = \frac{D}{1+r}$ and common ratio $x = \frac{1}{1+r}$. Since $r > 0$ we have $|x| < 1$, and the series converges. Applying the formula for an infinite geometric series (Section 2):

$$P = \frac{\frac{D}{(1+r)}}{1 - \frac{1}{(1+r)}} = \frac{\frac{D}{(1+r)}}{\frac{r}{(1+r)}}$$

$$P = \frac{D}{r} \quad (8.1)$$

This is precisely the perpetuity formula derived in Section 4.8. A stock paying a constant dividend forever is mathematically equivalent to a perpetuity.

From an intuitive standpoint, the convergence of this infinite series reflects the time value of money. A dividend received years from now is worth less than the same amount received today because money available today can be invested elsewhere to earn a continuous return. This opportunity cost is mathematically represented by the denominator in each term of the series. As the number of periods increases, the discount factor grows exponentially, causing the present value of dividends received in the distant future to shrink rapidly toward zero. This exponential decay explains why an infinite number of future payments yields a strictly finite present value, as dividends expected decades or centuries from now contribute almost nothing to the current price of the stock.

Example 8.1. A company is expected to pay a dividend of \$2.40 per share at the end of each year, indefinitely. An investor requires a return of 8% per year. What is the theoretical price of the stock?

Solution. Using formula (8.1) with $D = 2.40$ and $r = 0.08$:

$$P = \frac{2.40}{0.08} = \$30.00.$$

Equation (8.1) also provides a useful inversion: given the current market price P and dividend D , the implied required return is $r = \frac{D}{P}$. This ratio is called the dividend yield and is reported for many stocks in financial data services.

8.3 The Gordon Growth Model

In practice, companies that pay dividends tend to increase them over time as earnings grow. Assume that dividends grow at a constant rate g per period, so that

$$D_1 = D_0(1 + g), D_2 = D_0(1 + g)^2, D_3 = D_0(1 + g)^3, \dots$$

Here D_0 is the dividend just paid and $D_1 = D_0(1 + g)$ is the dividend expected at the end of the next period. The present value of all future dividends is

$$P = \frac{D_1}{(1 + r)} + \frac{D_1(1 + g)}{(1 + r)^2} + \frac{D_1(1 + g)^2}{(1 + r)^3} + \dots$$

Factor out $\frac{D_1}{1+r}$:

$$P = \frac{D_1}{(1 + r)} \cdot \left[1 + \frac{(1 + g)}{(1 + r)} + \frac{(1 + g)^2}{(1 + r)^2} + \dots \right]$$

The bracketed expression is an infinite geometric series with common ratio $x = \frac{1+g}{1+r}$. Provided $r > g$ (so that $|x| < 1$), the series converges to $\frac{1}{1-x} = \frac{1+r}{r-g}$. Substituting:

$$P = \frac{D_1}{(1 + r)} \cdot \frac{(1 + r)}{r - g}$$

$$P = \frac{D_1}{r - g} \quad (8.2)$$

This is the Gordon Growth Model (also called the Dividend Discount Model with constant growth), named after Myron J. Gordon who popularized it in the 1950s. Observe that when $g = 0$, equation (8.2) reduces to $P = \frac{D}{r}$, recovering formula (8.1).

Remarks:

- Note that the summation begins with D_1 rather than D_0 . In financial models, D_0 represents the dividend that was just paid at the present moment. An investor purchasing the stock today is buying the rights to all future cash flows, which commence at the end of the first period with D_1 . The previous owner retains D_0 . Therefore, D_0 is excluded from the present value calculation of future payments and serves only as the baseline from which to project D_1 using the assumed growth rate g .
- Formula (8.2) requires $r > g$. If $g \geq r$ the denominator is zero or negative, which signals that the model breaks down: a dividend stream growing at or faster than the discount rate has infinite present value under this model, which is not economically meaningful. In practice, no company can sustain $g \geq r$ forever, so the constant-growth assumption must eventually fail for high-growth companies.

Example 8.2. A company paid a dividend of \$3.00 per share this year. Dividends are expected to grow at 4% per year indefinitely. An investor requires a return of 10% per year. What is the theoretical price of the stock?

Solution. We have $D_0 = 3.00$, $g = 0.04$, and $r = 0.10$. The next dividend is

$$D_1 = 3.00 \times 1.04 = \$3.12.$$

Applying formula (8.2):

$$P = \frac{3.12}{0.10 - 0.04} = \frac{3.12}{0.06} = \$52.00$$

Example 8.3. Suppose the stock in Example 8.2 is currently trading at \$65.00. What required return does this price imply?

Solution. Rearranging (8.2) to solve for r :

$$r = \frac{D_1}{P} + g$$

$$r = \frac{3.12}{65.00} + 0.04 \approx 0.0480 + 0.0400 = 8.80\%.$$

So, a buyer willing to pay \$65.00 is implicitly accepting a required return of approximately 8.80% per year rather than 10%. This may reflect greater confidence in the company's prospects, or a lower tolerance for risk, compared with the investor in Example 8.2.

8.4 Required Rate of Return and Dividend Yield

Rearranging formula (8.2) gives an illuminating decomposition of the required return:

$$r = \frac{D_1}{P} + g \quad (8.3)$$

The two terms on the right have natural interpretations.

Dividend yield. The ratio $\frac{D_1}{P}$ is the dividend expected in the next period divided by the current price. It represents the income return — the return from cash dividends alone, analogous to the coupon yield on a bond.

Capital gains yield. Under the Gordon Growth Model, the stock price itself grows at rate g each period. This can be verified: applying (8.2) one period later gives

$$P_1 = \frac{D_2}{r - g} = \frac{D_1(1 + g)}{r - g} = P(1 + g)$$

Thus, the rate of price appreciation equals g , which is also the capital gains yield.

Hence, (8.3) states that the total return on a stock equals its dividend yield plus its capital gains yield.

Example 8.4. A stock trades at \$40. The most recent dividend was \$1.60, and dividends are expected to grow at 5% per year. Compute the dividend yield, the capital gains yield, and the total required return.

Solution.

$$D_1 = 1.60 \times 1.05 = \$1.68.$$

$$\text{Dividend yield} = \frac{1.68}{40.00} = 4.2\%.$$

$$\begin{aligned}\text{Capital gains yield} &= g = 5.0\%. \\ \text{Total return} &= 4.2\% + 5.0\% = 9.2\%.\end{aligned}$$

8.5 The Price-to-Earnings Ratio

Financial analysts frequently quote a stock's price-to-earnings ratio, or P/E ratio, defined as

$$\frac{P}{E} = \frac{\text{price per share}}{\text{earnings per share}} \quad (8.4)$$

The Gordon Growth Model sheds light on what the P/E ratio represents. If we assume that a company pays out all of its earnings as dividends (i.e., payout ratio = 100%), and dividends grow at rate g , then $D_1 = E_1$, and formula (8.2) gives

$$P = \frac{E_1}{r - g}$$

so that

$$\frac{P}{E_1} = \frac{1}{r - g} \quad (8.5)$$

Thus, the P/E ratio is the reciprocal of $(r - g)$. A high P/E ratio can reflect either a low required return r or a high expected growth rate g . The inverse $\frac{E_1}{P}$ is called the earnings yield and equals $r - g$ under these assumptions.

Author's remark: In practice, the payout ratio is rarely 100%. Companies retain a portion of earnings for reinvestment, and the relationship between earnings, dividends, and growth is more involved. Equation (8.5) should be understood as an idealized illustration of the forces driving P/E ratios, not a formula for computing them.

Example 8.5. Given the following (assume 100% payout of dividend)

- E_0 (most recent EPS) = 1.60
- g (growth rate; decimal per period) = 0.05
- r (required return; decimal per period) = 0.092
- P (stock price) = 40.00

Compute $E_1, \frac{P}{E_1}, \frac{E_1}{P}$

- $E_1 = E_0(1 + g) = 1.60 \times 1.05 = 1.68$
- $\frac{P}{E_1} = \frac{40.00}{1.68} = 23.81$ or $\frac{P}{E_1} = \frac{1}{r - g} = \frac{1}{0.042} = 23.81$
- Earnings yield $\frac{E_1}{P} = 4.2\%$

As a check, note that $P = \frac{E_1}{r - g} = \frac{1.68}{0.042} = \40.00

These values correspond to Example 8.4 when the firm pays out all earnings as dividends (so $D_1 = E_1$).

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To accommodate situations where a company retains a portion of their earnings for reinvestment, we introduce a variable for the payout ratio. Let x represent the fraction of earnings distributed as dividends.

The dividend expected in the next period is the earnings multiplied by the payout ratio:

$$D_1 = xE_1$$

Substituting this relationship into the Gordon Growth Model yields:

$$P = \frac{x E_1}{r - g}$$

Dividing both sides by E_1 provides a generalized formula for the price-to-earnings ratio:

$$\frac{P}{E_1} = \frac{x}{r - g} \quad (8.6)$$

This expanded formula shows that the price-to-earnings ratio is directly proportional to the payout ratio, provided the required return and growth rate remain constant.

8.6 Limitations of Dividend-Based Models

The dividend discount models in Sections 8.2 and 8.3 are elegant and mathematically clean, but they rest on several strong assumptions. Understanding these limitations is as important as knowing the formulas.

No dividends. Many companies — particularly young, high-growth technology firms — pay no dividends. They reinvest all earnings. For such companies, dividend-based models cannot be applied directly. Alternative approaches (such as discounting free cash flows or using relative valuation metrics) are needed.

Constant growth rate. The Gordon Growth Model assumes g is constant forever. In reality, growth rates change over time. A company may grow rapidly for several years, then slow to a mature rate. Multi-stage models address this by applying different growth rates in different phases, at the cost of additional assumptions.

Sensitivity to inputs. Formula (8.2) is highly sensitive to the spread $r - g$. Small changes in either r or g can produce large changes in P . For example, if $r = 10\%$ and $g = 8\%$, then $r - g = 2\%$ and $P = 50 \cdot D_1$. Raising g by just one percentage point to 9% doubles the price to $P = 100 \cdot D_1$. Modest errors in estimating g can result in very large pricing errors.

Market prices reflect more than dividends. Actual stock prices incorporate investor expectations about future earnings, macroeconomic conditions, interest rates, sentiment, and many other factors not captured by a simple dividend model. Market prices frequently diverge substantially from model values, sometimes for extended periods.

Dividends can be cut. Unlike bond coupons, dividends are discretionary. A company facing financial difficulty may reduce or eliminate its dividend, invalidating the assumption of steady growth.

Despite these limitations, the Gordon Growth Model remains a foundational tool in finance. It provides a clear, mathematically rigorous framework for thinking about how dividends, growth, and required returns interact to determine value — even when it cannot be applied mechanically to every stock.

8.7 Summary of Formulas

Table 47. Summary of formulas related to stocks

Formula	Expression	Reference
Constant dividend (perpetuity)	$P = \frac{D}{r}$	(8.1)
Gordon Growth Model	$P = \frac{D_1}{r - g}$	(8.2)
Required return decomposition	$r = \frac{D_1}{P} + g$	(8.3)
P/E ratio (full payout)	$\frac{P}{E_1} = \frac{1}{r - g}$	(8.5)
P/E ratio (fraction x of E as payout)	$\frac{P}{E_1} = \frac{x}{r - g}$	(8.6)

8.8 Supporting Program

Stock Valuation Calculator

File name: stock_valuation.ipynb

This calculator supports stock valuation under a **constant dividend**, the **Gordon Growth Model**, a **required-return decomposition**, and the **price-to-earnings ratio** (assuming 100% payout so that the next dividend equals next-period earnings). On each tab, choose what to solve for from the drop-down list, enter the remaining inputs, and click **Compute**. For growth models, enter only D_0 or E_0 ; D_1 and E_1 are computed as $D_0(1 + g)$ and $E_0(1 + g)$.

- **Compute price P** (constant dividend): constant dividend D , required return r per period as a decimal (e.g. 0.08 for 8% per period).
- **Compute required return r** (constant dividend): D , P .
- **Compute dividend D** (constant dividend): r , P .
- **Compute price P** (Gordon model): D_0 , growth rate g , r (decimals per period).
- **Compute required return r** (Gordon model): D_0 , g , P .
- **Compute growth rate g** (Gordon model): D_0 , r , P .
- **Compute prior dividend D_0** (Gordon model): g , r , P .
- **Compute return decomposition:** D_0 , g , P (reports dividend yield $\frac{D_1}{P}$, capital-gains yield g , and total return r).
- **Compute P/E and related quantities** (full payout): E_0 , g , r , P ; or solve for P , r , g , or E_0 from the other three.

How to use:

- Select a tab, choose the calculation from **Solve for**, enter the known values, and click **Compute**. Results appear in the text area below.
- Click **Clear** to reset inputs and results on that tab.
- Rates r and g must be entered as **decimals per period** (e.g. 0.08 for 8% per period), not as percentages (do not type 8 or 8%).

8.9 Exercises

1. A stock is expected to pay a constant annual dividend of \$1.80 per share. If the required rate of return is 9%, what is the theoretical price of the stock?

Answer: The Constant Dividend Model states that $P = \frac{D}{r}$.

Using a dividend of 1.80 and a required return of 0.09 yields the following.

$$P = \frac{1.80}{0.09} = 20.00 \text{ and so, the theoretical price of the stock is } \$20.00.$$

2. A stock currently priced at \$45 pays an annual dividend of \$2.25. What dividend yield does this represent? If the dividend is expected to grow at 3% per year, what is the total required return implied by the model?

Answer: Dividend yield is defined as the dividend expected in the next period divided by the current price, represented mathematically as $\frac{D_1}{P}$. Assuming the 2.25 dividend was just paid, the next expected dividend requires applying the 3 percent growth rate. So, $D_1 = 2.25 \times 1.03 = 2.3175$ and the Dividend yield $= \frac{2.3175}{45.00} = 0.0515$, or 5.15 percent. Total required return is the sum of the dividend yield and the capital gains yield and so, Total return $= 0.0515 + 0.03 = 0.0815$, or 8.15 percent.

3. A company just paid a dividend of \$4.00 per share. Dividends are expected to grow at 6% per year. An investor requires a return of 11%.
 - a. Compute the theoretical stock price.
 - b. If the stock is currently trading at \$90, is it overvalued or undervalued according to the model?

Answer: a. The Gordon Growth Model dictates $P = \frac{D_1}{r - g}$.

The next expected dividend is $D_1 = 4.00 \times 1.06 = 4.24$.

$$P = \frac{4.24}{0.11 - 0.06} = 84.80 \text{ and so, the theoretical stock price is } \$84.80.$$

b. The stock is overvalued because its current trading price of \$90.00 is higher than its theoretical model price of \$84.80.

4. A stock trades at \$60 and is expected to pay a dividend of \$2.40 next year. If the required return is 8%, what constant growth rate g is implied by the Gordon Growth Model?

Answer: Using the Gordon Growth Model formula $P = \frac{D_1}{r - g}$, we have that

$$60 = \frac{2.40}{0.08 - g} \Rightarrow g = .04$$

5. Show algebraically that under the Gordon Growth Model, the stock price grows at rate g each period. That is, if $P_0 = \frac{D_1}{r - g}$, show that $P_1 = P_0(1 + g)$, where P_1 is the price one period later.

Answer: The price at time zero is $P_0 = \frac{D_1}{r-g}$.

One period later, the price formula shifts forward to utilize the dividend expected in period two, meaning $P_1 = \frac{D_2}{r-g}$.

Because dividends grow at a constant rate, $D_2 = D_1(1 + g)$.

Substituting this relationship into the equation for the price one period later yields $P_1 = \frac{D_1(1+g)}{r-g}$. Factoring out the original price formula leaves $P_1 = P_0(1 + g)$.

6. A company has earnings per share of \$5.00, pays out 60% of earnings as dividends, and retains the rest for growth. If the required return is 10% and the expected dividend growth rate is 4%, compute
- the next dividend D_1 ,
 - the theoretical stock price P , and
 - the P/E ratio.

Compare your answer to formula (8.5) and explain any discrepancy.

Answer: a. The most recent dividend is 60 percent of current earnings. $D_0 = 5.00 \times 0.60 = 3.00$. The next expected dividend is $D_1 = 3.00 \times 1.04 = 3.12$

b. The theoretical stock price utilizes the Gordon Growth Model, i.e.,

$$P = \frac{3.12}{0.10 - 0.04} = \frac{3.12}{0.06} = 52.00$$

c. To compute the earnings expected for the next period, we apply the same growth rate, i.e., $E_1 = 5 \times 1.04 = 5.20$ and so, $\frac{P}{E_1} = \frac{52}{5.20} = 10$.

We could also use formula (8.6), i.e., $\frac{P}{E_1} = \frac{x}{r-g} = \frac{.6}{0.10-0.04} = \frac{.6}{.06} = 10$.

7. A stock currently pays no dividend but is expected to begin paying dividends of \$3.00 per share starting five years from now, growing at 4% per year thereafter. If the required return is 9%, what is the theoretical price of the stock today?

Answer: The price at year 4 can be calculated using the Gordon Growth Model because dividends grow at a constant rate from year 5 onward. The price at year 4 represents the present value of all dividends starting in year 5.

$$P_4 = \frac{3.00}{0.09 - 0.04} = \frac{3.00}{0.05} = 60.00$$

To find the theoretical price today, discount the year 4 price back four periods to the present using the required return.

$$P_0 = \frac{60.00}{(1.09)^4} = \frac{60.00}{1.41158} = 42.51$$

The theoretical price today is 42.51 dollars.

9 Derivatives and Structured Products

Finance is the art of passing money from hand to hand until it finally disappears.
— Robert W. Sarnoff

The market can remain irrational longer than you can remain solvent. — John Maynard Keynes

The financial instruments covered in previous chapters — loans, annuities, bonds, and stocks — are all direct claims: you either lend money and receive interest, or you own a share of a company and receive dividends. A **derivative** is a contract whose value is derived from the performance of something else: an underlying asset such as a single stock, an index of stocks, a bond, a commodity, or an interest rate. Derivatives do not represent ownership of the underlying asset; they are agreements about what will happen if the price of that asset moves in a certain way.

Derivatives are widely used for two broad purposes. **Hedging** means using a derivative to reduce an existing risk — a wheat farmer who locks in a price for the harvest, or a company that fixes its borrowing cost for the next five years. **Speculation** means using a derivative to take on risk in the hope of profit — betting that a stock will rise or fall without actually buying or selling the stock itself. The same instrument can serve either purpose depending on how it is used.

This chapter introduces the three main families of derivatives — options, futures and forwards, and swaps — and then discusses the structured credit products that became prominent in the years leading up to the 2008 financial crisis. The mathematics here is more accessible than it might appear: the forward pricing formula is nothing other than the compound interest formula from Section 3, and option payoffs require only elementary arithmetic. Swap valuation leans on the present-value ideas of Section 3 and the annuity formulas of Section 4.

Note: While an insurance policy might seem similar to a derivative because it pays out based on a future event, it is classified differently. Derivatives derive their value from tradable financial metrics, such as a stock price, whereas insurance policies pay out based on specific, non-financial events, such as an accident or death, and require the buyer to have a direct, legally recognized stake in the insured entity.

9.1 Options

An option is a contract that gives its holder the right, but not the obligation, to buy or sell an underlying asset at a specified price on or before a specified date. The key terms are:

- Underlying asset — the stock, index, commodity, or other item the option is written on.
- **Strike price** (or exercise price), K — the price at which the holder may buy or sell.
- Expiration date, T — the date on or by which the right must be exercised.
- Premium — the price paid upfront for the option contract.

9.1.1 Option Types and Payoffs

There are two basic types of option.

Call option. A call gives its holder the right to buy the underlying asset at the strike price K . If the asset's market price at expiration, S_T , exceeds K , the holder profits by buying at K and immediately

selling at the higher market price. If $S_T \leq K$, the option expires worthless and the holder simply loses the premium paid.

Put option. A put gives its holder the right to sell the underlying asset at the strike price K . If the market price S_T falls below K , the holder profits by selling at K while the asset is only worth S_T . If $S_T \geq K$, the option expires worthless.

The payoff at expiration (ignoring the premium paid) is:

$$\text{Payoff } f_{\text{call}} = \max(S_T - K, 0) \quad (9.1)$$

$$\text{Payoff } f_{\text{put}} = \max(K - S_T, 0) \quad (9.2)$$

Note that the payoff is always non-negative: an option holder can never be forced to exercise, so the worst outcome is that the option expires worthless.

Example 9.1. A call option on a stock has a strike price of $K = \$50$ and expires in one year. The option premium is \$4. Consider two scenarios:

- the stock price at expiry is $S_T = \$60$, and
- the stock price at expiry is $S_T = \$45$.

Solution.

- Payoff = $\max(60 - 50, 0) = \$10$. After subtracting the \$4 premium, the net profit is $\$10 - \$4 = \$6$ per share.
- Payoff = $\max(45 - 50, 0) = \$0$. The option expires worthless. The net loss is the premium: $-\$4$ per share.

9.1.2 The Mechanics of Trading Puts

Because a put option grants the right to sell an asset, a common point of confusion is how an investor can sell an asset they do not currently own. In financial markets, holding and profiting from a put option without owning the underlying stock is standard practice. Investors typically capture this profit through one of two methods.

The first method involves exercising the option. The contract does not require the holder to own the stock until the exact moment of exercise. If a stock drops to \$45 and the investor holds a put with a \$50 strike price, they can buy the shares on the open market for \$45 and immediately exercise the option to sell those same shares for \$50. This simultaneous transaction secures a \$5 profit per share, less the initial premium.

The second method involves selling the contract itself. This is how the vast majority of speculative trades are handled. The option contract functions as a transferable agreement. As the market price of the stock falls below the strike price, the intrinsic value of that agreement increases. Instead of buying the underlying stock to exercise the option, the investor simply sells the valuable contract to another participant on the open market, capturing the profit directly without ever touching the underlying asset.

9.1.3 Exercise Styles

Options also carry an important distinction based on when they may be exercised. A European option may only be exercised at expiration. An American option may be exercised at any time up to and including the expiration date. Most exchange-traded stock options are American; most

index options are European. (An index option is a financial derivative where the underlying asset is a market index, such as the S&P 500, the Dow Jones Industrial Average, or the Nasdaq.)

American options are at least as valuable as European options, because an American option can perfectly replicate a European option by the holder simply waiting until expiration, but a European option cannot replicate an American option. Therefore, the American contract must logically be worth greater than or equal to the European contract.

9.1.4 Put-Call Parity

There is a precise relationship between the prices of a European call and a European put that have the same underlying asset, strike price K , and expiration T . This relationship is known as put-call parity.

Consider two portfolios, each assembled today at time 0.

Portfolio A: buy one call option at price C , and invest the present value of K in a risk-free account at rate r . The present value of K is $\frac{K}{(1+r)^n}$, where n is the number of periods to expiration.

Portfolio B: buy one put option at price P , and buy one share of the stock at current price S .

At expiration, both portfolios produce an identical payoff regardless of whether $S_T > K$ or $S_T \leq K$. (you are asked to verify this in Exercise 2 of Section 9.7). Since two portfolios with identical future payoffs must cost the same today — otherwise an arbitrage opportunity would exist — their initial costs must be equal:

$$C + \frac{K}{(1+r)^n} = P + S \quad (9.3)$$

This is the put-call parity formula. It ties together the call price C , put price P , current stock price S , strike price K , risk-free rate r , and time to expiration n . Given any four of these quantities, the fifth is determined. In particular, if one option price is known, the other can be computed without any assumption about how the stock price moves — a remarkably powerful result.

Author's remark: Put-call parity holds for European options. For American options, the relationship becomes an inequality because early exercise is possible. Also observe that the term $\frac{K}{(1+r)^n}$ is simply the present value of the strike price — precisely the discounting operation from Section 3.

Example 9.2. A European call on a stock with $S = \$55$, $K = \$50$, $n = 1$ year, and $r = 5\%$ is priced at $C = \$8.00$. What must the European put price P be?

Solution. Rearranging (9.3):

$$P = C + \frac{K}{(1+r)} - S$$

$$P = 8.00 + \frac{50}{1.05} - 55.00$$

$$P = 8.00 + 47.62 - 55.00 = \$0.62$$

9.2 Futures and Forwards

A **forward contract** is an agreement between two parties to buy and sell an asset at a specified price F on a specified future date T . Unlike an option, a forward is an obligation: both parties must fulfill the contract at expiration, regardless of where the market price has moved. No money changes hands when a forward is entered into; the agreed price F is set so that the contract has zero initial value to both parties.

A **futures contract** is economically similar to a forward but is traded on an exchange, which introduces daily settlement (marking to market) and requires both parties to post margin. These operational details are important in practice but do not affect the core pricing logic.

9.2.1 Forward Pricing

What is the fair forward price F ? The answer follows directly from the no-arbitrage principle and the compound interest formula of Section 3.

(**No-arbitrage** means that the prices in a financial market are internally consistent, so that no investor can construct a risk-free trading strategy that requires no net investment and guarantees a positive profit. In practice, this principle is used to determine fair prices: if two investments have the same future cash flows and the same risk, then they should have the same present value. Otherwise, an arbitrage opportunity would exist.)

Suppose the underlying asset currently trades at price S and pays no income (no dividends, no coupons). An investor who wants to own the asset at time T has two equivalent strategies:

- Strategy 1: Enter a forward contract today and receive the asset at time T for price F .
- Strategy 2: Buy the asset today for S and hold it until time T .

Strategy 2 requires investing S today. At rate r over n periods, that investment grows to $S(1 + r)^n$ by time T . For the two strategies to cost the same — no arbitrage — the forward price must equal this future value:

$$F = S(1 + r)^n \quad (9.4)$$

This is the forward pricing formula. It is nothing other than the compound interest formula from Section 3.2, with the current asset price S playing the role of the principal. In the above formula, r represents the risk-free interest rate—such as the yield on a government Treasury bill—for a term matching the contract's expiration. It does not depend on the riskiness of the underlying asset itself, as that risk is already fully reflected in the current spot price S .

If the asset pays dividends or other income with present value I over the life of the contract, the forward price adjusts to:

$$F = (S - I)(1 + r)^n \quad (9.5)$$

Because owning the asset earns income worth I in present-value terms, the effective cost of carrying the asset to expiration is $S - I$ rather than S .

Example 9.3. A share of stock currently trades at \$80. It will pay a semi-annual dividend equal to 2% of its current price (\$1.60 per payment) over the next 3.5 years. The risk-free rate is 4% per year, and a forward contract expires in 3.5 years ($n = 3.5$). What is the fair forward price?

Solution. First, calculate the present value of the expected dividends I . Because the risk-free rate is 4% annually, the effective semi-annual discount rate is $\sqrt{1.04} - 1 \approx 1.98$. The 3.5 years result in 7 semi-annual dividend payments, which form an ordinary annuity. Using the present value of an annuity formula from Section 4:

$$I = 1.60 \times \frac{1 - (1.0198)^{-7}}{0.0198} \approx \$10.37$$

Next, use the forward pricing formula (9.5) to find the fair forward price, adjusting the initial stock price by the present value of the dividends and compounding it over the 3.5-year term:

$$F = (S - I)(1 + r)^n$$

$$F = (80 - 10.37)(1.04)^{3.5}$$

$$F = 69.63 \times 1.1475 \approx \$79.90$$

If the forward price in the market were higher than \$79.90, an arbitrageur could profit by selling the forward and buying the stock today with borrowed funds. If the market price were lower, the reverse trade would be profitable. Competition among traders keeps the forward price close to formula (9.4) or (9.5).

It might seem counterintuitive that the forward price (\$79.90) is almost identical to the current stock price (\$80.00) after 3.5 years. This illustrates the fundamental balance in forward pricing. The risk-free interest rate pushes the forward price upward, reflecting the time value of money and the cost of tying up capital. Conversely, the asset's dividends pull the forward price downward, because the forward buyer misses out on those cash flows while waiting for expiration. Because the annualized dividend yield in this example roughly matches the 4% interest rate, these two opposing forces almost perfectly offset each other.

9.3 Swaps

A **swap** is an agreement between two parties to exchange a series of cash flows over time. The most common type is the interest rate swap, in which one party pays a fixed interest rate and the other pays a floating rate — typically tied to a benchmark such as SOFR (the Secured Overnight Financing Rate, which replaced LIBOR in most markets) — both applied to the same notional principal N . The notional principal itself is never exchanged; it is simply the basis on which interest amounts are calculated.

In a plain vanilla (fixed-for-floating) interest rate swap, at each payment date the net cash flow from the fixed-rate payer to the floating-rate payer is:

$$\text{Net payment} = (r_{fix} - r_{flt}) \times N, \quad (9.6)$$

where r_{fix} is the fixed rate and r_{flt} is the floating rate for that period, both expressed as rates per payment period. If the net payment is negative, the fixed-rate payer receives money from the floating-rate payer.

The fixed rate in a swap is set at inception so that the present value of all fixed payments equals the present value of the expected floating payments — making the initial value of the swap zero to both parties. As interest rates change over the life of the swap, the market value of the position changes, creating gains for one party and losses for the other. Computing the market value at any

point requires discounting the remaining cash flows, exactly as in the bond and annuity calculations of Sections 4 and 6.

Why use a swap? A company that has borrowed at a floating rate but prefers certainty in its interest payments can enter a swap in which it pays fixed and receives floating. The floating receipts offset the floating payments on the loan, leaving the company with a net fixed rate. Conversely, a company that wants to benefit from falling rates can convert a fixed-rate obligation into a floating one.

Ultimately, every swap is a tug-of-war over future expectations. The buyer (who pays the fixed rate) is essentially purchasing protection, betting that variable interest rates will climb higher than their fixed obligation. Conversely, the seller (who pays the floating rate) is betting that interest rates will fall or remain stable, allowing them to collect the steady fixed payments while paying out a smaller variable amount in return.

Example 9.4. A company enters a two-year swap with annual payments, paying fixed at 5% and receiving floating on a notional of \$10,000,000. In Year 1 the floating rate turns out to be 4.2%; in Year 2 it is 5.8%. Compute the net payment each year.

Solution.

- Year 1: Net payment = $(0.05 - 0.042) \times 10,000,000 = 0.008 \times 10,000,000 = \$80,000$. The company pays \$80,000 (fixed rate exceeded floating).
- Year 2: Net payment = $(0.05 - 0.058) \times 10,000,000 = -0.008 \times 10,000,000 = -\$80,000$. The negative sign means the company receives \$80,000 (floating rate exceeded fixed).

Author's remark: Other common swap types include currency swaps (exchanging cash flows in different currencies) and commodity swaps (exchanging a fixed commodity price for a floating market price). The underlying logic is the same: two parties exchange cash flows to transform an exposure they have into one they prefer.

9.4 Structured Credit Products

The instruments discussed in Sections 9.1 to 9.3 — options, futures, and swaps — are straightforward in structure, even if their pricing can be mathematically involved. A second family of derivatives, known as structured credit products, bundles together existing debt obligations and repackages their cash flows. These products became central to the 2008 financial crisis, so a clear understanding of their mechanics is worthwhile.

9.4.1 Mortgage-Backed Securities

A mortgage-backed security (MBS) is created when a financial institution pools together a large number of individual mortgage loans — perhaps thousands of home loans from different borrowers — and sells claims on the combined cash flows to investors. The investors receive the monthly principal and interest payments made by the underlying borrowers, net of a servicing fee. The MBS transforms illiquid individual mortgages into a tradable security.

The appeal to investors is diversification: even if a few borrowers default, the pool as a whole continues to generate cash flows. The appeal to the originating institution is that it can remove the loans from its balance sheet and receive cash immediately, which it can use to make new loans. This process is called **securitization**.

The mathematics of MBS pricing is an extension of the loan amortization framework of Section 5. The cash flows from the pool are essentially a large annuity, subject to uncertainty because borrowers may prepay their mortgages (when interest rates fall) or default. Prepayment models and default probabilities must be incorporated to compute a fair price, making MBS valuation considerably more complex than a standard bond.

9.4.2 Collateralized Debt Obligations

A **collateralized debt obligation** (CDO) takes structured products one step further. Instead of pooling individual loans, a financial institution creates a CDO trust that pools a collection of debt securities, which may themselves include mortgage-backed securities. Rather than selling this entire pool as a single investment, the trust divides the resulting cash flows into distinct layers called **tranches**, from the French word for slice. Investors do not buy the CDO as a whole; they choose to invest in a specific tranche based on their risk tolerance.

When the underlying borrowers make their payments, that cash flows into the trust. The trust then distributes this money to the investors based on a strict hierarchy, often called a cash flow waterfall. Each tranche has a different priority claim on the cash flows and a different exposure to loss.

The tranche structure typically includes:

- Senior tranche — paid first. The trust pays these investors their promised return before any lower tier receives money. They absorb losses only after junior tranches are wiped out, earning them a AAA rating for highest credit quality.
- Mezzanine tranche — paid second. Once the senior investors are fully compensated, the trust pays this layer. They absorb intermediate losses and typically hold a BBB rating or similar.
- Equity tranche — paid last. These investors receive whatever cash remains at the bottom of the waterfall. They absorb the first losses if underlying borrowers default. Because they take on the highest risk, this unrated or below-investment-grade layer offers the highest potential return.

The tranching process can make highly rated securities from a pool of riskier underlying assets. The senior tranche of a CDO backed by BBB-rated mortgages might itself receive a AAA rating because the probability that losses are large enough to reach the senior tranche is estimated to be very low, provided the underlying assets are sufficiently diversified and the default probabilities are correctly modeled.

Author's remark: *In the years before 2008, CDOs were frequently built from pools of subprime mortgage-backed securities. Rating agencies assigned AAA ratings to senior tranches based on models that underestimated the degree to which mortgage defaults would be correlated: when house prices fell nationwide, defaults surged everywhere at once rather than being scattered independently across borrowers. The diversification benefits largely disappeared precisely when it was most needed.*

9.4.3 Credit Default Swaps

A credit default swap (CDS) is, despite its name, not a swap in the sense of Section 9.3. It is more accurately described as a form of insurance against the default of a particular borrower or debt instrument — the reference entity.

In a CDS contract:

- The protection buyer pays a periodic fee — the CDS spread or premium — expressed in **basis points** per year on a notional amount. (A basis point equals one-hundredth of a percentage point, making it a precise way to discuss even very small rate movements.)
- The protection seller receives this premium and, if a defined credit event (such as default, bankruptcy, or failure to pay) occurs, must compensate the buyer for the loss on the reference obligation.

For example, a bank holding \$10,000,000 of corporate bonds might buy CDS protection at a spread of 150 basis points (1.5%) per year. The annual premium is $0.015 \times \$10,000,000 = \$150,000$. If the company defaults, the bonds typically do not go to zero. Suppose the bankrupt company has enough assets left to repay 40 cents on the dollar to the bond holders. This means the bank suffers a loss of 60 cents on the dollar. The CDS contract is designed to make the bank whole by covering that exact shortfall, so the protection seller pays the bank $\$10,000,000 \times 0.60 = \$6,000,000$.

CDS contracts can also be used for speculation: an investor who believes a company is likely to default can buy CDS protection without owning any bonds, effectively betting on the default. This use of CDS as a speculative instrument — rather than a hedging one — was widespread in the pre-2008 period and significantly amplified the financial system's exposure to the housing market.

9.5 Opacity, Leverage, and the 2008 Financial Crisis

By 2007, the three instruments described in Section 9.4 had become deeply intertwined. Mortgage originators issued loans to subprime borrowers; those loans were securitized into MBS; the MBS were pooled into CDOs, which were sometimes pooled again into CDO-squared structures. CDS were written on all of these securities, creating webs of insurance obligations whose full extent was unknown to any single participant. Three systemic features made this accumulation of risk fragile.

Opacity. Each step in the chain - from individual mortgage to MBS to CDO - made it harder to assess what was actually inside the security. Buyers of CDO tranches, and even the credit rating agencies assigning grades to them, often could not verify the quality of the underlying mortgages. This informational opacity meant that risk was systematically mispriced across the entire market.

Leverage. Many participants funded their holdings of these securities with borrowed money, amplifying both gains and losses. An institution that borrows 90% of the cost of an asset (leverage ratio of 10 to 1) earns ten times the return if the asset rises but also suffers ten times the loss if the asset falls. When the housing bubble peaked and house prices simply stopped rising, many subprime borrowers could no longer refinance their loans, triggering a wave of initial defaults. Because of the immense leverage in the system, these localized defaults translated into catastrophic losses for the holders of subprime MBS and CDOs.

Interconnectedness. Because CDS had been written on essentially all of the relevant securities, the failure of one institution could immediately trigger losses at the CDS counterparties, which might in turn be unable to meet their own obligations. The insolvency of a single firm — like AIG, which had

sold enormous quantities of CDS protection — threatened to cascade through the financial system in a way that no individual actor had anticipated or had the information to foresee.

Ultimately, this structure transformed localized mortgage defaults into a global financial meltdown. When a relatively small cluster of subprime borrowers defaulted on their home loans, the extreme leverage in the system caused the value of the associated mortgage-backed securities and CDOs to plummet. This collapse triggered the credit default swaps written on those securities, legally obligating insurers to pay out billions of dollars simultaneously. Because these institutions lacked the cash to cover such a massive, synchronized payout, they faced immediate bankruptcy. This threatened the solvency of the global banks relying on those insurance contracts, turning what should have been a contained real estate correction into a worldwide chain reaction.

The lesson is not that derivatives are inherently dangerous. Options, futures, and plain vanilla swaps perform valuable economic functions and continue to be traded in enormous volumes. The lesson is that complexity, opacity, and leverage, when combined with inadequate understanding of how correlation behaves under stress, can transform individually comprehensible instruments into a collectively incomprehensible system.

9.6 Summary

Table 48. Derivatives and Structured Products

Instrument	Description	Key formula / concept
Call option	Right to buy at K	$\text{Payoff} = \max(S_T - K, 0)$
Put option	Right to sell at K	$\text{Payoff} = \max(K - S_T, 0)$
Put-call parity	Relationship between C, P, S, K, r, n	$C + \frac{K}{(1+r)^n} = P + S$
Forward/Futures	Obligation to buy/sell at F on date T	$F = S(1+r)^n$
Interest rate swap	Exchange fixed for floating payments	$\text{Net} = (r_{fix} - r_{flt}) \times N$
MBS / CDO / CDS	Structured credit products	Securitization; tranching; credit insurance

9.7 Exercises

1. A put option on a stock has a strike price of $K = \$40$ and an expiry of one year. The premium paid was \$3. Compute the payoff and the net profit or loss if (a) $S_T = \$33$ and (b) $S_T = \$44$.
2. Verify the put-call parity claim in Section 9.1.4, i.e., show that Portfolio A (one call + present value of K invested at rate r) and Portfolio B (one put + one share of stock) produce the same payoff at expiration for any value of S_T .
3. A European call on a stock with $S = \$100$, $K = \$95$, $n = 1$ year, and $r = 6\%$ is priced at $C = \$12.50$. Use put-call parity to find the price of the corresponding European put.
4. A commodity currently trades at \$200 per unit. The risk-free rate is 3% per year. Compute the fair one-year forward price. If storage costs of \$5 per unit are payable at the start of the year, how does the forward price change? (Note: For physical commodities like gold or crude oil, this represents the real-world expense of secure vault space, insurance, or industrial storage tanks required to hold the asset over the year.)
5. A company enters a three-year interest rate swap, paying fixed at 4.5% and receiving floating on a notional of \$5,000,000. The floating rates turn out to be 3.8%, 4.5%, and 5.2% in years 1, 2, and 3 respectively. Compute the net payment (or receipt) each year and the total net position over the life of the swap.
6. A CDO pool contains \$100,000,000 of mortgage-backed securities. The tranche structure is: 80% senior (AAA), 15% mezzanine (BBB), 5% equity. If defaults cause losses equal to 6% of the total pool, calculate the dollar loss absorbed by each tranche. Repeat the calculation for a loss rate of 20%.
7. A bank holds \$20,000,000 of corporate bonds rated BB. It buys CDS protection at a spread of 200 basis points per year.
 - a. What is the annual premium in dollars?
 - b. If the company defaults and the bonds recover 35 cents on the dollar, what payment does the bank receive from the CDS seller?
 - c. Compare the total premiums paid over five years with the protection payment.

Hints and Answers

- (a) Payoff = $\max(40 - 33, 0) = \$7$. Net profit = $\$7 - \$3 = \$4$.

(b) Payoff = $\max(40 - 44, 0) = \$0$. The option expires worthless. Net loss is the \$3 premium.
- Hint: Evaluate the payoff of both portfolios at expiration under two scenarios: $S_T > K$ and $S_T \leq K$.

If $S_T > K$, Portfolio A is $\max(S_T - K, 0) + K = (S_T - K) + K = S_T$. Portfolio B is $\max(K - S_T, 0) + S_T = 0 + S_T = S_T$.

If $S_T \leq K$, Portfolio A is $\max(S_T - K, 0) + K = 0 + K = K$. Portfolio B is $\max(K - S_T, 0) + S_T = (K - S_T) + S_T = K$.

In both scenarios, the payoffs match, proving parity.

- Using the formula $C + \frac{K}{(1+r)^n} = P + S$:

$$12.50 + \frac{95}{1.06} = P + 100$$

$$12.50 + 89.62 = P + 100$$

$$P = \$2.12$$

- Fair forward price = $200 \times 1.03 = \$206$.
With storage treated as negative income, the effective cost of the asset today becomes $200 + 5 = 205$. The new forward price = $205 \times 1.03 = \$211.15$.
- Year 1: $(0.045 - 0.038) \times 5,000,000 = \$35,000$ (company pays).
Year 2: $(0.045 - 0.045) \times 5,000,000 = \0 .
Year 3: $(0.045 - 0.052) \times 5,000,000 = -\$35,000$ (company receives).
Total net position over the life of the swap = \$0.

- The capacities for the tranches are: Senior \$80,000,000; Mezzanine \$15,000,000; Equity \$5,000,000.

For a 6% loss (\$6,000,000): Equity absorbs \$5,000,000 (wiped out), Mezzanine absorbs \$1,000,000, Senior absorbs \$0.

For a 20% loss (\$20,000,000): Equity absorbs \$5,000,000, Mezzanine absorbs \$15,000,000 (both wiped out), Senior absorbs \$0.

- (a) $0.02 \times 20,000,000 = \$400,000$.

(b) $20,000,000 \times (1 - 0.35) = \$13,000,000$.

(c) The total premiums paid over five years (\$2,000,000) are significantly less than the protection payout received upon default (\$13,000,000).

10 Case Studies in Financial Mathematics

The four most dangerous words in investing are: “This time it’s different.” — John Templeton

The only function of economic forecasting is to make astrology look respectable.
— John Kenneth Galbraith

The preceding chapters have built a toolkit of financial mathematics concepts: compound interest and present value (Section 3), annuities (Section 4), loan repayment (Section 5), bonds (Section 6), capital budgeting (Section 7), stock valuation (Section 8), and derivatives (Section 9). Real financial decisions rarely call on just one of these tools in isolation. This chapter presents three interconnected case studies that draw on the full range of techniques developed in this book.

The studies follow Maria and Anthony DeLuca, a married couple living in Wilkes-Barre, Pennsylvania. Both are attorneys. We meet them at three stages of their lives: buying their first home and planning their children's education; leaving a corporate employer to open their own law firm; and, finally, preparing for retirement. The numbers are representative of realistic scenarios and are chosen to keep the arithmetic tractable. Readers are encouraged to verify all calculations and to experiment with alternative assumptions.

Author’s remark: Throughout this chapter we work with annual interest rates unless stated otherwise, and we apply the formulas exactly as derived in the earlier sections. References such as “formula (4.3)” refer to the corresponding formula in that section.

10.1 Case Study 1: Buying a Home and Saving for Education

Maria is 31 years old and Anthony is 33. They have two children: Anna, age 4, and William, age 2. The family currently rents an apartment and has accumulated \$18,000 in savings. They have identified a house in Wilkes-Barre priced at \$240,000 and would like to purchase it in three years. At the same time, they want to start saving for their children's college education.

10.1.1 Accumulating the Down Payment

Their lender requires a down payment of 20% of the purchase price. The required down payment is therefore:

$$0.20 \times \$240,000 = \$48,000.$$

Their existing savings of \$18,000 are deposited in an account earning 4.0% per year, compounded monthly (monthly rate $i = \frac{0.04}{12} \approx 0.003333$). After three years (36 months) this account will have grown to:

$$FV_{existing} = 18,000 \times \left(1 + \frac{0.04}{12}\right)^{36} \approx \$20,284.$$

The shortfall they must bridge with additional monthly deposits is $\$48,000 - \$20,284 = \$27,716$. They will make level monthly deposits in the amount of R into the same account (4.0% per year, compounded monthly) for 36 months. The future value of this ordinary annuity must equal the shortfall. From formula (4.3):

$$FV_{annuity} = R \times \frac{(1 + i)^n - 1}{i}$$

Setting $FV_{annuity} = \$27,716$ with $i = 0.003333$ and $n = 36$:

$$R = \frac{27,716 \times 0.003333}{(1.003333)^{36} - 1} \approx \$727 \text{ per month. (10.1)}$$

Combined with the growth of their existing savings, monthly deposits of \$727 will bring their total to approximately \$48,000 at the end of three years.

Author's remark: This calculation illustrates the power of separating a savings goal into two components: the future value of current assets and the future value of new contributions. Both rely on the compound-interest and annuity formulas of Sections 3 and 4.

10.1.2 The Mortgage

At purchase, Maria and Anthony borrow $\$240,000 - \$48,000 = \$192,000$ at a fixed annual rate of 6.5%, compounded monthly, for 30 years (360 months). The monthly rate is $i = \frac{0.065}{12} \approx 0.005417$. The monthly payment is found from the loan repayment formula (Section 5):

$$R = \frac{P \cdot i}{1 - (1 + i)^{-n}}$$

$$R = \frac{192,000 \times 0.005417}{1 - (1.005417)^{-360}} \approx \$1,213 \text{ per month. (10.2)}$$

To see how each payment is allocated, we apply the amortization formula from Section 5. In the first month, the interest component is:

$$Interest_1 = 192,000 \times 0.005417 \approx \$1,040$$

Therefore, the principal repaid in the first payment is $\$1,213 - \$1,040 = \$173$. The outstanding balance after the first payment is $\$192,000 - \$173 = \$191,827$. After 12 payments (one year) the outstanding balance, computed by the prospective method (Section 5.4), is:

$$B_{12} = R \times \frac{1 - (1 + i)^{-348}}{i} \approx \$189,863$$

In the first year, they have repaid only $\$192,000 - \$189,863 = \$2,137$ of principal while paying $\$1,213 \times 12 - \$2,137 = \$12,419$ in interest. This illustrates the front-loading of interest discussed in Section 5: in the early years of a long mortgage, the vast majority of each payment covers interest rather than principal.

10.1.3 Education Savings for Anna and William

Maria and Anthony estimate that each child will need approximately \$100,000 at the start of college to cover four years of tuition, fees, and living expenses at a Pennsylvania university. They open separate 529 savings accounts for each child, earning 6.0% per year compounded annually.

(In the United States, a 529 plan is a tax-advantaged savings account designed to help families save for education expenses, including college, trade school, vocational school, and K-12 tuition. Earnings grow tax-free, and withdrawals are tax-free when used for qualified education expenses.)

Anna will begin college in 14 years (she is 4 now and will start at 18). To accumulate \$100,000 in 14 years with annual deposits of R_A , we solve the annuity formula below for R_A :

$$100,000 = R_A \times \frac{1.06^{14} - 1}{0.06}$$

$$R_A = \frac{100,000 \times 0.06}{1.06^{14} - 1} \approx \$4,759 \text{ per year } (\$397 \text{ per month}). \quad (10.3)$$

William will begin college in 16 years. A similar calculation gives:

$$R_W = \frac{100,000 \times 0.06}{1.06^{16} - 1} \approx \$3,895 \text{ per year } (\$325 \text{ per month}). \quad (10.4)$$

The additional two years of compounding for William's account means that somewhat smaller annual deposits are required, a direct application of the time-value principle.

Table 49 summarizes the family's monthly financial commitments after purchasing the house.

Table 49. Monthly financial commitments after home purchase

Item	Monthly amount
Mortgage payment	\$1,213
Anna's education savings	\$397
William's education savings	\$325
Subtotal (new commitments)	\$1,935

Author's remark: *The education savings deposits of \$397 and \$325 per month are modest in the context of the family's income, but the choice of a 6% growth assumption deserves scrutiny. A 529 account invested in stocks has historically averaged returns above 6% over long periods, but with significant year-to-year volatility. The calculation above assumes the average is achieved smoothly, which is a simplifying assumption. The actual accumulated amount will differ from \$100,000 unless the return is exactly 6% every year. This is a fundamental limitation of deterministic financial models.*

10.2 Case Study 2: Opening Deluca Law

Four years have passed. Maria is now 35 and Anthony is 37. The children are 8 and 6. Maria and Anthony have been working at a mid-sized regional firm, earning a combined salary of \$165,000 per year. They have grown increasingly confident in their client base and their ability to run a practice independently. They decide to open their own law firm, which they name Deluca Law.

Starting a law firm requires initial capital for office space, furniture, computer equipment, malpractice insurance, and operating reserves. Maria and Anthony estimate total startup costs of \$100,000. They contribute \$25,000 of personal savings and obtain an SBA (Small Business Administration) loan of \$75,000.

10.2.1 The Business Loan

The SBA loan carries an annual interest rate of 7.0%, compounded monthly, with a term of 7 years (84 months). The monthly payment is:

$$R = \frac{75,000 \times 0.005833}{1 - (1.005833)^{-84}} \approx \$1,132 \text{ per month} \quad (10.5)$$

where $i = \frac{0.07}{12} \approx 0.005833$ is the monthly interest rate. The annual loan service cost is $\$1,132 \times 12 \approx \$13,584$.

Their estimated annual operating costs (excluding their own compensation) are:

Table 50. Estimated annual operating costs for Deluca Law

Cost item	Annual amount
SBA loan service	\$13,584
Office rent	\$36,000
Staff (one paralegal)	\$52,000
Insurance, utilities, misc.	\$48,416
Total operating costs (exc. compensation)	\$150,000

10.2.2 Net Present Value Analysis

To evaluate whether starting the firm is financially worthwhile, Maria and Anthony perform a capital budgeting analysis using the NPV method from Section 7. The relevant cash flow is the incremental income relative to their existing employment income of \$165,000 per year. That is, they compare what the firm will pay them to what they would have earned as employees.

Based on their client relationships and industry knowledge, they project the following revenue trajectory for the firm's first seven years. Their combined compensation from the firm equals revenue minus operating costs:

Table 51. Projected revenue and incremental income for Deluca Law (years 1–7)

Year	Revenue	Net Comp. = Revenue - Operating Costs	Incremental vs. \$165K
1	\$210,000	\$60,000	−\$105,000
2	\$265,000	\$115,000	−\$50,000
3	\$315,000	\$165,000	\$0
4	\$360,000	\$210,000	+\$45,000
5	\$395,000	\$245,000	+\$80,000
6	\$425,000	\$275,000	+\$110,000
7	\$450,000	\$300,000	+\$135,000

The incremental cash flows in column 4 represent the gain (or loss) from opening the firm relative to staying employed. The initial outlay of personal capital is \$25,000. Using a discount rate of $r = 8\%$ per year (their estimated opportunity cost of capital), the NPV of the investment is:

$$NPV = -25,000 + \sum_{t=1}^7 \frac{C_t}{(1+r)^t}$$

Substituting the values from Table 51 with $r = 0.08$:

$$NPV = -25,000 + \frac{-105,000}{1.08} + \frac{-50,000}{1.08^2} + \dots + \frac{135,000}{1.08^7} \approx \$70,523.70 \quad (10.6)$$

The positive NPV indicates that, under these projections and at an 8% discount rate, opening Deluca Law creates approximately \$70,527 of value relative to remaining employed. This figure represents the present value (in today's dollars) of the incremental lifetime compensation from the firm, net of the \$25,000 personal investment.

Author's Remark: The NPV calculation above evaluates the incremental cash flows, directly isolating the marginal financial impact of opening the firm. Alternatively, this decision can be modeled as a comparison of two mutually exclusive projects. In that approach, you calculate the absolute present value of remaining employed and subtract it from the absolute present value of operating the new firm, accounting for the initial outlay. Because the mathematics of discounting are linear, subtracting the present values of the two distinct paths yields the identical net present value of +\$70,527. While the incremental approach is standard in capital budgeting, analyzing the total present value of each option independently is a mathematically equivalent and equally valid perspective.

10.2.3 Payback Period

The NPV analysis tells us about value creation over the full seven-year horizon. Maria and Anthony are also interested in a simpler question: when will the firm's cumulative incremental income turn positive? This is the payback period (Section 7.3).

Summing the incremental cash flows, we have the following table:

Table 52. Cumulative incremental income (undiscounted)

Year	Annual increment	Cumulative
0	-\$25,000 (initial)	-\$25,000
1	-\$105,000	-\$130,000
2	-\$50,000	-\$180,000
3	\$0	-\$180,000
4	+\$45,000	-\$135,000
5	+\$80,000	-\$55,000
6	+\$110,000	+\$55,000
7	+\$135,000	+\$190,000

The cumulative total turns positive between Year 5 and Year 6. Interpolating linearly:

$$\text{Payback period} \approx 5 + \frac{55,000}{110,000} = 5.5 \text{ years.} \quad (10.7)$$

Using discounted cash flows instead (the discounted payback period) shifts this to approximately 6 years, since early negative cash flows are discounted less than later positive ones.

Author's remark: The NPV and payback analyses both rest on projected cash flows that are inherently uncertain. Revenue in the first few years could be lower than projected if client acquisition is slower than expected, or higher if Maria and Anthony bring a larger book of business. Sensitivity analysis — repeating the calculation under pessimistic and optimistic scenarios — is essential before making a decision of this magnitude. Section 7.5 discusses sensitivity analysis in the context of capital budgeting.

10.3 Case Study 3: Retirement Planning

Twenty-four years have passed since Case Study 2. Maria is now 59 and Anthony is 61. The children, Anna (32) and William (30), have long since graduated from college and are established in their careers. Deluca Law has grown into a successful practice, allowing Maria and Anthony to use the strong cash flow to make additional principal payments and pay off the mortgage on the Wilkes-Barre house a few years ago. Maria and Anthony are thinking seriously about retirement, with Anthony expecting to retire at 65 and Maria at 63.

Over their careers they have made systematic contributions to tax-advantaged retirement accounts (IRAs and a SEP-IRA through the firm). We model their savings in two phases: the years when they were building their practice (lower contributions) and the later years of peak earning (higher contributions).

10.3.1 Retirement Savings Accumulation

For the seven years of Case Study 2 (ages 35–42 for Maria, the benchmark), combined annual retirement contributions averaged \$15,000 per year. For the following 21 years (ages 42–63), contributions averaged \$35,000 per year. All funds earn 7.0% per year compounded annually.

The future value at retirement (age 63 for Maria) of the first phase, computed as the future value of an ordinary annuity grown forward an additional 21 years:

$$FV_{\text{phase 1}} = 15,000 \times \frac{1.07^7 - 1}{0.07} \times 1.07^{21} \approx \$537,487 \quad (10.8)$$

The future value of the second phase (21 years of contributions, computed at retirement):

$$FV_{\text{phase 2}} = 35,000 \times \frac{1.07^{21} - 1}{0.07} \approx \$1,570,281. \quad (10.9)$$

The total retirement nest egg at the time of Maria's retirement is approximately:

$$\text{Total savings} = \$537,487 + \$1,570,281 = \$2,107,768 \approx \$2.1 \text{ million.}$$

10.3.2 How Much Do They Need?

Maria and Anthony estimate that they will need \$90,000 per year in retirement income, in today's dollars. They expect inflation to average 2.5% per year over their retirement, and they plan for a retirement horizon of 30 years. They will also receive combined Social Security benefits of approximately \$42,000 per year (in today's dollars), so the portfolio must supply the remaining \$90,000 – \$42,000 = \$48,000 per year in real terms.

Author's remark: This model treats the Social Security benefit as a constant real cash flow, meaning it implicitly assumes the nominal payout increases at the exact rate of inflation every year to preserve its purchasing power. This aligns with the historical structure of statutory cost of living adjustments for these benefits. Maintaining a constant real value allows for the direct subtraction of the benefit from the total retirement need to find a level shortfall. This shortfall can then be

discounted using a constant real interest rate. If the benefit were fixed in nominal terms instead, its real value would decline over time, which would require a more complex model featuring a progressively increasing real withdrawal from the investment portfolio.

We assume the portfolio earns a real return of 3.0% per year (that is, 3% above inflation) during retirement. The present value of a 30-year annuity of \$48,000 per year at a 3.0% real rate gives the required nest egg:

$$PV_{needed} = 48,000 \times \frac{1 - 1.03^{-30}}{0.03} \approx \$940,000. \quad (10.10)$$

Their projected savings of \$2.1 million exceeds this requirement by a substantial margin. Even after setting aside funds for contingencies, bequests, and large one-time expenses such as long-term care, the DeLucas appear to be in a strong financial position.

Author's remark: *The figure of \$940,000 is the amount needed to support the portfolio-funded portion of spending. The full \$2.1 million nest egg is not "surplus" — it also needs to generate inflation-adjusted income, and the real return of 3% is not guaranteed. A more conservative 1.5% real return would raise the required nest egg to about \$1.2 million, still well within reach. Financial planners typically run scenarios across a range of real return and inflation assumptions to stress-test a retirement plan.*

10.3.3 Portfolio Allocation and the Gordon Growth Model

Maria and Anthony plan to retire with a portfolio split 40% in investment-grade bonds and 60% in a diversified equity index fund. We examine each component.

Bond portfolio. The \$840,000 bond allocation ($40\% \times \$2.1\text{M}$) is invested in bonds yielding 4.0% per year. Annual income from this allocation: $\$840,000 \times 0.04 = \$33,600$.

Equity portfolio. The \$1,260,000 equity allocation ($60\% \times \$2.1\text{M}$) is invested in a broad stock index. The index currently offers a dividend yield of 2.0% and dividends have historically grown at approximately 5.0% per year. We can apply the Gordon Growth Model from Section 8.3 to infer the total return the market is pricing in:

$$r = \frac{D_1}{P} + g = 0.02 + 0.05 = 7.0\%. \quad (10.11)$$

Thus, a 2% dividend yield plus a 5% expected capital gains yield implies a total market return of 7%. This is consistent with the 7% assumption used in equations (10.8) and (10.9) to compute savings accumulation — a reassuring internal consistency. The annual dividend income from the equity allocation is approximately $\$1,260,000 \times 0.02 = \$25,200$.

Combined income from the portfolio:

Table 53. First-year portfolio income at retirement

Source	Annual income
Bond interest ($40\% \times \$2.1\text{M}$ at 4%)	\$33,600
Equity dividends ($60\% \times \$2.1\text{M}$ at 2%)	\$25,200
Total portfolio income (year 1)	\$58,800

The \$58,800 of portfolio income (year 1) plus \$42,000 of Social Security gives a total first-year income of \$100,800 — modestly above their \$90,000 target. The equity dividends are expected to

grow at 5% per year, providing a natural inflation hedge for a portion of the income. The bond income is fixed in nominal terms and will lose purchasing power over time, which is one reason a portfolio with a substantial equity component is generally recommended for retirees with a long time horizon.

Author's remark: *The equity income of \$25,200 per year is based on dividends alone, setting aside capital appreciation. Over a 30-year retirement, the portfolio is not simply consumed; the equity portion grows in value as share prices rise, providing a buffer. A full retirement income analysis would model systematic withdrawals — a topic closely related to the annuity mathematics of Section 4.*

10.4 Concluding Remarks

The three case studies trace a complete financial life cycle, from a young family accumulating a down payment to a retired couple drawing on decades of savings. Several themes emerge.

Compounding is the engine. Every major calculation — savings accumulation, mortgage amortization, education funding, business loan repayment, retirement nest egg, and stock valuation — reduces ultimately to the compound interest formula derived in Section 3. Time and interest rate interact to produce effects that are often larger than intuition suggests.

The time value of money imposes discipline. Discounting future cash flows to the present — the logic of NPV — forces Maria and Anthony to weigh the sacrifices of the early firm years against the later gains. Without this framework, the large negative cash flows in years 1 and 2 might appear prohibitive. The NPV of +\$70,527 shows that those losses are outweighed by future gains, properly valued.

Mathematics provides structure, not certainty. Every number in this section is a projection. Revenue could be lower, inflation higher, investment returns worse. The value of the formulas lies not in their ability to predict the future with precision, but in their ability to make explicit which assumptions drive which conclusions — and thus to identify the most important uncertainties to monitor.

The models connect. The Gordon Growth Model from Section 8 gives a required return of 7% for a diversified equity portfolio in Case Study 3, which is exactly the rate used to compound retirement savings in equations (10.8) and (10.9). The annuity formula used to set Anna's education savings target in equation (10.3) is the same formula used to compute the SBA loan payment in equation (10.5) — the same mathematics, applied to different contexts. Financial mathematics is a unified discipline.

Maria and Anthony's story is, of course, a simplification. Real financial planning involves taxes, estate planning, insurance, behavioral biases, and a great deal of uncertainty. But the mathematical skeleton traced in these pages — present value, annuities, amortization, NPV, and dividend valuation — is real. Mastering it is an essential step toward understanding the financial decisions that shape a lifetime.

10.5 Exercises

1. In Case Study 1, Maria and Anthony have \$18,000 in savings earning 4.0% per year compounded monthly. Verify that this account grows to approximately \$ 20,290.89 after 36 months. What monthly deposit R is required so that the combined balance reaches exactly \$48,000 at the end of three years? Compare your answer to equation (10.1).
2. In Section 10.1.2, the monthly mortgage payment was computed as approximately \$1,213. Using the amortization method of Section 5, compute
 - a. the interest and principal components of the 60th payment (i.e., after five years), and
 - b. the outstanding balance at the end of year 10. How do these figures compare to the year-1 values reported in the text?
3. Suppose Maria and Anthony can afford to deposit only \$300 per month toward Anna's education (instead of \$397), at the same 6.0% annual rate. How much will Anna's account contain when she starts college in 14 years? What single lump-sum deposit today (at the same 6.0% rate) would produce \$100,000 in 14 years?
4. Reproduce the NPV calculation in equation (10.6) by discounting each of the seven incremental cash flows in Table 51 at 8% per year and summing.
 - a. Confirm the total of approximately +\$70,527.
 - b. Recalculate the NPV using a discount rate of 12% instead of 8%.
 - c. At approximately what discount rate does the NPV equal zero? (This is the internal rate of return, IRR.)
5. In Section 10.2.3, the undiscounted payback period was computed as approximately 5.5 years. Compute the discounted payback period by discounting each annual increment at 8% per year and finding the year in which cumulative discounted cash flows (including the initial \$25,000 outlay) first become positive.
6. In Case Study 3, equation (10.8) gives the future value of Phase 1 contributions (years 1–7) as approximately \$537,487 at retirement. Verify this calculation, then confirm equation (10.9) for Phase 2. What would the total nest egg be if contributions in Phase 2 averaged \$40,000 per year instead of \$35,000?
7. In Section 10.3.2, the required nest egg was computed at a 3.0% real rate of return.
 - a. Recalculate the required nest egg using a 1.5% real rate.
 - b. Recalculate using a 4.5% real rate.
 - c. Comment on what these results tell you about the importance of managing investment costs and inflation during retirement.
8. In Section 10.3.3, the Gordon Growth Model is used to infer a 7% required return for equities from a 2% dividend yield and 5% growth rate.
 - a. If the dividend yield rose to 2.5% (because stock prices fell) but the growth rate stayed at 5%, what would be the implied required return?
 - b. Interpret this result in terms of the price-return relationship discussed in Section 8.4.

Acronyms and Symbols

Only when the tide goes out do you discover who's been swimming naked. — Warren Buffett

AER – Annual Equivalent Rate

APY – Annual Percentage Yield

CDO – Collateralized Debt Obligation

CDS – Credit Default Swap

CSV – Comma-Separated Values

DPP – Discounted Payback Period

EIR – Effective Interest Rate

EMI – Equated Monthly Installment

FV – Future Value

GUI – Graphical User Interface

IRA – Individual Retirement Account

IRR – Internal Rate of Return

LIBOR – London Interbank Offered Rate

MBS – Mortgage-Backed Security

MIRR – Modified Internal Rate of Return

NPV – Net Present Value

P – typically used to represent Principal

P/E – Price-to-Earnings Ratio

PI – Profitability Index

PV – Present Value

SBA – Small Business Administration

SEP-IRA – Simplified Employee Pension Individual Retirement Account

SOFR – Secured Overnight Financing Rate

TV – Terminal Value

YTM – Yield to Maturity

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Prediction is very difficult, especially about the future. — often attributed to Niels Bohr

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